

MAGNETOM

Replacement of Parts System

Magnet System

Symphony, A Tim System

Document Version

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This chapter provides an overview of the field replaceable units (FRUs) for the magnet system.



Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" (e.g. compressor) and the protective earth wire connected to the "main component" (e.g. SEP).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

Tab. 1 Magnet

FRU (Magnet / Page 13)	Time	Instructions	Final Checks	Tools
Burst disk for quench valve	120 min Additional ramping time	(→ Installing a new bursting disk / Page 13)	Leakage Check	Torque wrench 0.8 Nm Allen key 4 mm, 3 mm sockets Socket wrench Helium detector Heat gun Vacuum grease Thread locking compound (e.g., Loctite) Tissue paper
Baffle, current lead, syphon seal configuration	45 min	(→ Baffle; Current Lead; Syphon Seal Configuration / Page 23)	Leakage Check	Allen keys
Pressure gauge	45 min	(→ Replacing the pressure gauge / Page 24)	Leakage Check	Non-magnetic tool kit
Smart syphon	45 min	(→ Replacing the smart syphon / Page 25)	Leakage Check	Non-magnetic tool kit

Tab. 2 Refrigerator

FRU (Refrigerator / Page 27)	Time	Instructions	Final checks	Tools
Displacer	5 hours	(→ Replacing the Displacer - Leybold system / Page 27) or (→ Replacing the Displacer - APD systems / Page 55)	■ Cooling temperatures ■ Leakage Check ■ Helium pressure of the refrigerator system	■ Displacer exchange kit – 10119990 ■ Displacer exchange tool kit (Trio system) order number 8395753 ■ 1 x 50 liter (200 bar) cylinder of helium gas 99.999% purity. ■ Helium leak detector, capable of detecting leaks to 1×10^{-4} mbar l/s. ■ Cotton gloves ■ Dust mask (material number 07014868) ■ Lint-free paper towelling. ■ Petroleum ether or alcohol-based cleaning fluid which leaves no oily residue ■ Cold Head Service tool ■ Pressure regulator 200/50 bar

FRU (Refrigerera- tor / Page 27)	Time	Instructions	Final checks	Tools
10K Cold head	160 min (Ramping time is not included)	(→ Replacing the 10K Cold Head / Page 94)	■ Leakage check ■ Efficiency test	■ Standard tool kit ■ Cold head exchange kit (10119989) ■ 3600 Magnet Power Supply kit (08395613) ■ Apiezon N grease (10099740) ■ Torque wrench(5 - 20 Nm) ■ 10, 13, 19 mm wrench/ spanner ■ 5 and 6 mm hexagon Allen key (5 mm hexagon ball driver) ■ Flat blade screwdrivers. ■ Bottle 99.999% (5.0) helium gas and high pressure regulator ■ Cotton gloves ■ De-greasing fluid ■ Protective clothing ■ Lint free paper towelling ■ O2 monitor (room or personal). (Personal monitor material number 10142116)

FRU (Refrigerator / Page 27)	Time	Instructions	Final checks	Tools
Compressor APD For the correct konfiguration refer to the SPC.	60 min	(→ Replacement of the Compressor (APD) / Page 119)	■ Check helium pressure ■ Leakage Check	Screwdriver Open-ended wrench: 13 mm Open-ended wrench: 1"; 1 3/16"
Compressor F70 For the correct konfiguration refer to the SPC.	120 min	(→ Replacing the F70 Compressor with Tooling kit / Page 122)	■ Check helium pressure ■ Leakage Check	■ Transportation lifting kit (material number 11060700, F70 compressor kit) ■ 14-mm and 17-mm open-ended wrench ■ 17-mm socket ■ Torque wrench (0–20 Nm) ■ 8-mm Allen (Hex) key ■ 7-mm socket and ratchet

Tab. 3 Magnet Supervisory

FRU (Magnet Supervisory Unit / Page 166)	Time	Instructions	Final checks	Tools
MSUP	90 min	(→ Replacing the Magnet Supervisory Unit (MSUP) / Page 166) or (→ Replacing the Magnet Supervisory Unit (MSUP 2657B) / Page 173)	ERDU test	Non-magnetic tool kit ERDU test load
MSUP battery	40 min	(→ Replacing the MSUP battery / Page 192)	n.a.	Non-magnetic tool kit Screwdriver, 10 mm nut, 5 mm Allen key.

Tab. 4 Alarm Box

FRU (Alarm Box / Page 195)	Time	Instructions	Final checks	Tools
Alarm box	60 min	(→ Replacing the Alarm box and cables with the Magnet at Field / Page 195)	- Check LEDs - ERDU test	Non-magnetic tool kit ERDU test load

Tab. 5 Shim Filter Unit

FRU (Shim power supply / Page 199)	Time	Instructions	Final checks	Tools
Shim power supply	60 min	(→ Replacing the Shim Power Supply (SPSU) / Page 199)	■ Phantom Shim ■ Shim Check	Standard tool set
Shim Filter Unit	60 min. additional ramping time	(→ Replacing the Shim Filter Unit / Page 201)	■ Phantom Shim ■ Shim Check	Standard tool set

2.1 Installing a new bursting disk



WARNING

Risk of asphyxiation

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogenes.

- » Helium gas must not be vented into the magnet room.
- » Ensure that the turret venting kit is connected to the quench pipe.



CAUTION

Magnet field and pressurized helium vessel

If not observed, serious injury or property damage can occur

- » The magnet must be de-energized and depressurized before attempting this procedure



Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" (e.g. compressor) and the protective earth wire connected to the "main component" (e.g. SEP).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

2.1.1 Required Tools

- Torque wrench capable of measuring 0.8 Nm
- Allen key with 4 mm, 3 mm sockets
- Socket wrench
- Vacuum grease
- Thread locking compound (e.g., Loctite)
- Detector (leakage meter)
- Tissue paper

2.1.2 Spare parts

- Bursting disk (20 psi)
- O-ring

- Fiber gasket

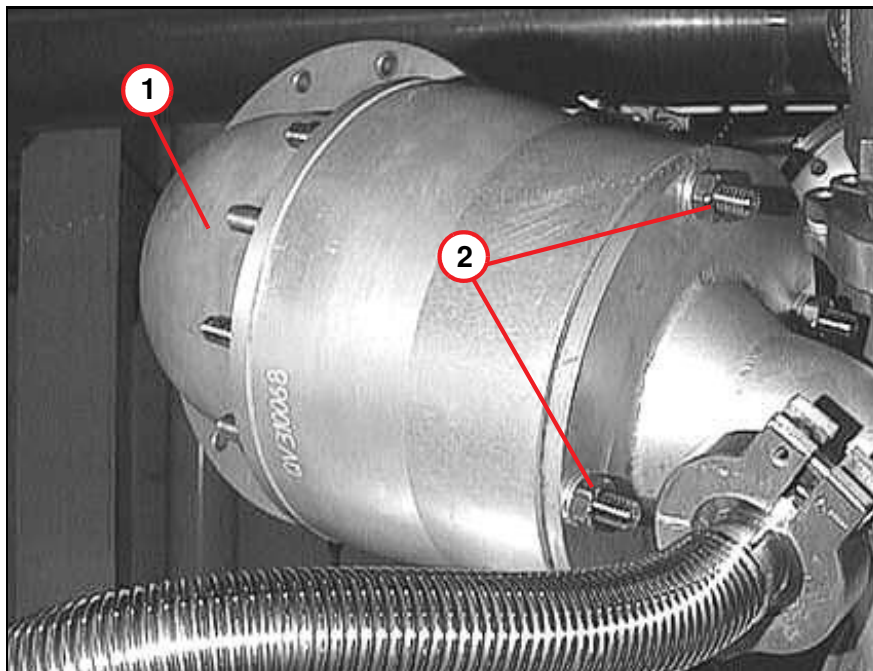
2.1.3 Replacement procedure

- Close the depressurization valve on the venting system
- Remove quench tube from 90° elbow
- Remove 90° elbow from quench valve
- Remove quench valve from turret. Record its orientation



Remove the 4 screws uniformly;
The valve is spring-loaded

Fig. 1: Quench valve



- (1) Elbow
(2) Screw

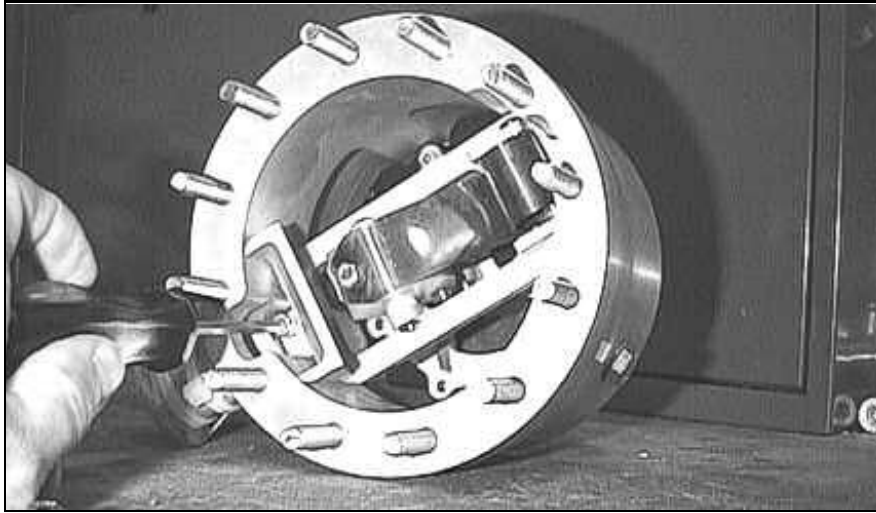
- Remove the quench valve +
- Remove the gasket



While removing the valve, the turret flange must be temporarily blanked using tissue paper. The following steps should be performed quickly to keep air from entering.

- Unscrew and remove the two M5 screws

Fig. 2: Quench valve



- Remove the spring plate assembly and the bursting disk assembly

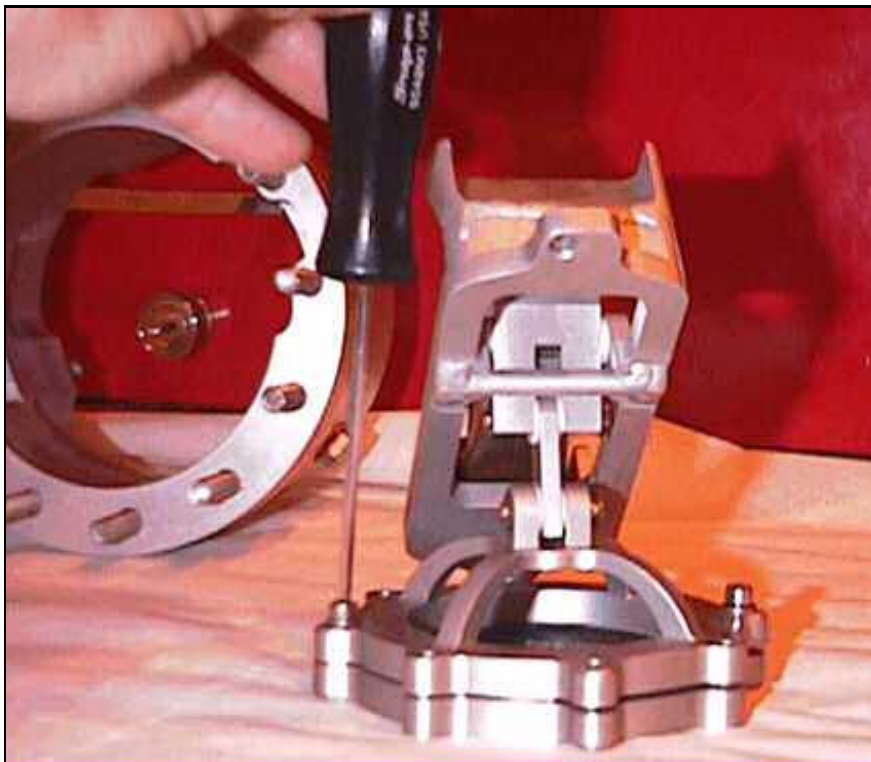
Fig. 3: Spring plate assembly



- Inspect the burst disk. Ruptured or cracked disks must be replaced.

- Unscrew and remove the eight M4 screws

Fig. 4: Burst disc assembly



- Remove broken burst disk

Fig. 5: Broken burst disc

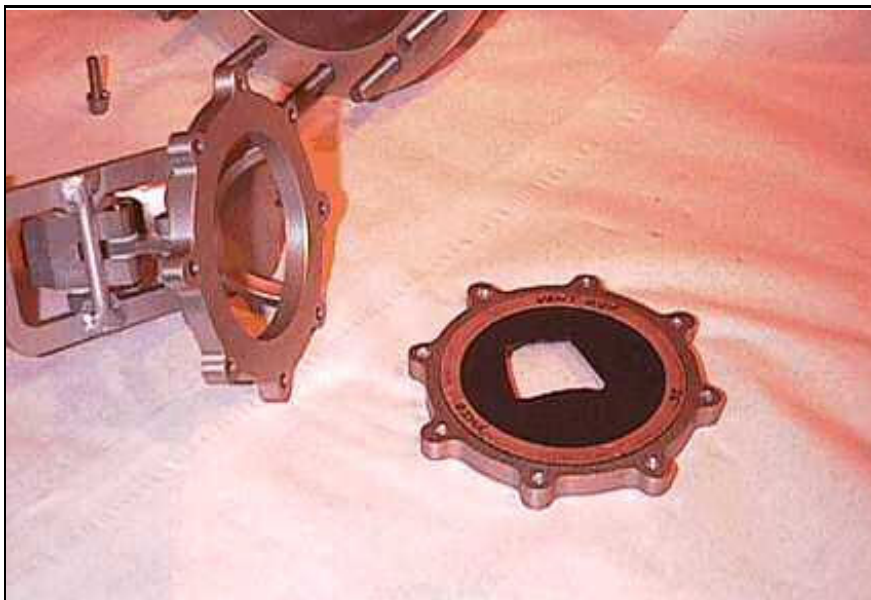
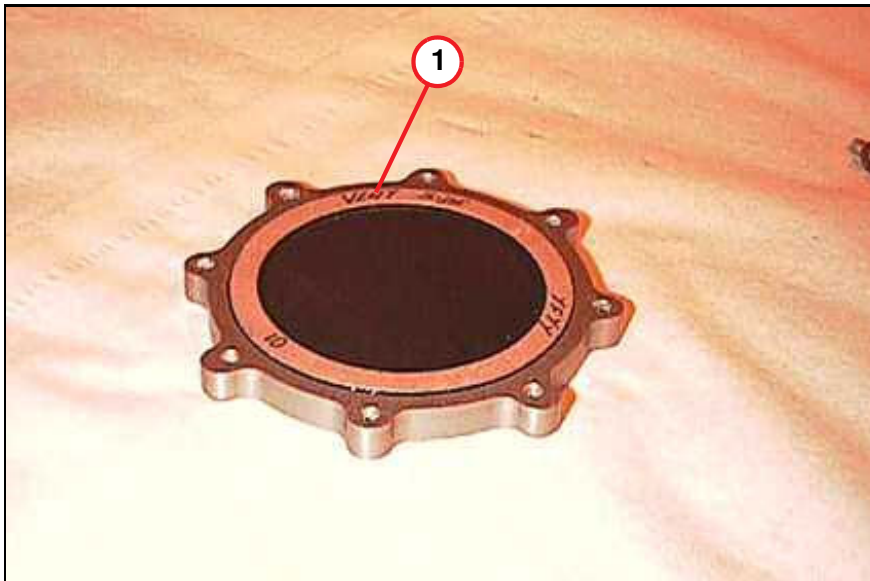


Fig. 6: Broken burst disc



- Fit the new burst disc into the burst disc assembly as specified.
The words "Vent side" written on the burst disk gasket should be on top.

Fig. 7: Burst disc assembly



(1) Vent side

- Reassemble the burst disk assembly.

The eight M4 screws and plain washers should be torqued to 0.8 Nm and locked with Perma-bond

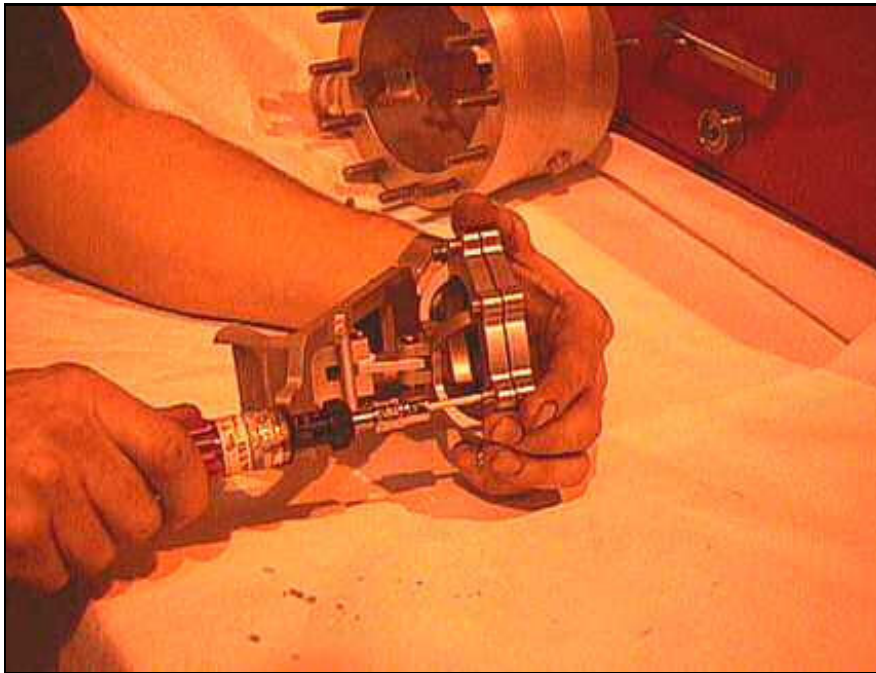
Fig. 8: Burst disc assembly



Fig. 9: Thread locking compound

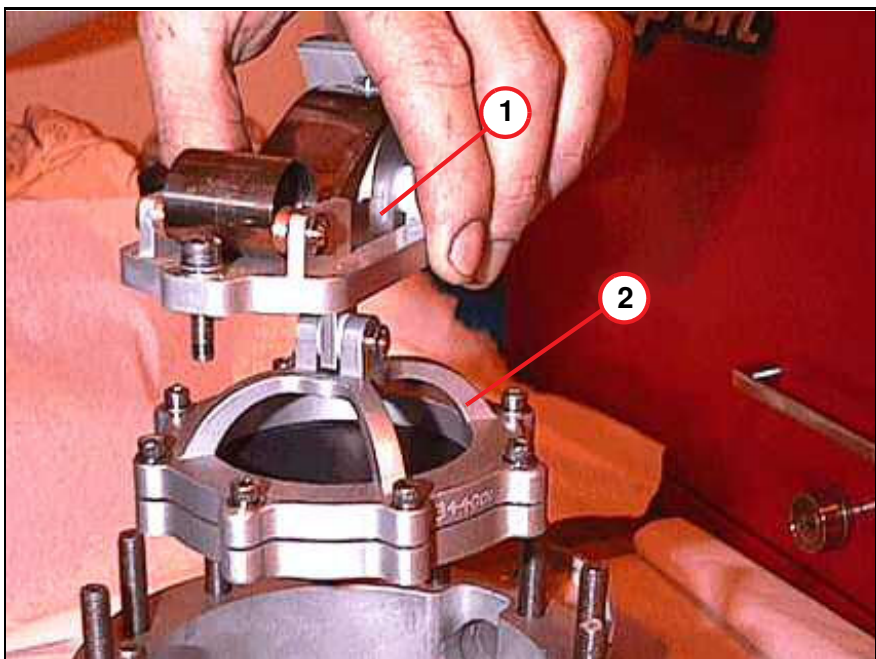


Fig. 10: Torque wrench



- Refit the spring plate assembly and burst disk assembly into the quench valve body with the two M5 screws and washers. Torque evenly to 5 Nm

Fig. 11: Mk1 Refitting the spring plate



- (1) Spring plate
- (2) Burst disk assembly

- Before refitting the quench valve assembly to the turret flange, apply a **thin film** of vacuum grease to the turret flange, which fits into the fiber gasket.

Inspect the O-ring seal in the bursting disk assembly

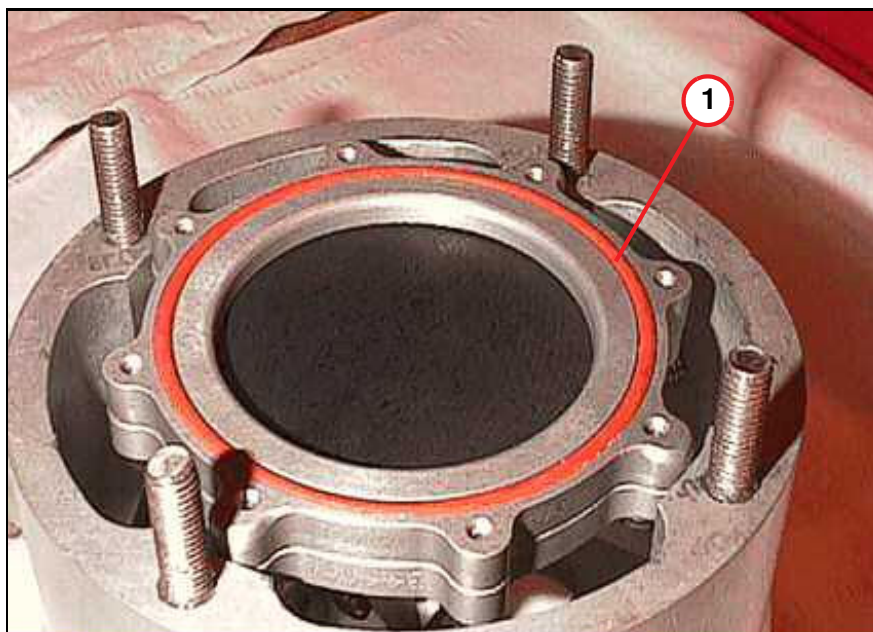
Inspect the fiber gasket that fits to the turret flange

If either seal appears damaged, it should be replaced



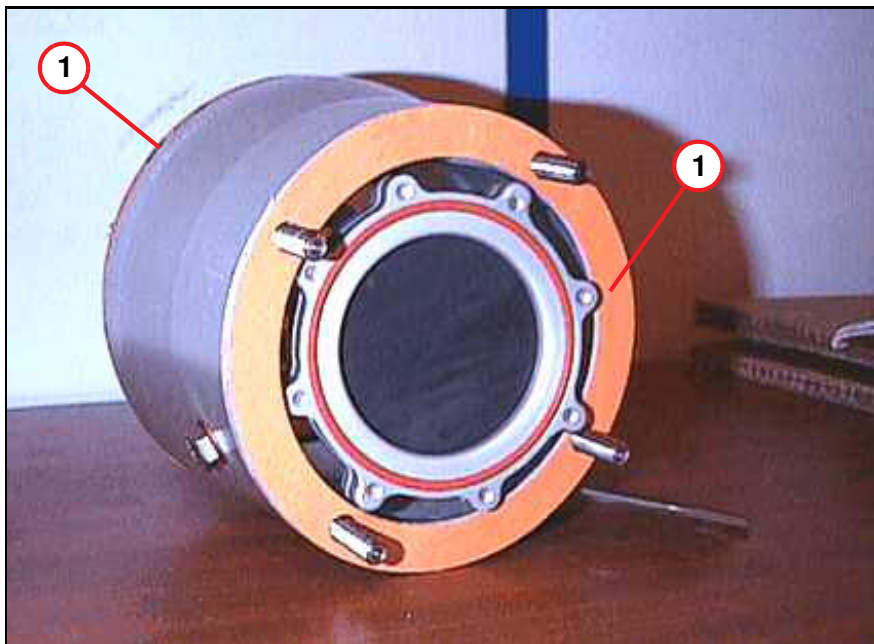
Do not use vacuum grease for the O-ring and the mating turret flange. The O-ring and the fiber gasket are part of the magnet spare kit

Fig. 12: Inspecting the O-ring



(1) O-ring

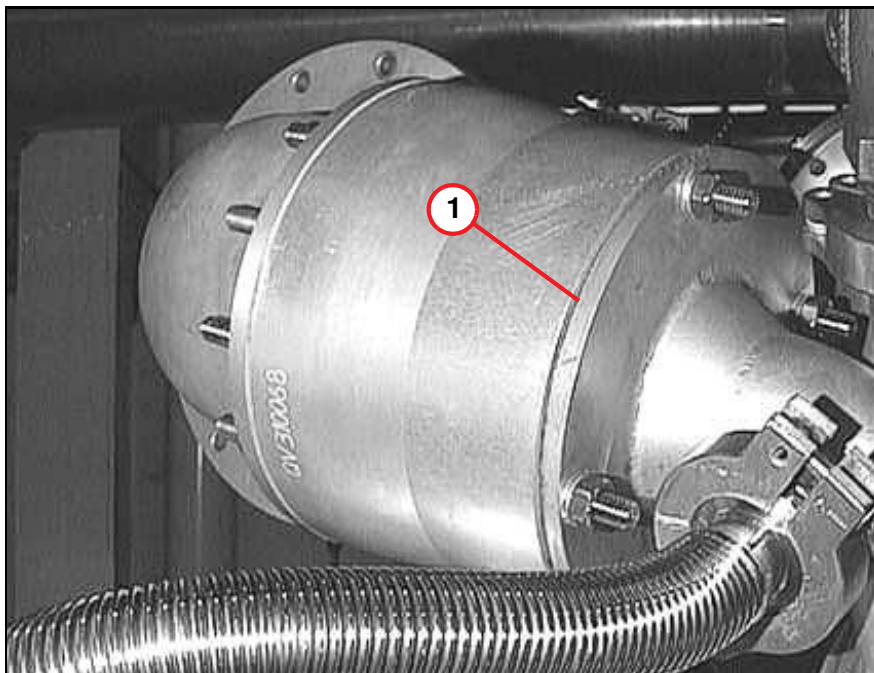
Fig. 13: Mk1 Inspecting the fiber gasket



(1) Fiber gasket

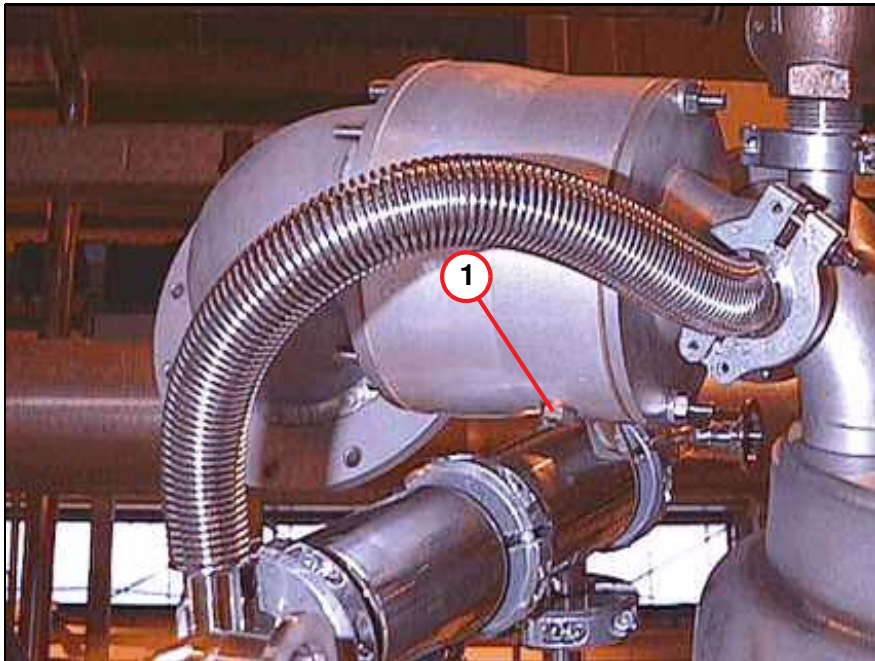
- Remove the tissue paper used to temporarily seal the turret flange, and ensure that the O-ring sealing surface is clean before refitting the quench valve assembly
- Refit the quench valve assembly to the turret flange
Ensure correct orientation
Torque to 0.8 Nm

Fig. 14: Quench valve



(1) Turret flange

Fig. 15: Orientation of the quench valve



(1) Drain port

2.1.4 Final check

- Wait for the system to pressurize slightly, then check integrity of the valve using a helium detector, or check the pressure gauge. If the pointer stays in the yellow/green range, the valve is tight.

If a leak is present, the quench valve should be removed, the turret flange sealed, and the burst disk assembly disassembled and reassembled again, with particular focus on the sealing faces.

- Reconnect the venting elbow and the venting line to the quench valve body
- Replace the identification tag of the broken disk, which is fitted to the turret flange, with the identification tag of the new bursting disk

2.2 Baffle; Current Lead; Syphon Seal Configuration

Fig. 16: Baffle plug seal configuration

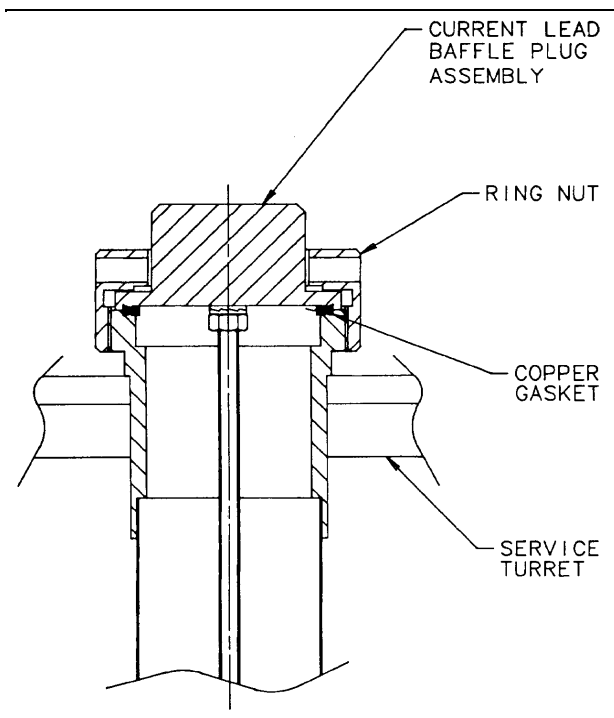


Fig. 17: Current lead seal configuration

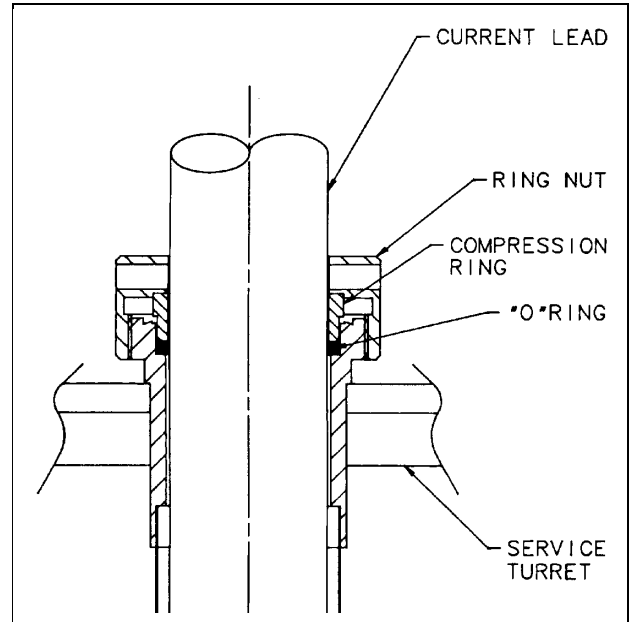
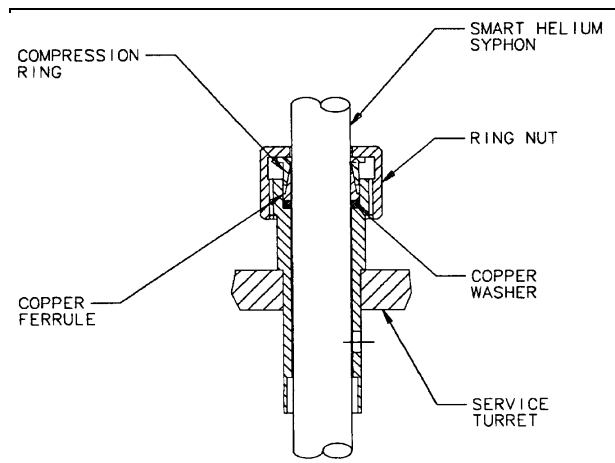


Fig. 18: Helium syphon seal



2.3 Replacing the pressure gauge



CAUTION

Pressurized helium vessel

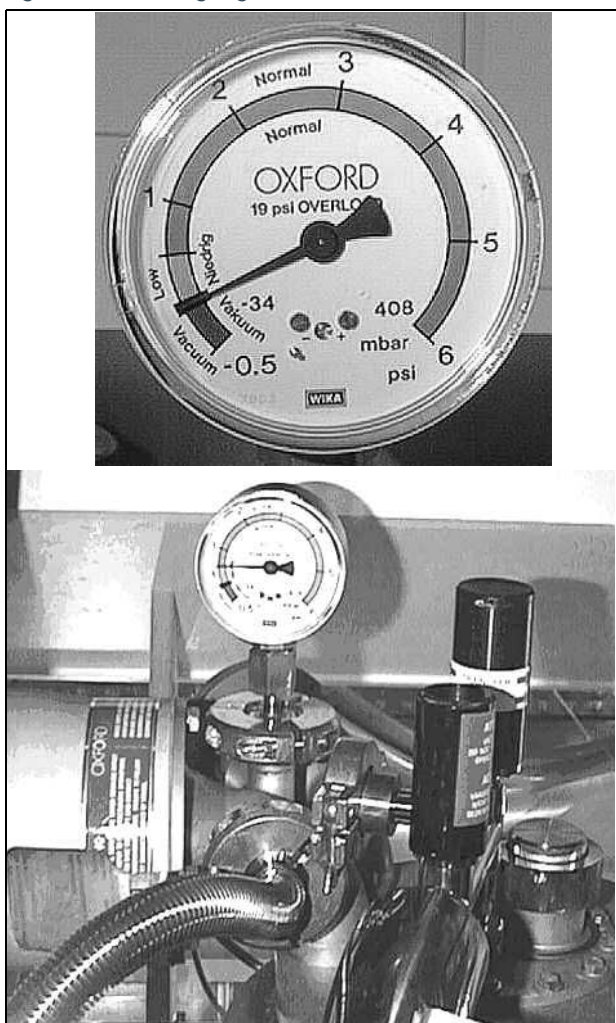
If not observed, serious injury can occur

- » Do not undo any of the fittings without first depressurizing the helium vessel



To avoid icing in the neck, do not leave the system depressurized for more than the minimum time necessary.

Fig. 19: Pressure gauge



- Depressurize the helium tank via the bypass valve until the gauge reads 0 psi
- Remove clamp
- Remove pressure gauge
- Close bypass valve
- Attach the new pressure gauge using the clamp. Ensure that seal is clean and dry.
- Check for leaks after the pressure has built up in the helium tank. (Handy Tector)

2.4 Replacing the smart syphon



CAUTION

Pressurized helium vessel

If not observed, serious injury can occur

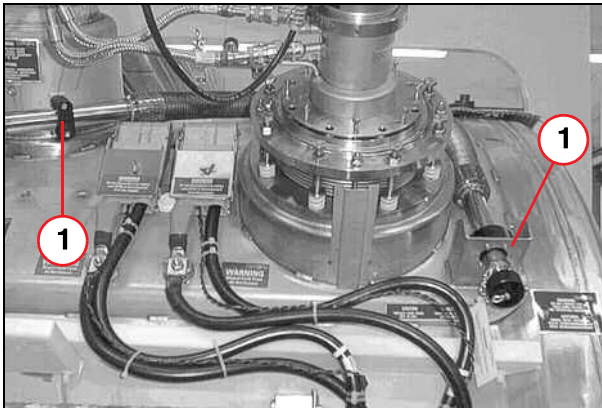
» **Do not undo any of the fittings without first depressurizing the helium vessel**



To avoid icing in the neck, do not leave the system depressurized for more than the minimum time necessary

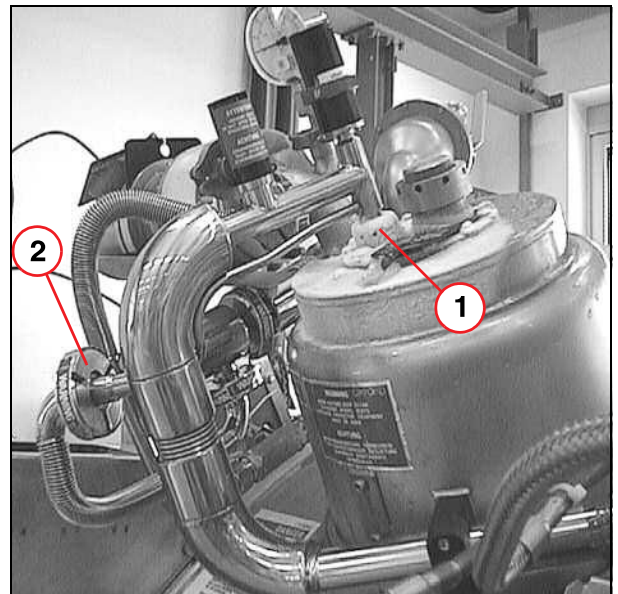
- Depressurize helium tank via the bypass valve until the gauge reads 0 psi .

Fig. 20: Smart syphon



(1) Mounting brackets

Fig. 21: Smart syphon



(1) Ring nut
(2) Clamp with high pressure seal

- Remove the screws from the mounting brackets
- Remove the clamp and discard the old high pressure seal
- Remove the ring nut; use the ring nut spanner
- Pull the smart syphon out
- Plug the syphon port with the baffle to prevent air entering the helium reservoir



Make sure that the smart syphon is dry before inserting into the magnet. Purge the syphon with helium gas from a gas bottle.

- Insert the new syphon into the port. Removing the baffle prior to inserting the new syphon. Use the correct seals. Use a new copper washer.
- Tighten the ring nut using the ring nut spanner
- Reconnect the venting line to the syphon, use a new high pressure seal

- Close the bypass valve



The system should start to re-pressurize

- Check for leaks

3.1 Replacing the Displacer - Leybold system



Servicing the APD cold head and replacing the displacer is different from the Leybold cold head. Training of service personnel on the APD cold head is offered by CS TC.



These instructions are intended as a supplement to the "Refrigerator " course.



Instructions for displacer exchange (Leybold system and APD system) are shown in a video.

The video is a supplement to the SMT Refrigerator course.



Only the LICH II displacer unit may be installed on Harmony/Symphony systems with a Leybold cold head. The Double Damped Displacer as well as the Single Damped Displacer are obsolete. For magnets on Impact/Expert/Vision/SP systems, the Single Damped Displacer must continue to be used for technical reasons.

3.1.1 Auxiliary tools and measurement devices

- Displacer exchange kit - 10119990
- Cold Head Service tool (→ Fig. 25 Page 29), (→ Fig. 26 Page 29); order number 4765 611 KE999
- Helium gas 50 l / 200 bar, grade 5.0 (99.9990%)
- Pressure regulator 200/50 bar
- Helium irrigation and filling configuration (→ Fig. 27 Page 30).

3.1.2 Spare parts

Tab. 6 Spare part

	Displacer	LICH II displacer with control disk 74 61 333
Harmony / Symphony		x



The displacers are packed air-tight in a container filled with gaseous helium.

- Control disk; order number 1562 516 KE030, not for the LICH II. The control disk for LICH II displacer is part of the displacer order number
- Displ. kit (O-rings, rotating valve)
- The Displacer exchange kit includes **all tooling and components required** within the one kit, **Displacer exchange kit - 10119990**.

A collection of various hand tools, including screwdrivers, wrenches, and sockets, arranged on a black foam mat. The tools are organized in rows, with some tools having yellow handles and others having silver or black handles. The tools include several screwdrivers of different sizes, a few open-end wrenches, and a set of sockets. The tools are laid out on a black foam mat, which is placed on a light-colored surface.

This image shows a complete hydraulic test kit for a 1000 psi system, organized in a black carrying case. The kit includes a pressure gauge, various fittings, hoses, a pump unit, and a control valve.

Fig. 24: Displacer exchange kit - Bottom tray

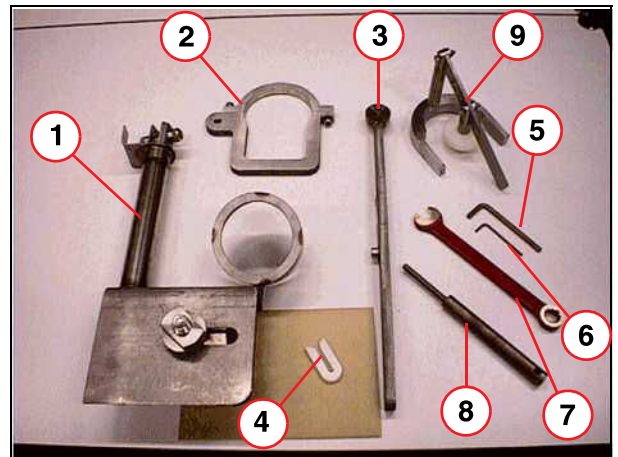


The cold head service tool allows for displacer exchange with the magnet at field.

Fig. 25: Cold head, service tool kit



Fig. 26: Cold head, service tool kit

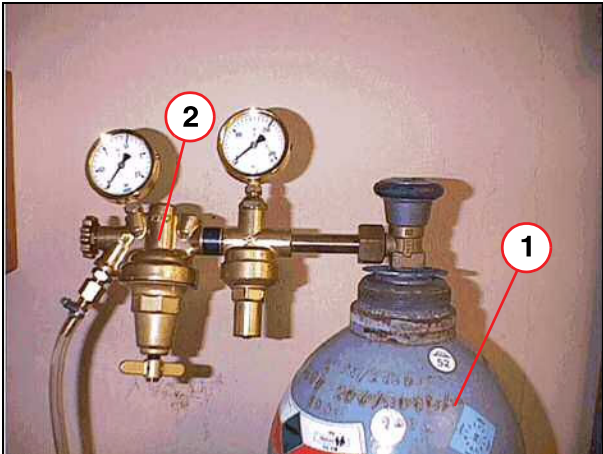


- (1) Cold head tool base
- (2) Motor clamp
- (3) Motor clamp arm
- (4) Iron shield clip
- (5) M6 Allen key
- (6) M3 Allen key
- (7) M17 combination spanner
- (8) Socket spanner for motor clamp
- (9) Control disk insertion tool



M6 and M3 Allen keys are made from beryllium copper, use the Allen keys from the "Non-magnetic Tool Set MR".

Fig. 27: Irrigation configuration



- (1) Gas bottle filled with 5.0 (99.999 %) helium
- (2) Pressure regulator

Fig. 28: Irrigation configuration



- (1) Shut-off valve
- (2) Adapter Aeroquip1/2"
- (3) High pressure hose (length 10 m)

3.1.3 Source of the supplier for the high pressure hose

REGENT HOSE & HYDRAULICS	Tel.: 0993 778278/9
UNIT 2 MINISTER INDUSTRIAL	Fax: 0993 779018
DOWNS ROAD MINSTER LOVELL	
WITNEY	
OXON OX8 5QS	
England	

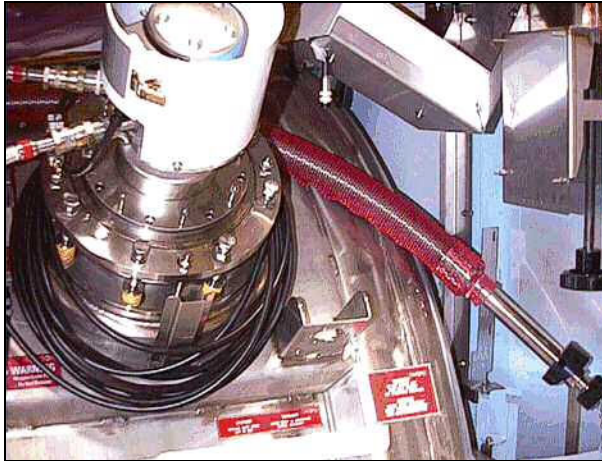
Order number	
Pressure hose:	RH/OMT/02515 SS
Adapter:	RH/OMT/S2-8/S5-4SP

3.1.4 Installation of the Cold Head Service Tool

- Remove the upper longitudinal support bar from the magnet cover frame to allow clear access to the cold head.
- Remove the 4 nuts securing the frequency converter to the magnet frame. Place the converter in a safe position.
- Remove the loudspeaker.

- Remove the 4 M4 screws which secure the helium syphon. Remove the syphon end cap and pass the syphon back through the bracket. Refit the end cap (→ Fig. 29 Page 31).

Fig. 29: Cold head

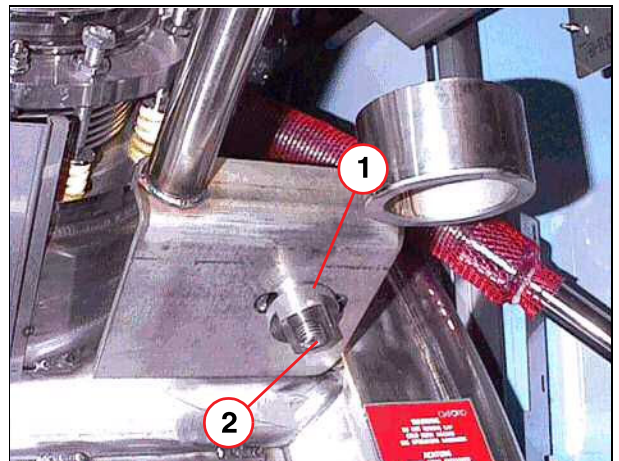


- Pass bolt "A" through the syphon bracket and manually attach the tool base to the bracket with nut "B" (→ Fig. 30 Page 31), (→ Fig. 31 Page 31)

Fig. 30: Cold head tool base



Fig. 31: Tool base



- (1) Nut "B"
(2) Bolt "A"

3.1.5 Symphony systems only



CAUTION

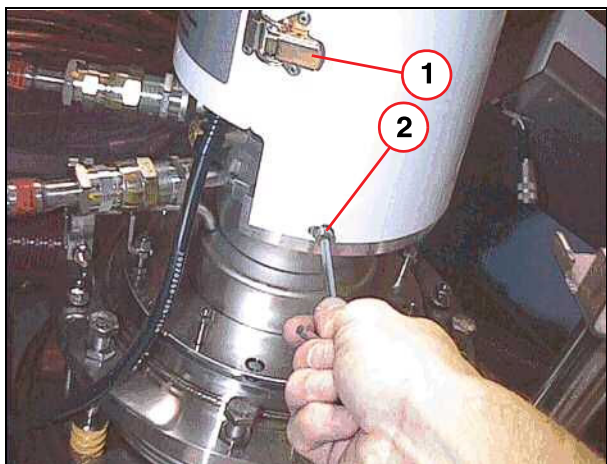
The cold head shield is magnetic

If not observed, serious injury or property damage can occur

- » Do not attempt to remove the cold head iron shield with the magnet at field

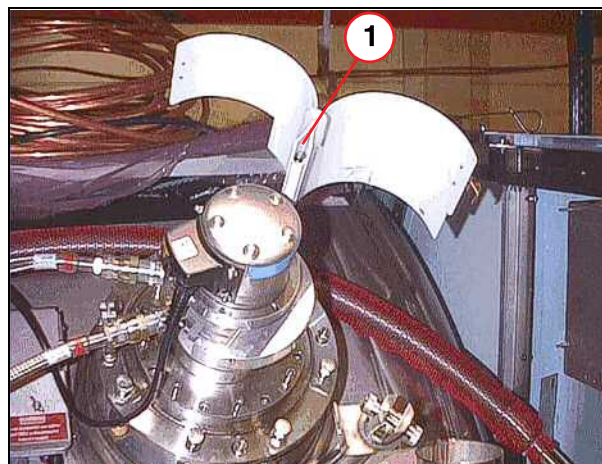
- Remove screw "C" of the iron shield, using the non-magnetic M6 Allen key (→ Fig. 32 Page 32).
- Release clasp "D" and open the wings of the iron shield (→ Fig. 32 Page 32). Hold the wings together using the plastic clip (→ Fig. 33 Page 32).
- Keep the iron shield in position.

Fig. 32: Iron shield



- (1) Clasp "D"
- (2) Screw "C"

Fig. 33: Iron shield wings

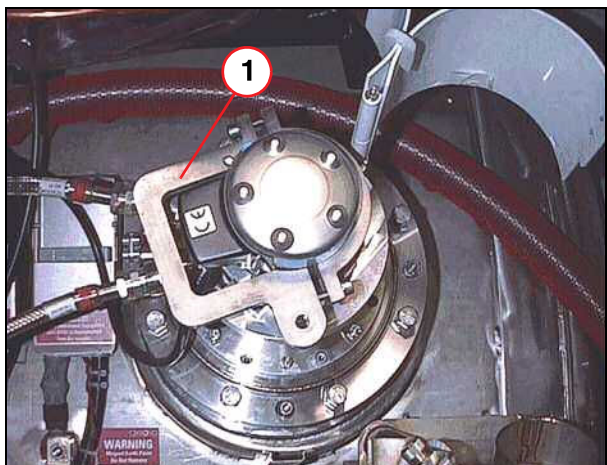


- (1) Plastic clip

3.1.6 Harmony and Symphony

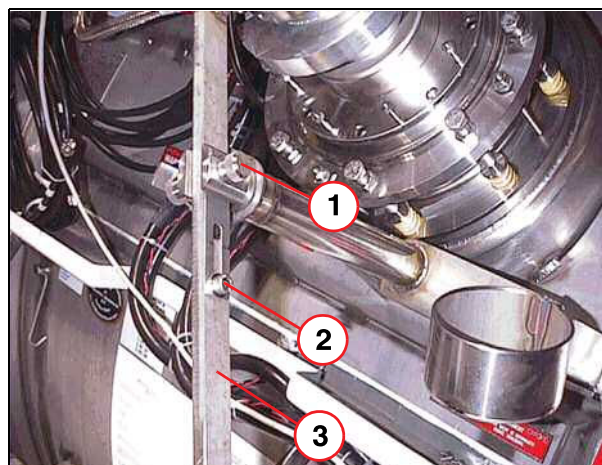
- Fit the motor clamp around the middle of the cold head motor and secure it by hand. Using the M6 Allen key (→ Fig. 34 Page 32).

Fig. 34: Motor clamp



- (1) Motor clamp

Fig. 35: Motor clamp arm

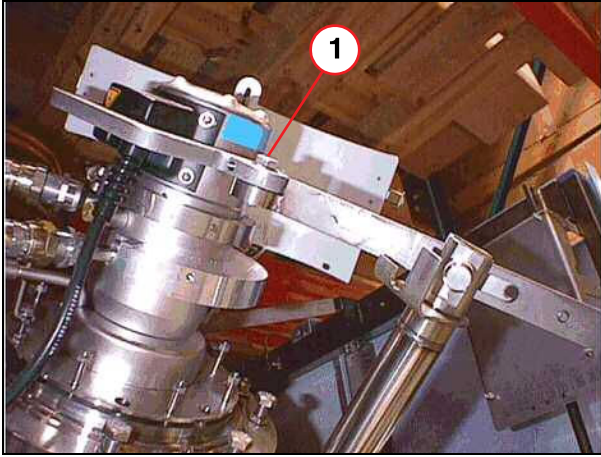


- (1) Pivot bolt
- (2) Bolt "E"
- (3) Motor clamp arm

- Place the motor clamp arm in the slot of the tool base and loosen locking plate bolt "E" (→ Fig. 35 Page 32).
- Fix the motor clamp arm with the pivot bolt (→ Fig. 35 Page 32).

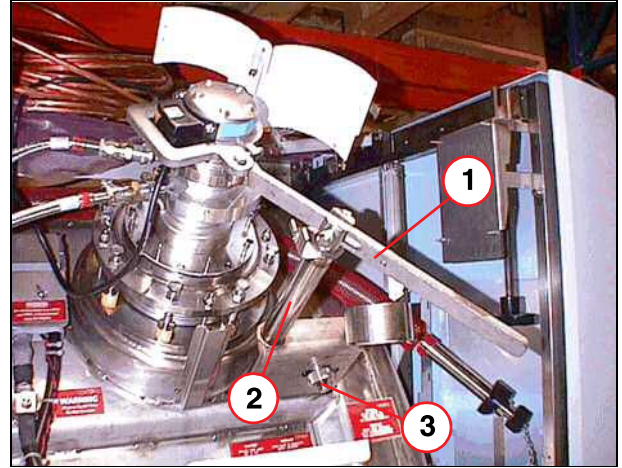
- Slide the motor clamp arm forward and attach it hand-tight to the motor clamp with bolt "F" (→ Fig. 36 Page 33). It will be necessary to loosen locking plate bolt "E" to align all parts.

Fig. 36: Cold head tool



(1) Bolt "F"

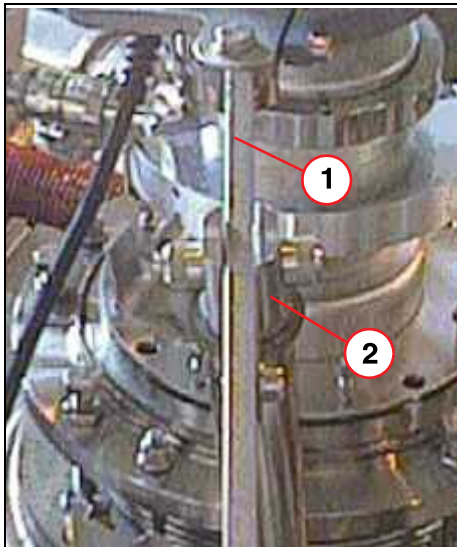
Fig. 37: Cold head tool



(1) Motor clamp arm
(2) Tool base
(3) Nut "B"

- When all parts have been aligned, secure bolt "F" (→ Fig. 36 Page 33).

Fig. 38:



(1) Motor clamp arm
(2) Tool base



The motor clamp arm must be aligned with the tool base. Ensure that the clamp arm is not tilted.

- Secure the tool base with nut "B", use the 1 3/16" non-magnetic wrench (→ Fig. 37 Page 33).
- Secure locking plate with bolt "E" (→ Fig. 35 Page 32).

- Secure motor clamp (→ Fig. 34 Page 32)

3.1.7 Replacing the Displacer

Safety recommendations



The cold head, compressor, elbows, adapters, and pressurized hoses are filled with helium under high pressure. The static pressure is approximately 12 - 16 bar. The operating pressure will be approximately 19 - 22 bar.



CAUTION

Check if all service tools are non-magnetic.

If not observed, serious injury or property damage can occur

- » Use only non-magnetic tools

3.1.8 Order for changing the displacer

1. Preliminary steps
2. Disconnecting the helium pressure lines
3. Depressurizing the cold head
4. Removing the cold head motor and the spacer
5. Removing the control disk
6. Removing the displacer
7. Replacing the rotating valve
8. Preparing the control disk
9. Heating and cleaning the cold head
10. Replacing the displacer

3.1.9 1) Preliminary steps

Establishing the irrigation configuration

Establishing the configuration for irrigating and refilling the cold head with helium.

The configuration is shown in (→ Fig. 27 Page 30).

Purge the pressure hose.

**CAUTION**

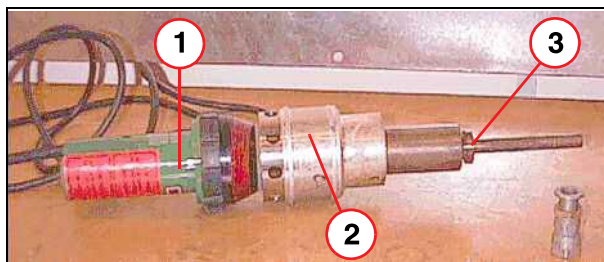
Helium gas bottle is magnetic.

If not observed, serious injury or property damage can occur

» Place it outside of the magnet room.

3.1.10 Assembling the heat gun

Fig. 39: Heat gun



- (1) Heat gun
- (2) Adapter ring
- (3) Adjustment screw



The heat gun contains magnetic parts.



The flow of the heated air to the two stages can be regulated with the adjustment screw on the hot air cylinder.

3.1.11 2) Disconnecting the helium pressure lines from the cold head

Tools

1 3/16" - 1 1/8" wrench

- Switch off the compressor.
Switch off F3 in the RCA
- Disconnect both pressure lines from the cold head.

	RGD 5100
1/2" = high pressure connection	Red = high pressure connection
1/2" = low pressure connection	Green = low pressure connection

3.1.11.1 Working with self-sealing couplings

Fig. 40: Attaching and detaching self-sealing couplings

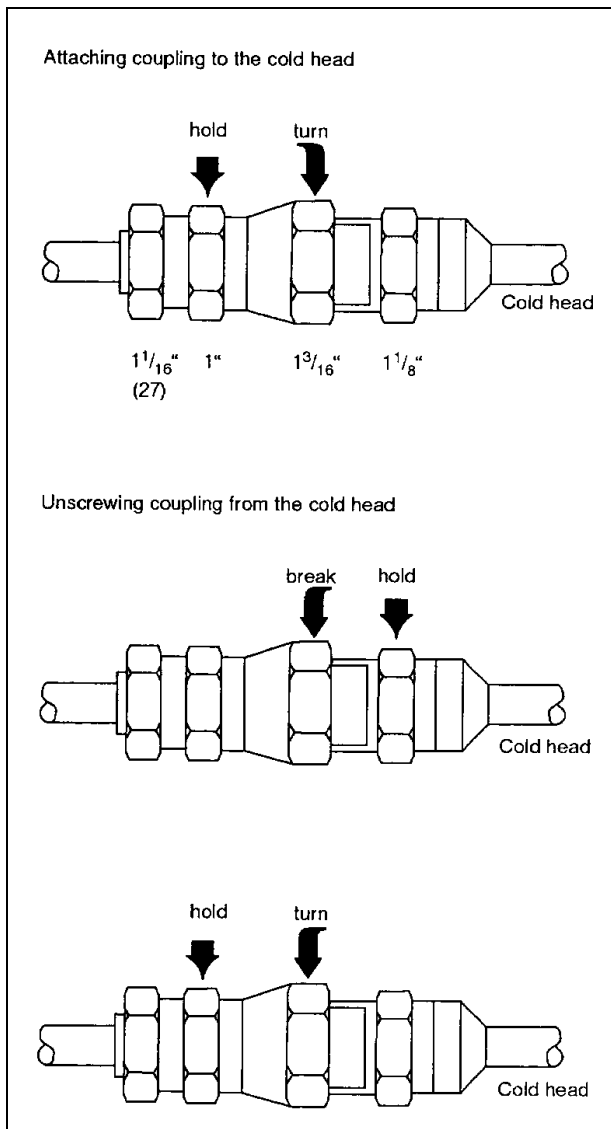
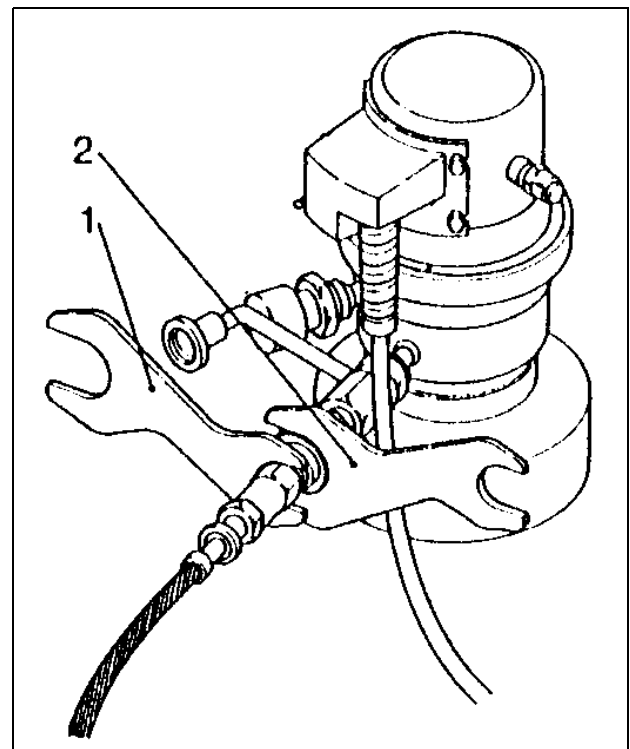


Fig. 41: Disconnecting the helium pressure hoses



Do not twist the pressure hose. Use a second wrench when removing the Aeroquip couplings

3.1.12 3) Depressurizing the cold head

Tools

Adapter, Aeroquip 1/2" f/DN 16 small flange

Wrench 1 3/16" - 1"



The Aeroquip adapter contains magnetic parts.

Fig. 42: Depressurizing the cold head



- Connect the Aeroquip adapter 1/2" f/DN 16 small flange to the high pressure connection of the cold head. It should open up slightly allowing the helium to escape.
- Connect the Aeroquip adapter 1/2" f/DN 16 small flange to the low pressure connection of the cold head.
- Connect the hose for the helium irrigation and filling configuration to the high pressure connection. Ensure that the self-sealing mechanism of the Aeroquip does not open up.

3.1.13 4) Removing the motor unit

Tools from the Cold Head Service Tool Kit

10 mm socket wrench (→ 8/Fig. 26 Page 29)

Wrench 1 3/16" - 1"

Fig. 43: Removing the cold head motor



- Remove clamp with socket wrench
- Remove the motor from the cold head by gently pushing down on the motor clamp arm. Park the clamp arm in the cradle of the tool base
- Remove the spacer

3.1.14 5) Removing the control disk

Tools

Extractor for control disk

Fig. 44: Removing the control disk



- Position the extractor on the housing of the cold head.
- Insert the two knurled screws up to the end stop into the threaded holes of the control disk(→ Fig. 44 Page 38).
- Turn the threaded spindle in the center of the extractor to unseat and pull out the control disk.

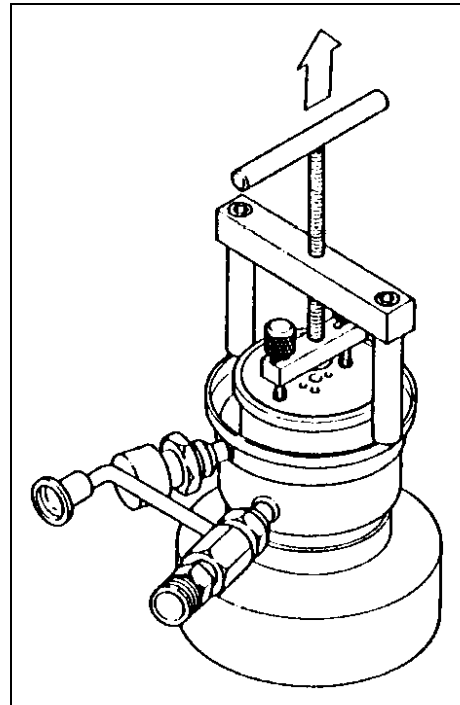
3.1.15 6) Removing the displacer

Tools
Extractor for displacer or
Extractor for guide sleeve

Fig. 45: Removing the displacer with extractor



Fig. 46: Removing the displacer with extractor for control disk



- Install the extractor
- Pull out the displacer

3.1.16 7) Replacing the control valve

Tools
File
Power supply for motor

Material:
Paper towels, cleaning fluid
New control valve

Fig. 47: Removing the control valve



- Remove the old control valve and don't lose the spiral spring.
- Ensure that the new control valve is leveled using a hairline ruler. Clean the valve and reattach it to the motor shaft.
- Check whether the valve moves smoothly in its support in both the lateral and the vertical direction when pressure is applied to the spiral spring. If necessary, widen the slot in the valve with a file.
- Check the direction of rotation of the valve. Disconnect the motor cable from the filter plate and connect it to the power supply. Switch on the power supply. The valve should turn in clockwise direction.

3.1.17 8) Prepare the new control disk

Material:
Paper towels and cleaning fluid
O-rings

- Clean the control disk

- Provide the new control disk with new O-rings

Fig. 48: Control disk

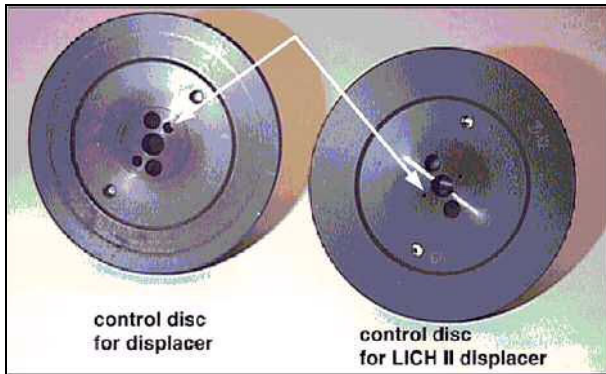
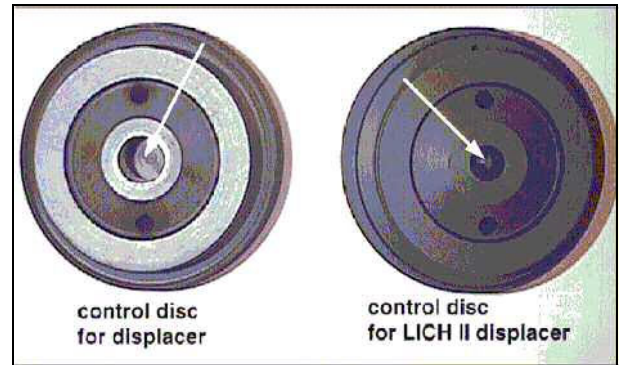


Fig. 49: Control disk



Diameter of the drill holes are different between both control disks

3.1.18 9) Heating and cleaning the housing of the cold head

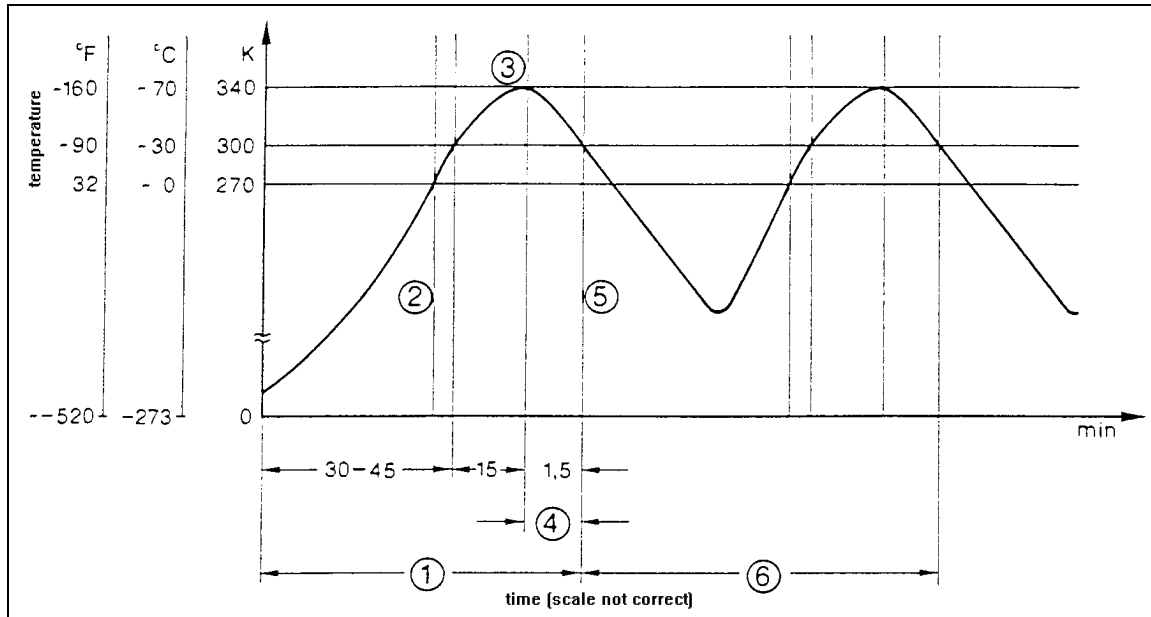
Tools
Heat gun
Thermo sensor
Flashlight

Tools
Pair of long tweezers

Material
Paper towels and cleaning fluid

3.1.18.1 Working with the heat gun

Fig. 50: Diagram for working with the cylinder heater



3.1.19 Explanations for the diagram

1. Time interval required to reach the operating temperature of the cold head

T < 100 Kelvin (= -170° Celsius) and the cooling period after the critical temperature

T = 300 Kelvin (= 30° Celsius), (one cycle)!

2. T = 270 Kelvin (= 0° Celsius). Melting point for ice. Wipe off the water droplets using a cloth or let them evaporate. You can now clean the cylinder.

When the temperature falls below 270 Kelvin (= 0° Celsius), the drying and cleaning process must be repeated because the temperature inside the cylinder will reach the dew point for humidity.

3. T = 340 Kelvin (+70° Celsius). Maximum temperature threshold which should not be exceeded to protect the copper band and the cold head.

4. **Important step:**

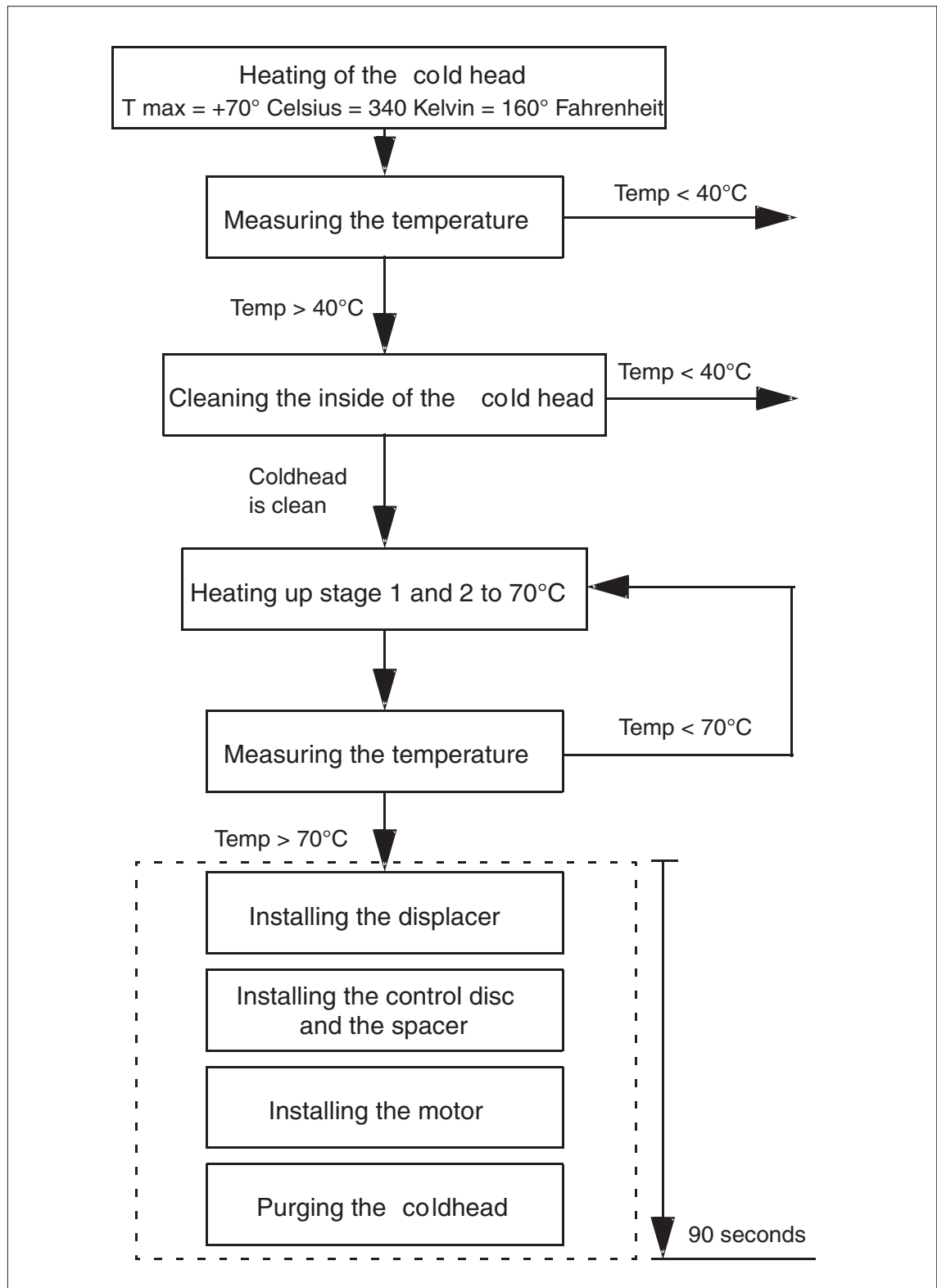
After cleaning the cylinder, you have approximately 90 seconds to reinsert the displacer, reattach the control disk, spacer and the motor, close the cold head, and irrigate it with 5.0 helium.

Perform these steps quickly and carefully. In case of delays, impurities will freeze to the walls of the cylinder.

5. T = 300 Kelvin (+ 30° Celsius). Critical temperature level. If the temperature falls below this level, water will begin to condense at the cylinder walls. The cylinder will need to be dried again.
6. Item **one** (time interval ...) can be repeated as required.

3.1.20 Order of installation for the new displacer

Fig. 51:



3.1.20.1 Heating of the cold head

The heat can be set with the control knob at the heat gun.

- Set the heat to power stage "seven" and switch on the gun. Subsequently, reduce the setting to power stage "four" or lower. Both stages should show the same warm-up temperature. Measure the temperature with the thermo sensor at least once every minute.

Tmax = +70° Celsius = 340 Kelvin = 160° Fahrenheit.

Fig. 52: Cleaning the displacer cylinder



Overheating will destroy the housing of the cold head. It will also damage the cryostat which in turn will cause helium boil-off.

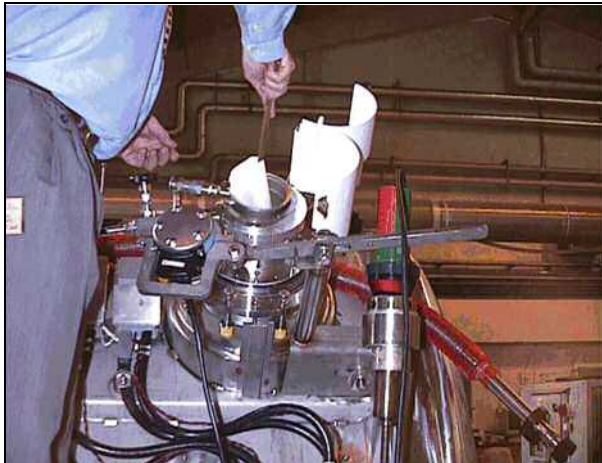
3.1.20.2 Cleaning the inside of the cold head

- Use paper towels to clean the inside wall of the housing of the cold head. Subsequently, clean the entire inside wall by wrapping a cloth soaked in cleaning fluid around a pair of long tweezers.



Do not use a towel containing plastic material. It will melt due to the high temperature (+40 - +70 °C) causing severe contamination of the surface.

Fig. 53: Cleaning the displacer cylinder



- Let the residual film of cleaning fluid evaporate.
- To remove the lint, purge the inside wall with helium gas.



Do not scratch the contact surface of the cold head. The honed inside wall of the housing of the cold head is highly sensitive to scratches.

3.1.20.3 Installing the new displacer

Tools
Ring compressor for control piston
Ring compressor for first stage
Ring compressor for second stage

Material
Container with new displacer
Cotton gloves
Irrigation and fill configuration
Helium gas 5.0

Establish the purging configuration before heating the cold head.

- Connect the Aeroquip coupling to the high pressure connection.

Fig. 54: New displacer



Fig. 55:



3.1.20.4 Preparing the displacer

- Unpack the new displacer.

The transport container protects the new displacer against humidity. The displacer is packed in helium gas under atmospheric pressure.

- For this reason, the container should remain closed when not in use. Wear cotton gloves when removing the displacer from the transport container or when replacing the old displacer. Expose the container only briefly to the ambient air to avoid contamination.
- Check the seals (control piston, first and second stage) at the new displacer. Press the seals down by using the respective ring compressors.



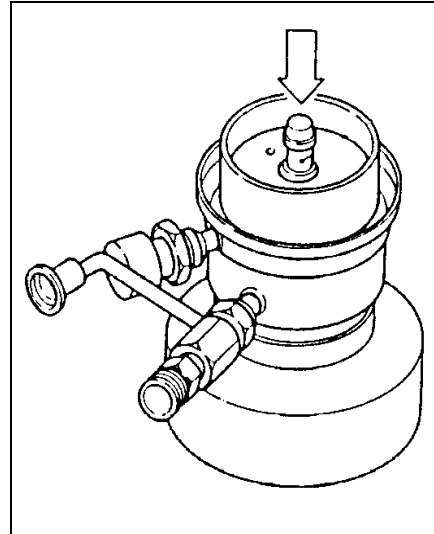
Do not use force when sliding the ring compressor over the seals.

- Hold the ring compressor with the cone-shaped side pointing away from you. Turn the ring compressor slightly and slide it over the first and second stage as well as the control piston so that the seals are pressed down. Subsequently, remove the ring compressor from the second stage. Turn the displacer slightly while inserting it quickly and carefully into the housing of the cold head (→ Fig. 56 Page 47).

Fig. 56: Inserting the displacer (RGD 5100)



Fig. 57: Removing the ring compressor



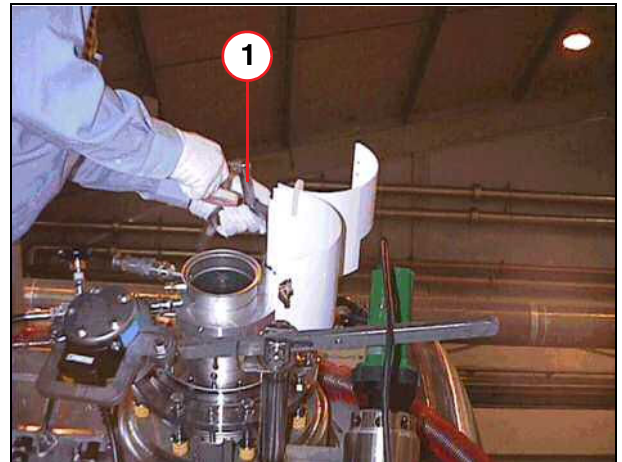
- Ensure that the seals of the second stage do not move out from the displacer. The ring compressors of the first stage and the control piston should not be removed until the displacer can be pressed up to the end stop into the housing (→ Fig. 56 Page 47).

3.1.20.5 Installing the control disk

Tool from the Cold Head Service Tool Kit

Control disk insertion tool

Fig. 58: Inserting and pressing down the control disk (RGD 5100) Fig. 59:



(1) Insertion tool

- Press down on the clean control disk with both thumbs and carefully insert it into the housing of the cold head (up to the end stop). Use the insertion tool for the control disk.
- Check that the gas bores are clear.

3.1.20.6 Installing the motor

Tools from the Cold Head Service Tool Kit

10 mm socket wrench



The flange of the motor and the cold head will be hot from the hot air of the heat gun that heats the cold head.

Fig. 60: Motor clamp



- Insert the rubber spacer with the O-ring. Make sure that one of the four holes of the spacer is placed between the two helium gas connections (high and low pressure).
- Swing the motor toward the cold head.
- Slide the clamp over it and tighten it.



Attach the motor so that the connection cable is routed between the high pressure and low pressure connection.

3.1.20.7 Purging and filling the cold head with helium

Tools

Wrench

Adapter, Aeroquip 1/2" f/DN 16 small flange

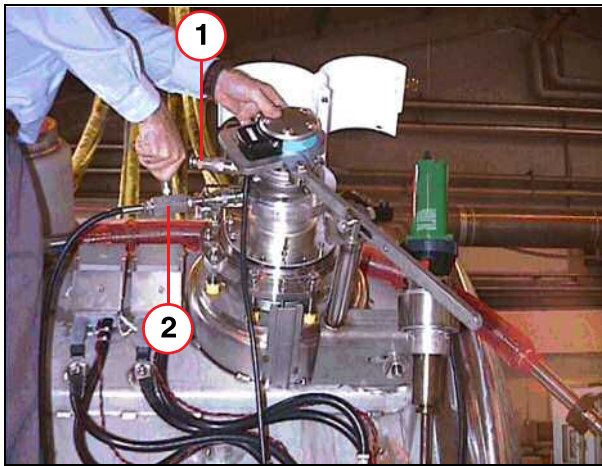
Material

Irrigation and fill configuration

Helium gas 5.0

Establish the purging configuration before heating the cold head

Fig. 61: Purging the cold head



- (1) Adapter
- (2) Shut-off valve

- Attach the Aeroquip coupling 1/2" (→ Fig. 60 Page 48) to the high pressure hose of the fill configuration to the red high pressure connection of the cold head. Slowly open the bottle valve, the pressure regulator and the shut-off valve. Slowly fill the cold head up to 5 bar with helium.
- Attach Aeroquip adapter 1/2" f/DN 16 small flange to the green low pressure connection to allow helium to flow from the cold head.
- Activate the switch at the power supply for the motor.
- The helium escapes in pulses from the cold head via the green low pressure connection. After approximately 5 pulses, close the shut-off valve (→ Fig. 61 Page 49) while the motor is running. After the helium of the last pulse has escaped, open the shut-off valve again. Keep it open for another 5 pulses, then close the valve.
- Open and close the shut-off valve approximately 5 times.
- Disconnect the Aeroquip adapter from the low pressure connector of the cold head, while the motor is running and the gas escapes.
- Switch off the motor power supply.
- Fill the cold head to 14 - 15 bar, close the shut-off valve. Disconnect the high pressure hose from the cold head.
- The replacement procedure for the displacer is now completed.

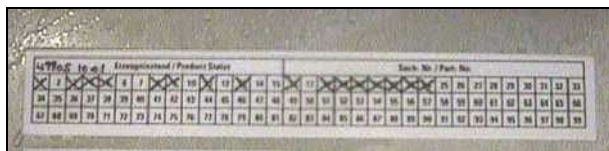
3.1.20.8 Start-up of compressor

- Reconnect both pressure helium lines to the cold head, high pressure first
- Reconnect the motor cable
- Switch on F3
- Switch on the compressor
- Check the helium fill pressure

3.1.20.9 Final check if LICH II displacer was installed

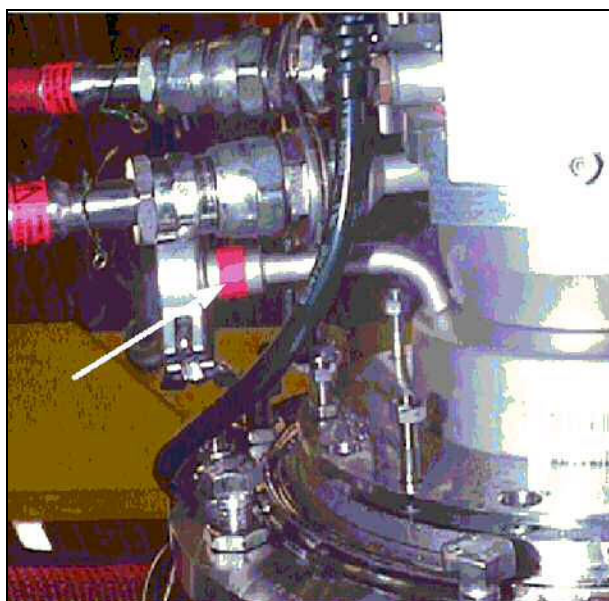
- Mark the revision level if necessary
ES 22 for OR70 (1.5 Tesla)
ES 21 for OR24 (1.0 Tesla)

Fig. 62: Revision level



- Mark the cold head with red tape at the vacuum pumping port

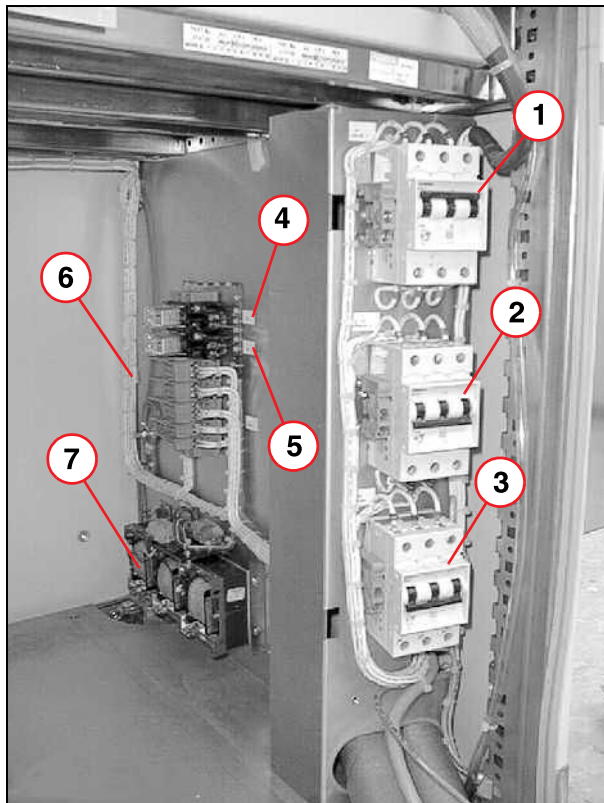
Fig. 63: Vacuum port



3.1.21 Electrical components and their location in the RCA

The electrical components (transformer, relays) are located to the right of the helium compressor. You can access them only after the compressor has been removed.

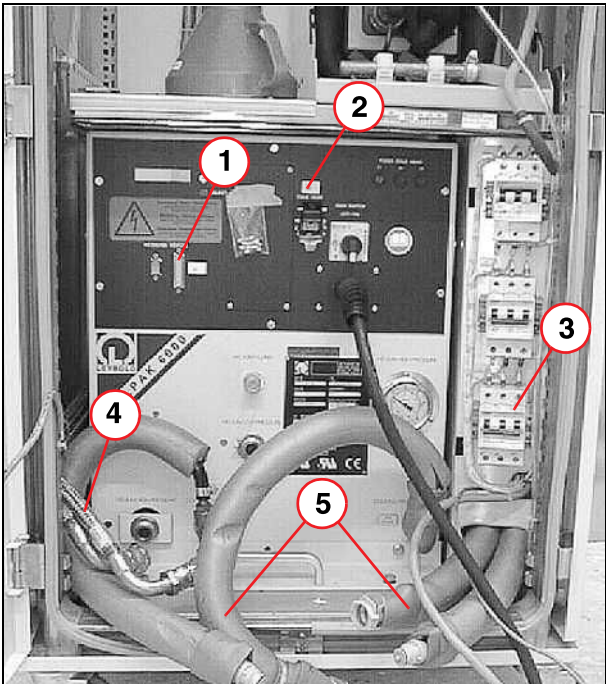
Fig. 64: Electrical components



- (1) F1
- (2) F2
- (3) F3
- (4) K1
- (5) K2
- (6) X102
- (7) T1

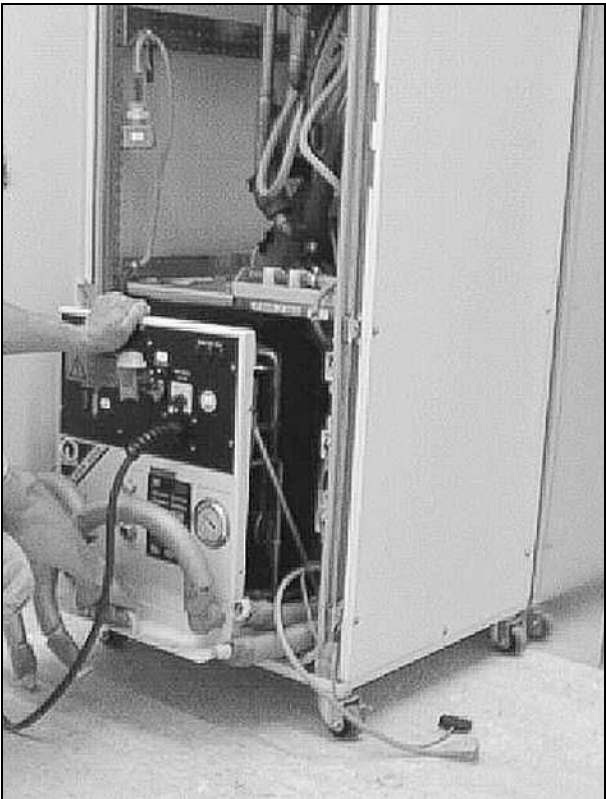
3.1.22 Replacement of the Compressor (COOLPAK)

Fig. 65: Cool pack



- (1) X7
- (2) X5
- (3) F3
- (4) Flex lines
- (5) Water hoses

Fig. 66: Cool pack



3.1.22.1 Removing the COOLPAK

Step	Instructions	Required Tools
1	Switch off compressor Switch off F3 Switch off F4 on top of CCA	
2	Disconnect the water lines	
3	Disconnect the high and low pressure helium flexlines	Wrench 1 3/16" and 1"
4	Remove cover from circuit breaker Only necessary if the power cable is not pluggable	
5	Unplug the power cable or disconnect the power cable from F3 Notice: Input voltage on top Output voltage at the bottom	Screwdriver
6	Disconnect cable X7 and X5	

Step	Instructions	Required Tools
7	Unscrew compressor from the base plate	Wrench
8	Pull out the compressor (weight approx. 100 kg) The front of the compressor should be placed on top of a sturdy support.	

3.1.22.2 Installing the COOLPAK

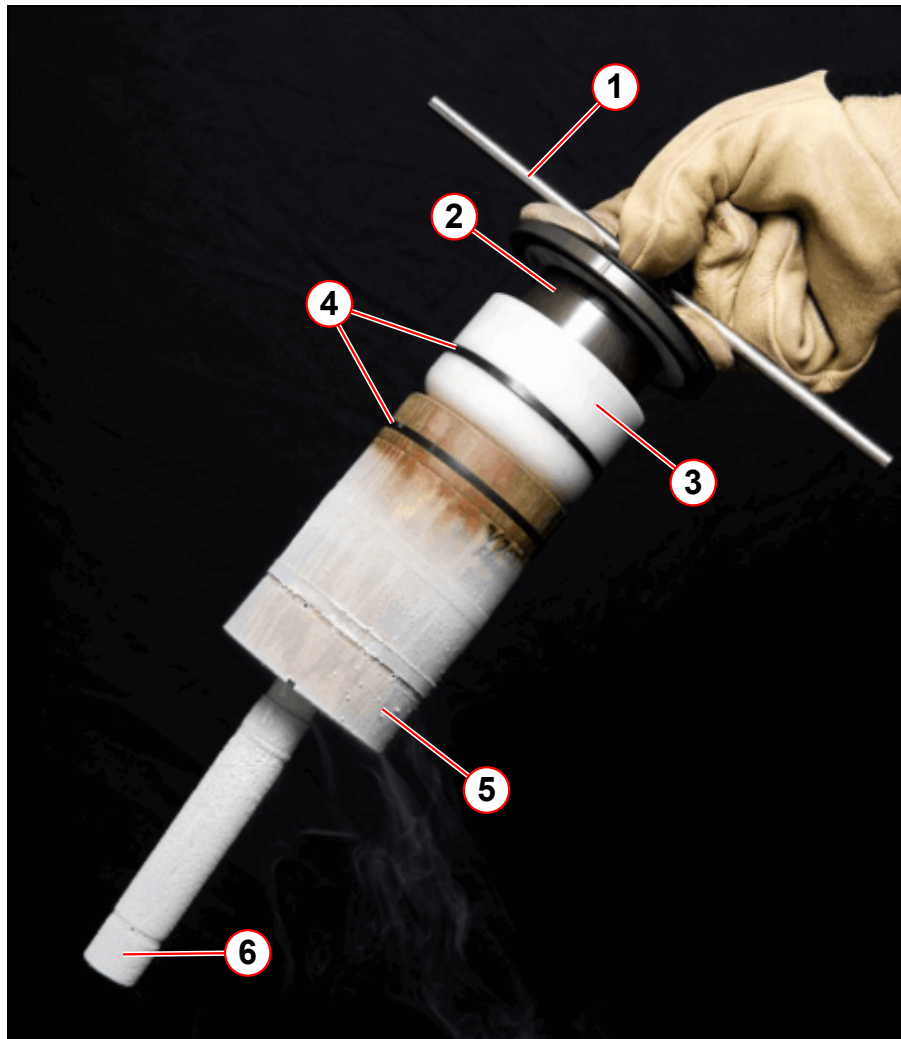
Step	Instructions	Required Tools
1	Slide compressor into the cabinet	
2	Tighten the compressor on the base plate	Wrench
3	Connect the line voltage cable Verify that the direction of rotation is correct	
4	Install cover for circuit breaker	
5	Connect the high and low helium flexlines 1. High-pressure (red) 2. Low-pressure (green)	Wrench 1 3/16" and 1"
6	Connect the water lines	
7	Connect cable X7 and X5	
8	Check the helium pressure on the pressure gauge (compressor off) Operation at 50 Hz: Nominal value: 15 bar (218 psig) Minimum value: 14 bar (203 psig) Operation at 60 Hz Nominal value: 14 bar (203 psig) Minimum value: 13 bar (189 psig)	
9	Switch on F4 Switch on F3 Switch on the compressor Typical operating pressure (compressor on) Operation at 50 Hz: 20 - 21 bar (290-305psig) Operation at 60 Hz: 19 - 20 bar (276-290psig)	



Drain the water from the pipes before returning the replaced unit

3.2 Replacing the Displacer - APD systems

Fig. 67: Displacer APD



- (1) Extraction bar 7.5 mm
- (2) Valve stem
- (3) Slack cap
- (4) Gas seals
- (5) 80K stage
- (6) 20K stage

3.2.1 General

This procedure describes the displacer exchange for the APD (Whisper Plus) cold head fitted to the 10K magnet types Harmony, Symphony, Sonata, Trio (upgrade to syngo) and Impact/Expert from Vision, manufactured by Siemens Magnet Technology.



Read the procedure and check serviceability of replacement parts before carrying out the replacement process.



This displacer replacement procedure must only be performed by Certified Engineers, who have completed the relevant training course(s).

3.2.2 Safety recommendations



The cold head, compressor, elbows, adapters, and pressurized hoses are filled with helium under high pressure. The static pressure is approximately 12 - 16 bar. The operating pressure will be approximately 19 - 22 bar.



WARNING

Pressurized Helium Vessel.

Failure to observe the following safety information may result in physical injury.

- » Do not disconnect any of the turret fittings without first depressurizing the helium vessel.
- » Only trained and accredited engineers must perform this procedure.



WARNING

Risk of Asphyxiation.

Failure to observe the following may result in dizziness and loss of consciousness.

- » Do not vent helium gas directly into the an enclosed place, with no ventilation.
- » Restrict unauthorised access to the Imaging room.
- » Personal Oxygen monitors must be worn if a fixed room monitor has not been installed.



CAUTION

Risk of Cold Burns.

Failure to observe the following safety information may result in physical injury.

- » Do not touch cold surfaces with bare hands.
- » Personnel safety equipment must be worn.

**CAUTION**

Risk of Hot Burns.

Failure to observe the following safety information may result in physical injury.

- » Do not touch hot surfaces with bare hands.
- » Personnel safety equipment must be worn.

**WARNING**

Strong Magnetic Fields.

Failure to observe the following safety warning may result in physical injury.

- » Only non-magnetic equipment must be taken into the magnet room, when a magnetic field is present.

**CAUTION**

Flammable Cleaning Fluids.

Failure to observe the following safety caution may result in burns.

- » Protective clothing, gloves and eye protection must be worn.
- » Follow the manufactures instructions when handling flammable cleaning fluids.

**WARNING**

Danger of LEAD CONTAMINATION during the heating process.

Dispersed lead dust may be inhaled and could result in personal injury.

- » Wear a respirator mask to a minimum filter standard of P2 during the cold head heating process.

The following Safety precautions **must** be followed.

- Wear a protective work apron and protective gloves (especially if there are open wounds on the hands).
- It is prohibited to eat, drink and smoke while working with lead/ displacers.
- Never touch the eyes or the mouth with your hands while working with lead/ displacers.
- Thoroughly wash your hands with soap and water after completion.
- Hazardous materials containing lead may not be placed or stored in containers which, due to their shape or labeling, may be confused with food containers..
- Do not dispose of lead scrap (lead remnants and production resources contaminated with lead) into waste receptacles. Collect the lead in a separate container and have it

collected up as special refuse by the responsible department according the local legal requirements.

Personal inhalation protection **MUST** be worn during this procedure.

- **Lead dust** can escape during the heating process.
- The **2nd stage** of the displacer is filled with **lead balls**.

The **European** standards applicable are listed below. Any equipment procured in the **United States** must adhere to **OSHA** standards and be **NIOSH / MSHA** certified as suitable for the task specified.

Fig. 68: Example of appropriate mask



Fig. 69: Mask (part number 70 14 868



Tab. 7 European Standards

Description	European Standard
3M - Model 6000 Half Mask	EN 140
3M – 2135 (P2) Filter	EN 140

3.2.3 Spare parts

The replacement **APD displacer kit** contains the displacer and valve stem assembly in a sealed bag. The kit also contains all the replacement O-rings, rotary valve and seals required.

- APD replacement displacer kit order number **08395738**.

Fig. 71: Displacer exchange kit - Middle tray

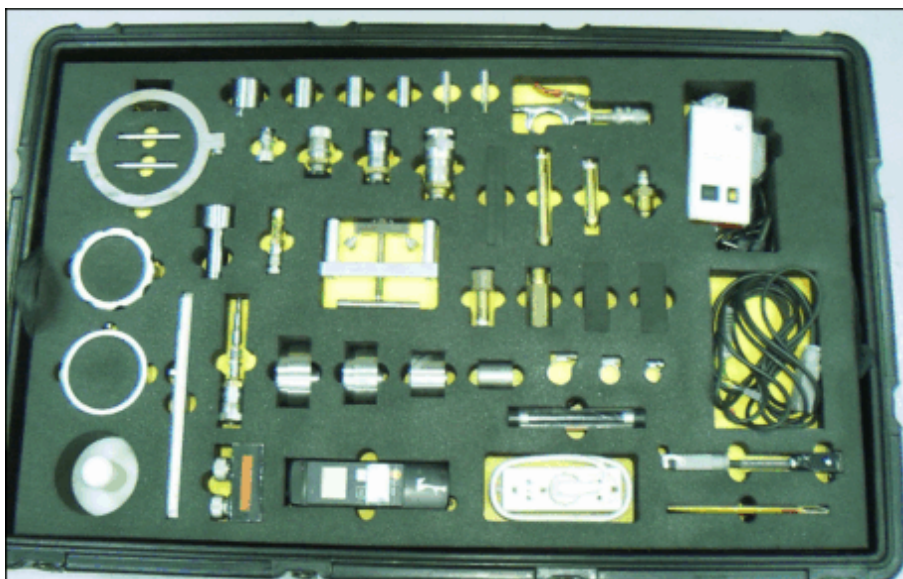


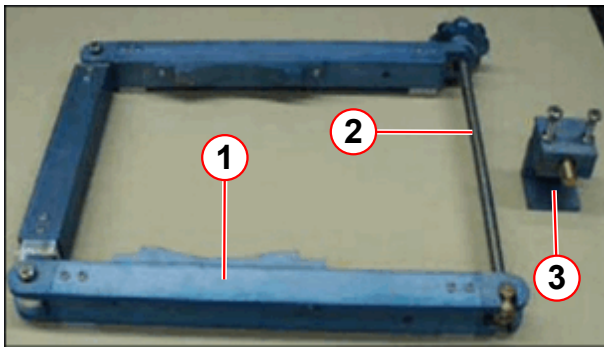
Fig. 72: Displacer exchange kit - Bottom tray



3.2.6 Displacer exchange tool kit (Trio magnets)

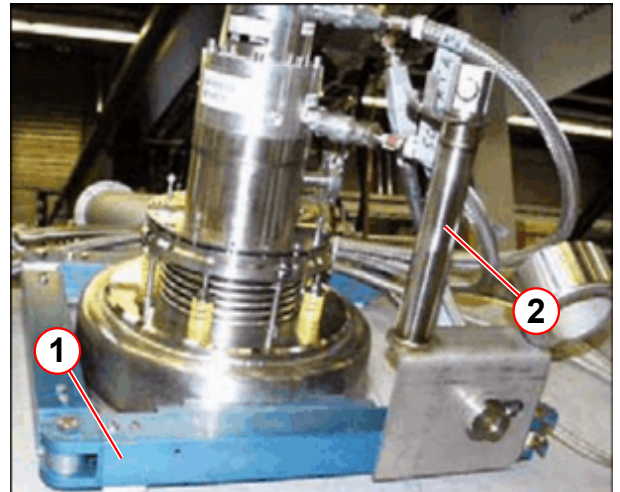
An additional tool kit is necessary if exchanging the displacer **at field**. Order part number **8395753**.

Fig. 73: Displacer exchange tool kit - Trio system



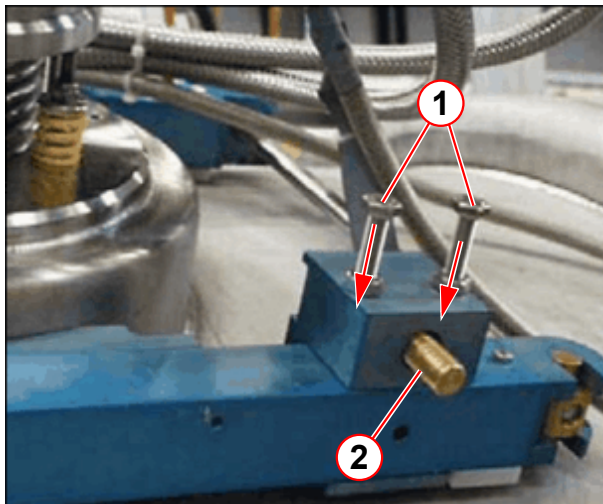
- (1) Frame
- (2) Locking bar
- (3) Clamp

Fig. 74: Displacer tooling - Trio systems



- (1) Trio exchange tooling
- (2) Motor removal tooling

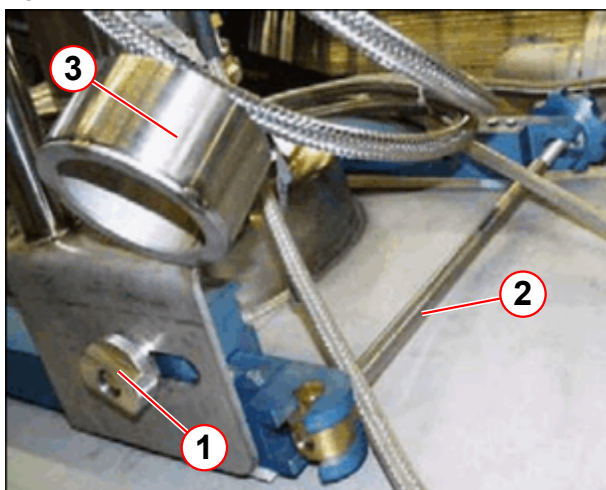
Fig. 75: Fit the clamp to the frame



- (1) Screws and nuts (2 off)
- (2) Mounting stud

- Fit the clamp to the frame using the 2 off screws.

Fig. 76: Fit the tool base to the frame

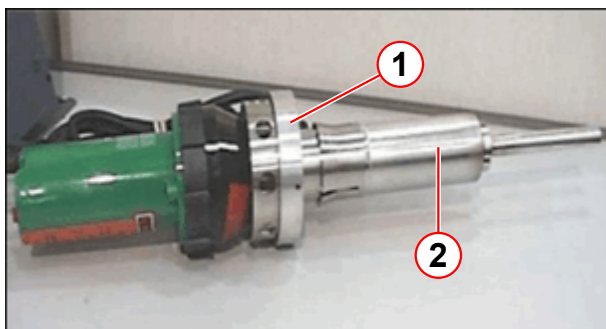


- (1) Tool base locking nut
- (2) Locking bar
- (3) Tool base and heat gun holder

- Align the mounting **stud** with the slot in the tool tray.
- Secure with the tool base **locking nut** (pos1).
- Tighten the **locking bar** (pos 2).

3.2.7 Heat gun assembly

Fig. 77: Heat gun tooling



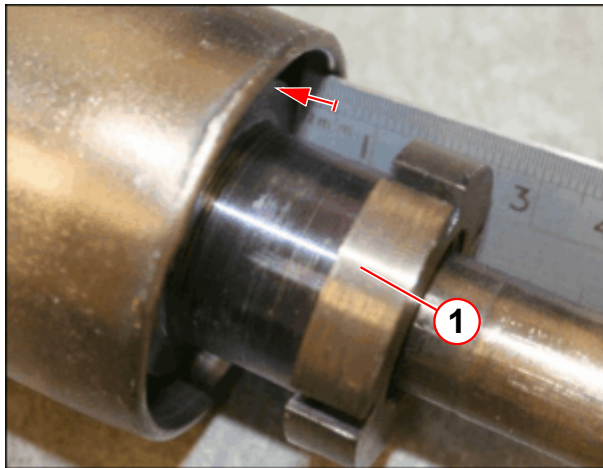
- (1) Adaptor ring
- (2) Air cylinder

- **Assemble** the adaptor **ring** and **air cylinder** onto the heat gun.



The heat gun contains magnetic parts.

Fig. 78: Set the gap to 5 mm



(1) Air flow adjustment screw

- Set the initial measurement to by turning the **air flow adjustment screw**.
 - » set the gap to **5 mm**.



The flow of the heated air to the two stages can be regulated with the adjustment screw on the hot air cylinder.

3.2.8 Magnetic fields



WARNING

Strong Magnetic Fields.

Failure to observe the following safety warning may result in personal injury.

- » Only non-magnetic equipment must be taken into the magnet room.



Low ceiling heights restrict the ability to work safely.

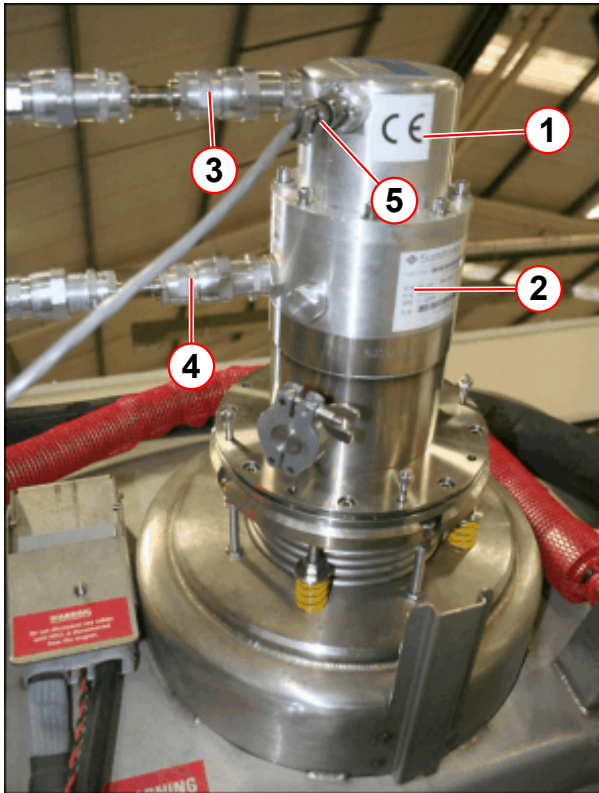
3.2.9 Displacer exchange preparation

- **Remove** the required **magnet covers** and **equipment** necessary to allow **clear access** to the **cold head**.
- **Position** the helium **gas cylinder**, in a **safe position** as required.

OR70 only

- To allow **clear access** to the cold head:
 - **Ensure** that the **power to the frequency converter is OFF** otherwise the magnetic field can **destroy** the frequency converter.
 - **Remove** the **4 off nuts** securing the **frequency converter** and place it in a safe location.
 - **Remove** the loud speaker.

Fig. 79: APD Cold head - OR70



- (1) Motor housing
- (2) Valve stem housing
- (3) Supply helium gas line
- (4) Return helium gas line
- (5) Cold head motor power cable

- Switch **off** and electrically **isolate** the **compressor**.
- **Disconnect** the motor **cab**le from the **cold head** (pos 5).
- **Disconnect** the **Supply** and **Return** gas lines, using three **Aeroquip** wrenches (pos 3,4).



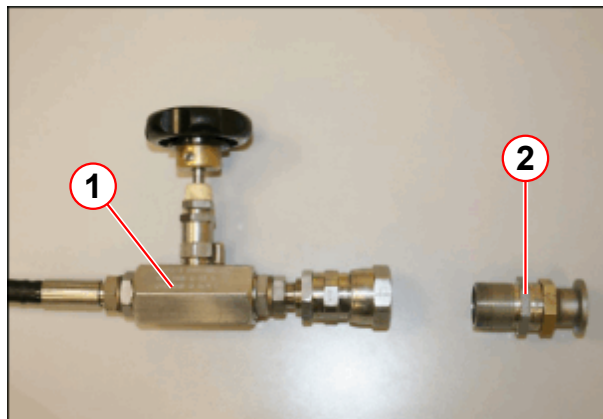
To prevent movement of the fixed Aeroquip couplings, always use three wrenches when removing the connections.

- **Connect** the **high pressure regulator** to the **gas cylinder**.
- **Connect** the **gas supply hose** to the **regulator**.



Do not connect to the cold head at this time.

Fig. 80: Gas line purge tool



- (1) Shut-off valve
(2) Male purge adaptor

- **Purge** the gas line using a **male purge adaptor** (pos 2).
- It is recommended that the **gas line** is **purged** for a minimum of **30 seconds** with a **low flow** of helium gas.

Fig. 81: Mark the orientation of the motor



- (1) Orientation mark

- Mark the **orientation** of the motor to the **cold head body** using a marker pen.

3.2.10 Fitment of the cold head motor tooling (at field)



CAUTION

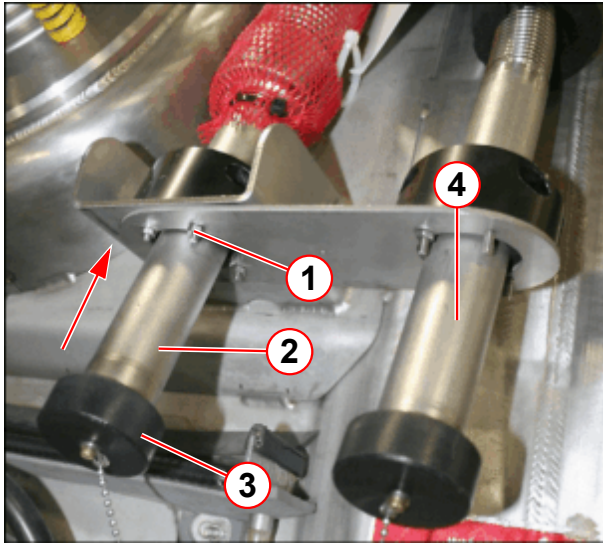
Check if all service tools are non-magnetic.

If not observed, serious injury or property damage can occur.

» Use only non-magnetic tools.

If the displacer exchange is being performed with the **magnet at field**, the following tooling **must be fitted** to allow **safe removal of the motor**.

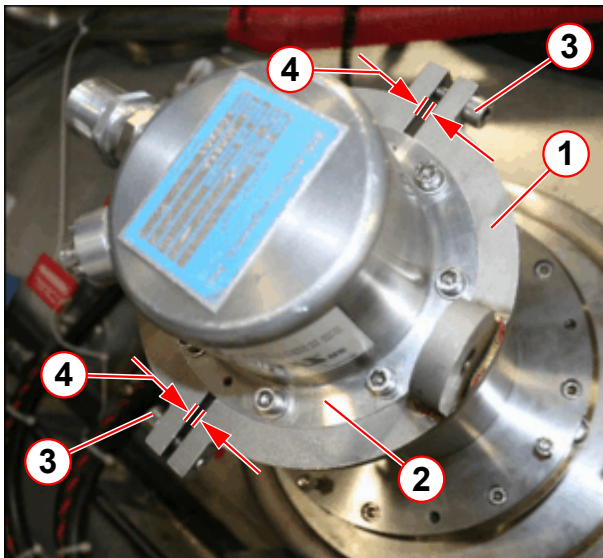
Fig. 82: Removing the helium syphon - OR70



- (1) Securing nuts
- (2) Helium syphon
- (3) Syphon end cap
- (4) Vent line

- Remove the **securing nuts** securing the helium **syphon** (pos 1).
- Remove the **syphon end cap** and pass the syphon back through the **bracket**.
- **Immediately refit the end cap** and move the syphon aside.

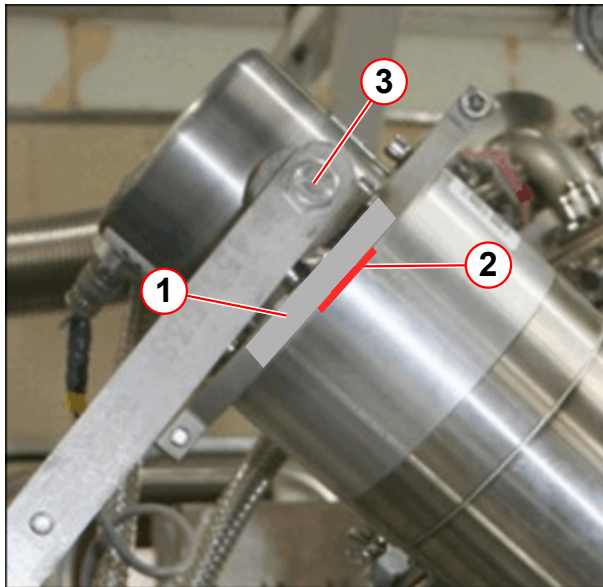
Fig. 83: Fit the motor clamp ring



- (1) Motor clamp ring
- (2) Motor flange
- (3) Screws (2 off)
- (4) Maintain equal spacing

- Place the **motor clamp ring** around the housing.
- **Assemble the screws by hand.** (pos 3)
 - » **do not tighten** at this stage.
- Maintain **equal spacing** when assembling the **clamp ring** (pos 3).

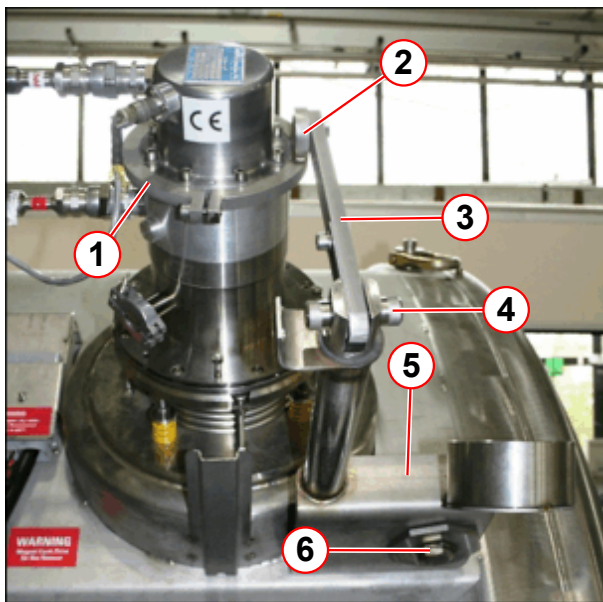
Fig. 84: Align the motor clamp ring



- (1) Motor clamp ring gap
- (2) Valve stem housing
- (3) Pivot arm

- Connect the **pivot arm** (pos 3).
- Ensure the **motor clamp** does not touch onto the **valve stem housing** (pos 2).

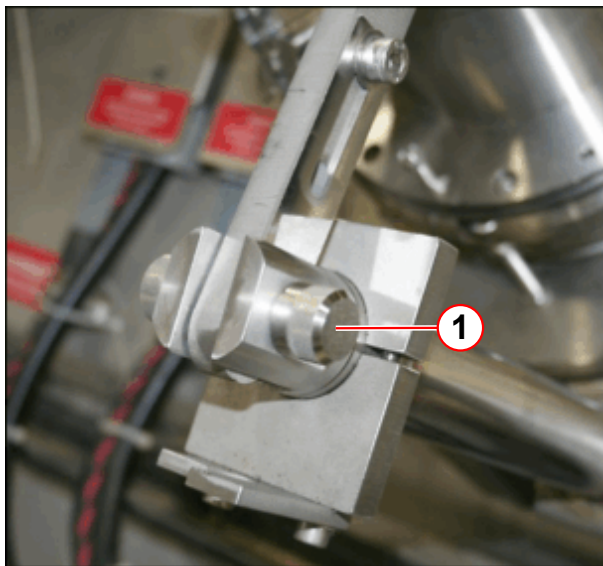
Fig. 85: Motor removal tooling (at field)



- (1) Motor clamp ring
- (2) Pivot arm connection to motor clamp ring
- (3) Pivot arm
- (4) Pivot bolt
- (5) Tool base and heat gun holder
- (6) Locking nut

- Attach the **tool base** to the **syphon bracket**.
- **Hand tighten** the **locking nut** (pos 6).

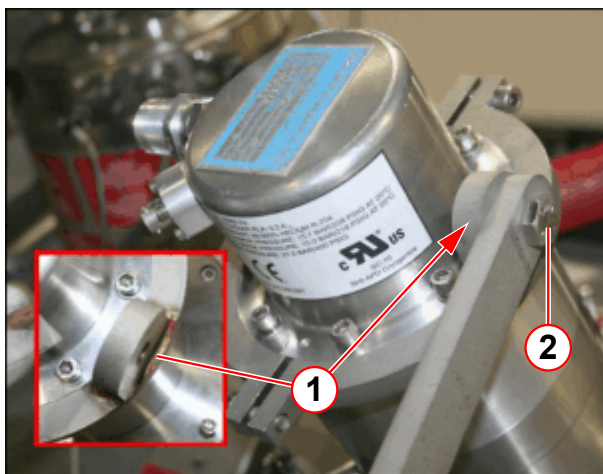
Fig. 86: Tighten the pivot bolt



(1) Pivot bolt

- Place the **pivot arm** in the **slot** of the tool base pillar.
- **Secure** the arm by **tightening** the **pivot bolt**.

Fig. 87: Fit the clamp arm to the motor housing



(1) Motor clamp ring
(2) Set screw

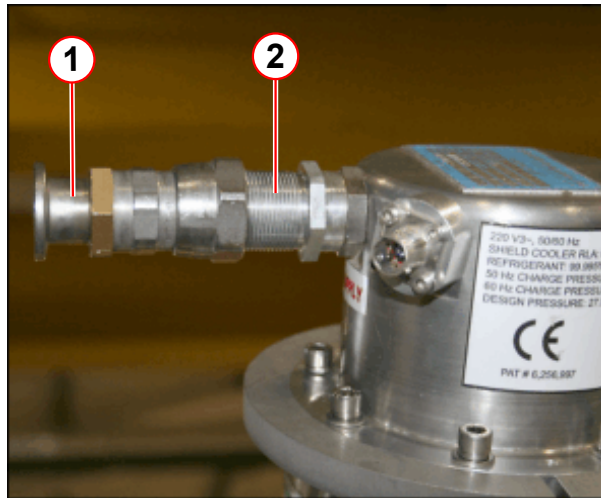
- Slide the arm forward and **attach** it to the **motor clamp**. **Tighten** the set screw **by hand**.
 - To allow for movement do not fully tighten the set screw.
- When all parts are **aligned**, tighten the:
 - set screw
 - pivot bolt
 - locking nut

- **Connect** the **heat gun** to mains power.
- Place the heat gun in the **cradle**.

3.2.11 Displacer exchange

Depressurise the cold head

Fig. 88: Depressurize the cold head



- (1) Female Aeroquip purge adaptor
- (2) SUPPLY coupling

- Screw the **female Aeroquip purge adaptor** securely onto the **supply coupling**, until **gas is heard** to escape.
- When **no more gas** is heard escaping, **remove the purge adapter**.
- Screw the **female Aeroquip purge adaptor** securely onto the **return coupling**, until **gas is heard** to escape.
- When **no more gas** is heard escaping, **remove the purge adapter**.



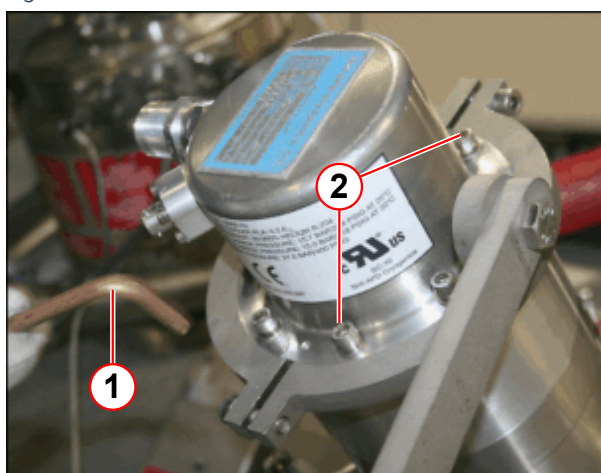
CAUTION

Risk of Air Ingress and Ice forming.

Failure to observe the following caution may result in damage to the cold head and displacer before it is disassembled.

- » Once the gas pressure within the cold head has subsided immediately remove the female Aeroquip adapter to prevent air ingress into the cold head.

Fig. 89: Remove the M5 bolts



- (1) Allen key (non magnetic)
- (2) M5 x 90 Allen bolts (8 off)

- Remove the **8 off M5 x 90 allen bolts**, securing the **motor and valve stem housing** to the **cold head**.

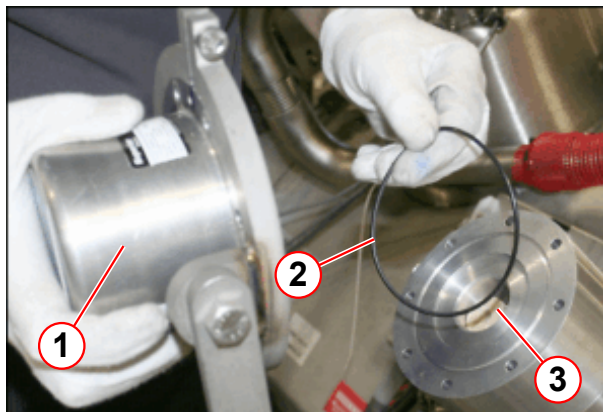
**CAUTION**

Risk of Air Ingress and Icing.

Failure to observe the following caution may result in damage to the new displacer.

- » Once the 8 off M5 x 90 Allen screws have been removed complete the rest of the removal procedure as quickly as possible in a single operation.

Fig. 90: Remove the rotary valve



- (1) Motor housing
- (2) O-ring
- (3) Rotary valve

- Lift away the **motor housing** and remove the **valve stem housing O-ring**.
 - » In case of accidental damage, retain the **old** O-ring as a backup.
- Remove the **rotary valve** (pos 3).
 - the rotary valve can sometimes **remain in the valve stem housing** when the motor housing is removed.

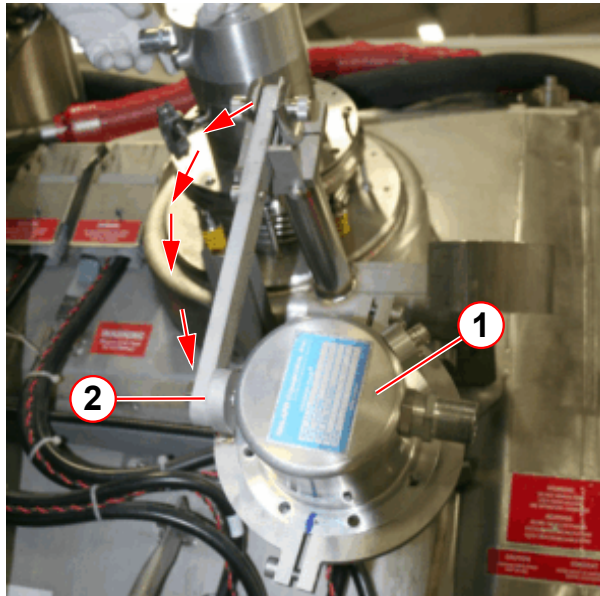
**WARNING**

Risk of Cold Burns.

Failure to observe the following safety information may result in physical injury.

- » Do not touch the cold displacer surfaces with bare hands.

Fig. 91: Magnet at field - Support the motor housing in the service tool cradle



- (1) Motor housing
- (2) Pivot arm

- Swing the **cold head motor** away, **using the cradle** of the cold head service tool.



Whisper Plus cold heads prior to serial number 30, have 2 off M3 x 60 screws that secure the motor and valve stem housing together.

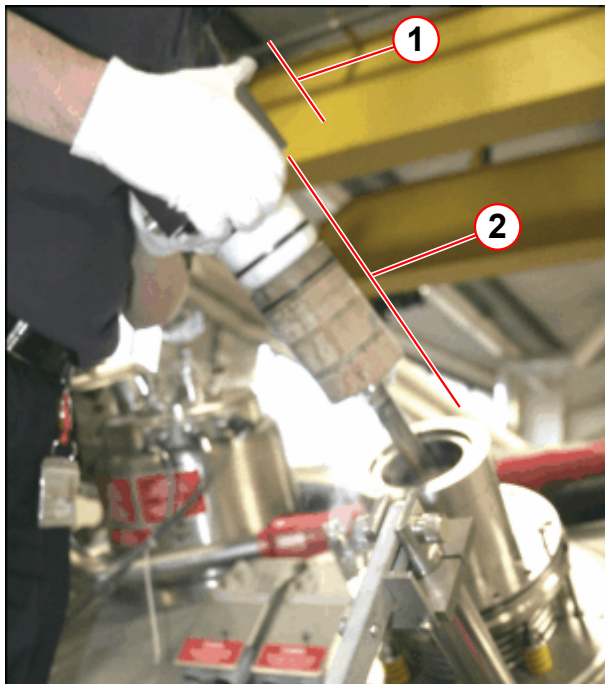
For these units remove the motor and valve stem housing as a complete assembly. After the assembly is removed, the 2 off screws can be discarded.

If the cold head is prior to serial number 30:

- **Remove the 2 off M3 x 60 screws**, securing the **motor** and **valve stem housing** together.
- **Remove the O-ring** from the **cold head**.

Remove the displacer

Fig. 92: Removing the valve stem and displacer



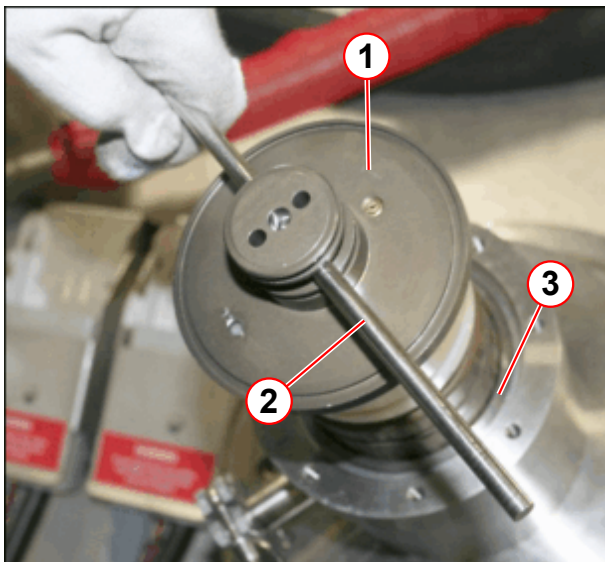
- (1) Valve stem housing
- (2) Displacer

Valve stem separation

It is possible for the **valve stem** to **stick in the valve stem housing** when being removed from the cold head.

In this case, the **valve stem** and **displacer** will be extracted as **one unit**, when the valve stem housing is removed.

Fig. 93: Valve stem extractor



- (1) Valve stem
- (2) 7.5 mm extraction bar
- (3) Cold head O-ring

If the **valve stem** and **displacer assembly** have **remained** in the **cold head**:

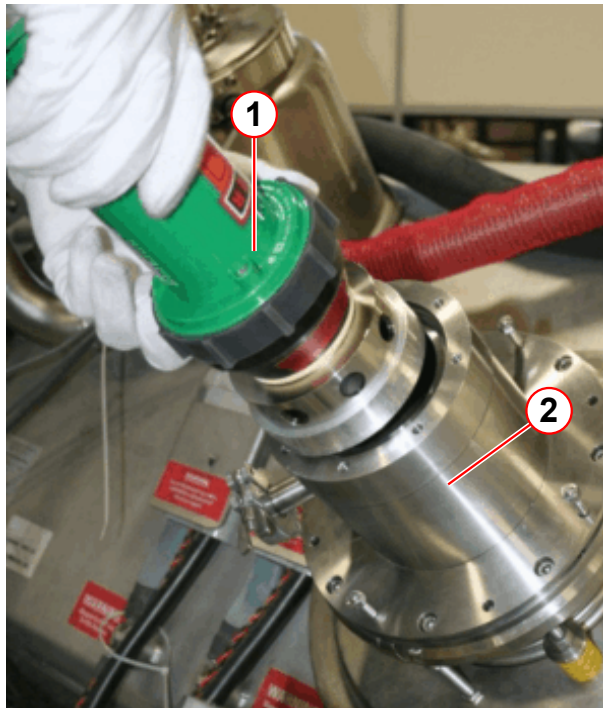
- Insert the **7.5 mm extraction bar** through the hole in the valve stem.
- Pull **upwards**, and position safely as these components will be **very cold**.
 - » Wear **protective gloves** during exchange procedure.

Fig. 94: Clean the cold head cylinder



- Remove the O-ring from the **cold head cylinder**.
- **Clean** the upper part of the **cold head cylinder**, (where the O-ring is located).
- **Do not clean inside the cold head.**
 - » Due to the very cold temperatures **any cleaning materials will freeze.**

Fig. 95: Insert the heat gun into the cold head cylinder

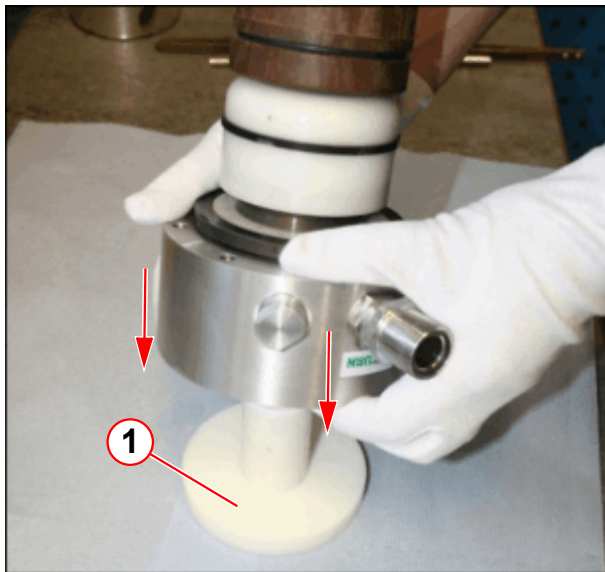


- (1) Heat gun
(2) Cold head cylinder

- **Insert** the **heat gun** into the cold head. **Select control setting '3'.**
 - The heat gun can be **left on** (at this low setting), during the next preparation stages.

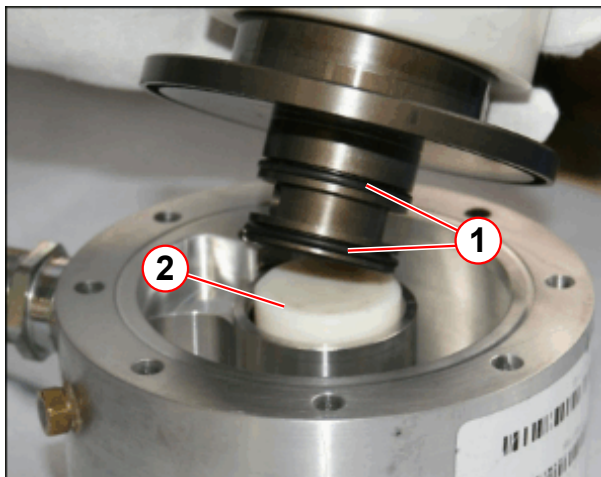
3.2.11.1 Separation of the valve stem

Fig. 96: Separate the valve stem housing and displacer



(1) Stem extraction tool (large diameter shown)

Fig. 97: Displacer and valve stem



(1) O-rings

(2) Extraction tool (small diameter)

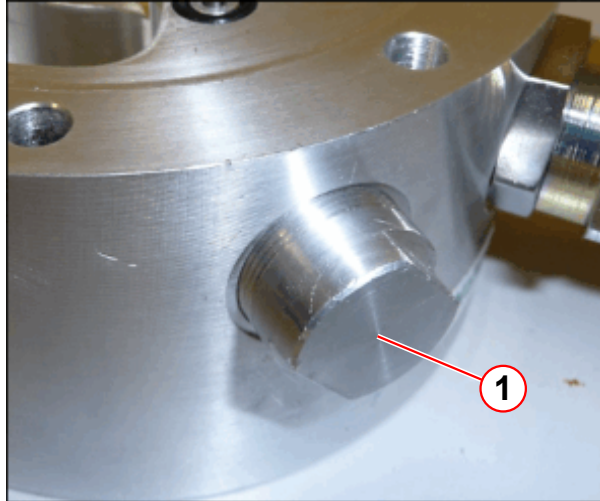
- The **displacer** can **remain** in the **valve stem housing** during removal.
- Use the **valve stem extraction tool** to separate the displacer.
 - Place the **large diameter** of the **extraction tool** on a smooth flat surface.

- Fit the **smaller diameter** into the **valve stem housing** and gently push down on the housing as shown.

3.2.12 Preparing the new displacer

3.2.12.1 Variable orifice cleaning and preparation

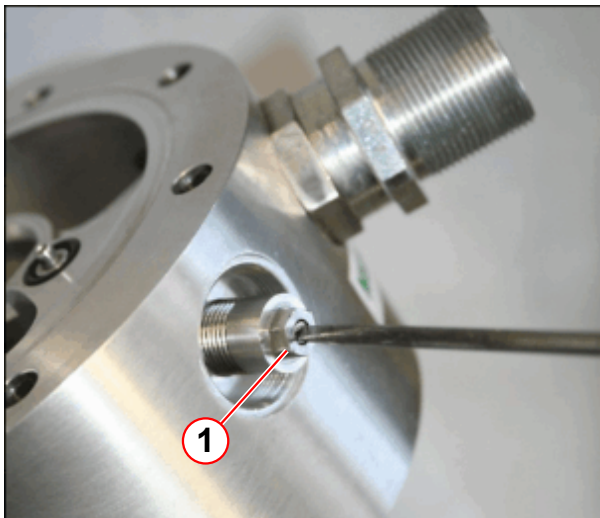
Fig. 98: Variable orifice cover



(1) Orifice cover

- Remove the variable **orifice cover**, using a **17 mm** wrench.

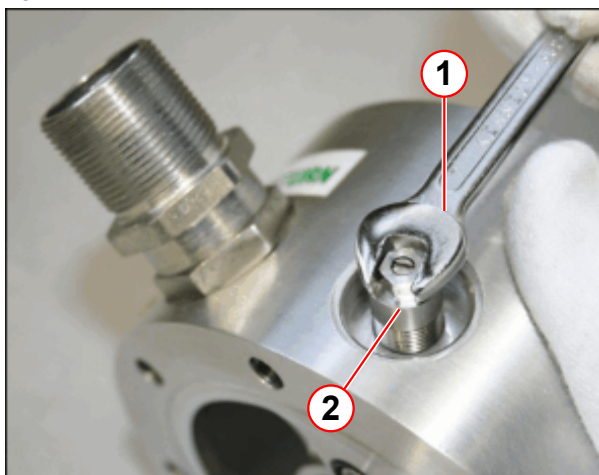
Fig. 99: Lock the variable orifice screw



(1) Turn the screw until level with the nut

- Use a **3 mm** flat bladed screwdriver to turn the variable orifice screw counter-clockwise, until it locks in the variable orifice nut:
 - » **count** the number of **turns** until the screw locks tight and is **level** with the **end of the nut**.
- **Record** the number of **turns**.

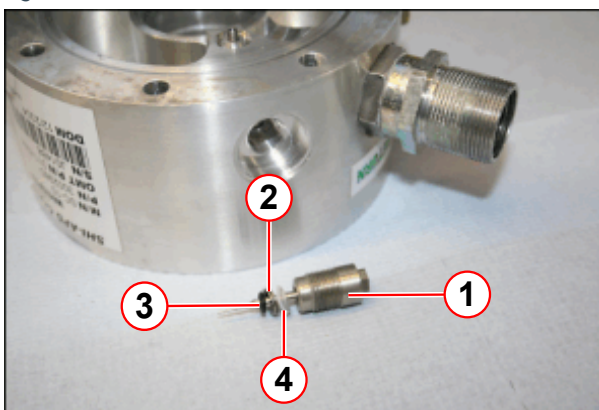
Fig. 100: Remove the variable orifice nut



(1) 7 mm combination wrench

- Use a **7mm** wrench to **slacken** the **variable orifice**.

Fig. 101: Remove the variable orifice needle valve



(1) Needle valve assembly
(2) St. Steel washer (2 off)
(3) O-ring (1 off)
(4) PTFE washer (1 off)

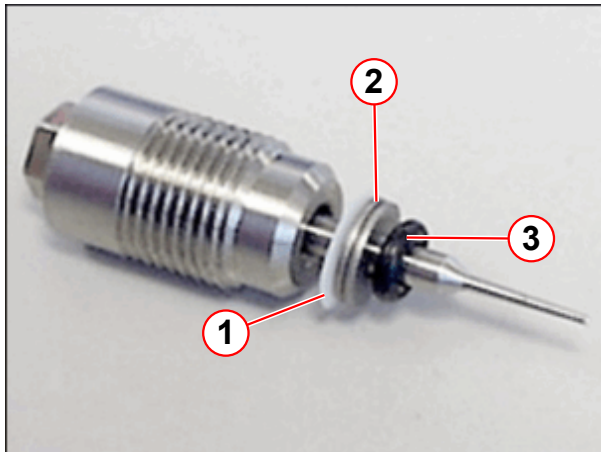
- **Unscrew** and **remove** the variable orifice **needle valve** (pos 1).
- **Remove:**
 - 1 off **O-ring** (pos 3),
 - 2 off **St Steel washer** (pos 2),
 - 1 off **PTFE washer** (pos 4),
- **Discard** the removed O-ring and washers.

- **Clean** the variable orifice **needle valve**.
- Blow **dry** with **helium gas**.



DO NOT apply GREASE to the O-ring.

Fig. 102: Fit new washers and O-ring

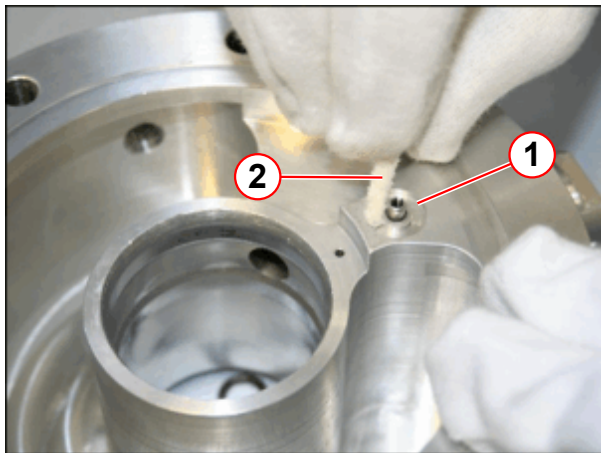


- (1) PTFE washer (1 off)
- (2) St. steel washer (2 off)
- (3) O-ring (1 off)

- **Fit the new O-ring and washers** (from the new displacer kit), to the **shaft of the needle valve**.

- PTFE washer (1 off)
- St Steel washer (2 off)
- O-ring (1 off)

Fig. 103: Clean the valve seat

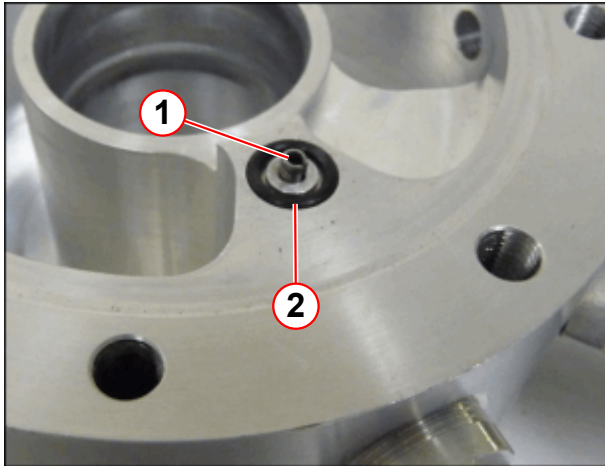


- (1) Pipe cleaner and cleaning fluid

- Using **pipe cleaners** and the **cleaning fluid**, clean the **valve seat** in the valve stem housing.

- **Insert the needle valve assembly** into the valve stem **housing** and screw down **by hand**.
- **Tighten** using the **7mm wrench**.
- Use the **3mm** flat bladed screwdriver to **screw the needle valve** into the valve stem **housing**. **Apply the number of turns** recorded during removal.
- **Recheck the tightness** of the **needle valve** with the **7 mm wrench**.
- **Refit the variable orifice cover**, using the **17 mm wrench**.

Fig. 104: Roll pin with the O-ring fitted

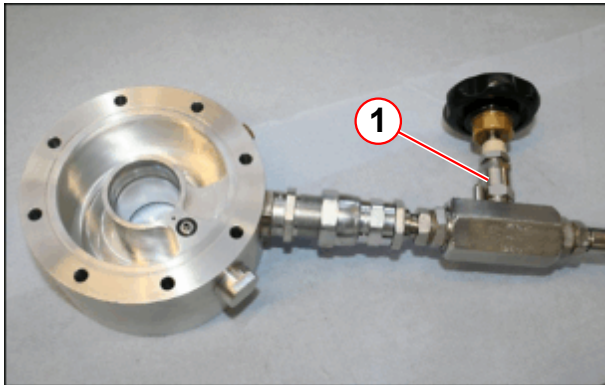


- (1) Roll pin
(2) O-ring

- **Clean all surfaces** of the valve stem housing to **remove** any **contamination**.
- Fit the **new valve seat O-ring**, at the roll pin (pos 2).

3.2.12.2 Prepare the valve stem housing

Fig. 105: Connect the He gas supply to the housing



- (1) Shut off valve

- **Prepare** the valve stem **housing** for assembly of the new displacer.
- **Connect** the He **gas supply**.
 - **do not open** the gas shut off valve at this time.

- **Clean** the motor of all **contamination** from the old displacer.
- Check the **motor shaft roll pins** are **centered** on the **shaft** and that the pins and valve spring are undamaged.
- Screw the female **Aeroquip purge adapter** to the supply coupling of the **cold head motor**.
 - Be sure **gas** is able to **flow** through the **purge adapter**.
 - This will allow gas to **escape** through the **motor** when it is **installed** onto the valve stem assembly.

Prepare the new rotary valve



The rotary valve is a loose fit to the motor shaft and may fall off during assembly.

- Rub away the **rotary valve surface** with the **sharpening stone**.
- This will reduce the possibility of a high pitched sound coming from the new rotary valve once installed and operating.

Checking and preparing the motor for purging using the Leybold simulator

Fig. 106: Leybold simulator



- Connect the **Leybold simulator** to a suitable 220 volt AC supply.
- Connect the **cold head power cable** removed from the **filter panel** to the **simulator**.
- Switch on the **Leybold Simulator** to run the **cold head motor**.



Using the Leybold simulator tool with an APD motor, the motor rotation will be counter clockwise due to a change in the phases from Leybold to APD motors.

This is adequate for the testing and purging procedure.

Displacer preparation

Fig. 107: Displacer kit

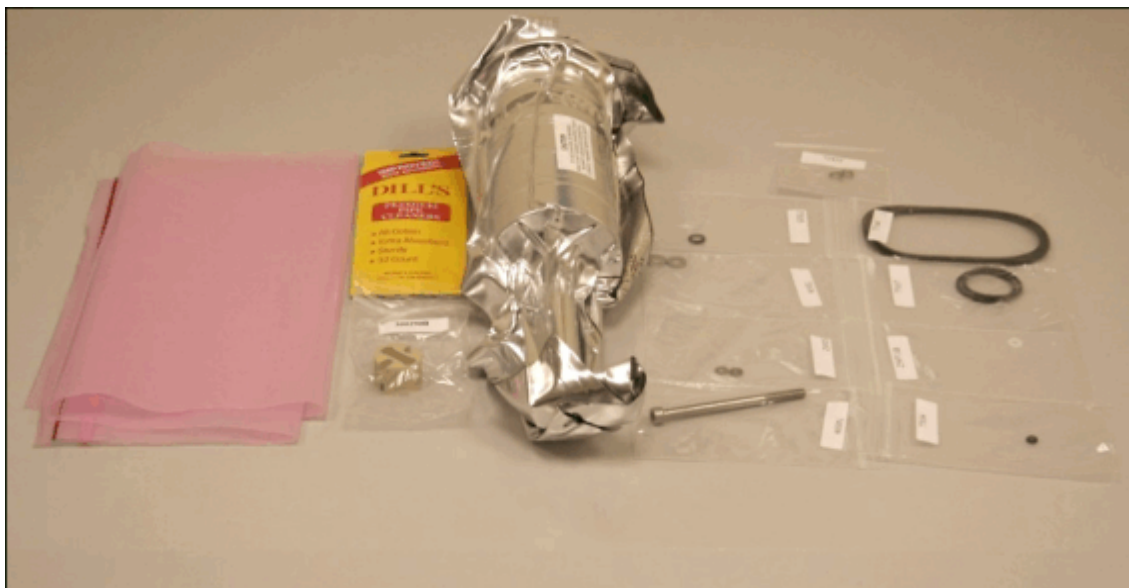


Fig. 108: Seal compression tools



- **Clean the inner surfaces** of the displacer **seal compression tools**.
- Use **lint free** paper towelling, lightly soaked in cleaning fluid



CAUTION

Contamination of the Displacer.

The displacer body is made of Hygroscopic material and performance will be compromised if contaminated.

- » Avoid prolonged atmospheric exposure.
- » Avoid grease.
- » Wear cotton gloves during exchange procedure.
- » Place the displacer in a helium background whenever possible throughout this procedure.



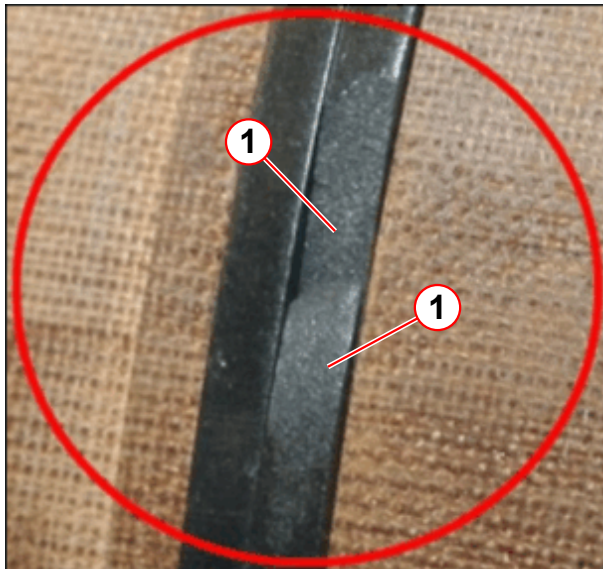
Wear cotton gloves when handling the displacer.

- Remove the **displacer** from the sealed packaging.

Fig. 109: New Displacer and packaging



Fig. 110: Displacer seals - tapered ends at 180 deg to each other



(1) Tapered ends

- Inspect the **displacer seals** to ensure the tapered seal **ends** are at **180° to each other**.

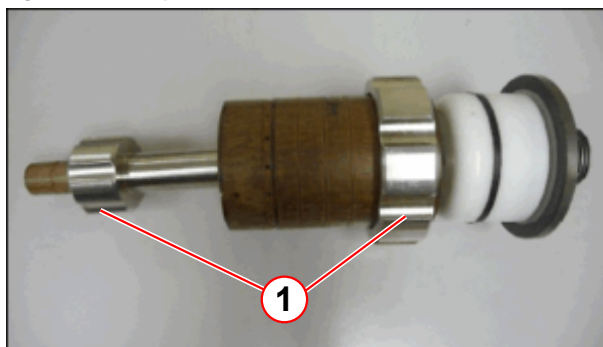
» This will **prevent** gas flow past **the seals** when installed.



The seal compression tools have an internal taper. The large end of the taper must be passed over the seal first, continue to slide the tool onto the seal until the seal is fully compressed into its seat.

Repeat for all seals. **DO NOT FORCE** the tools. Excessive force will result in damage to the seal.

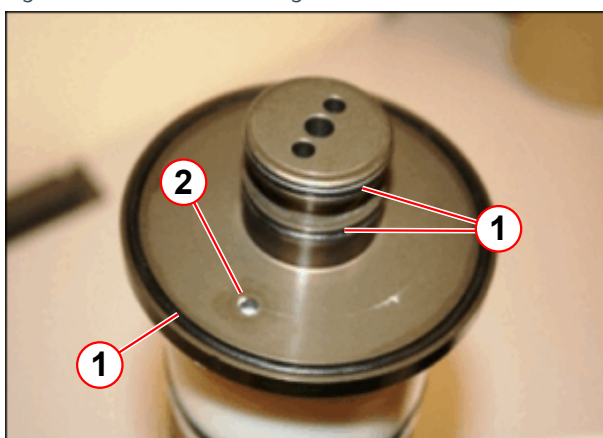
Fig. 111: Compression tools fitted



(1) Compression tools

- **Carefully** press the **seals** into their **seats** using the displacer seal **compression tools**

Fig. 112: Valve stem O-rings



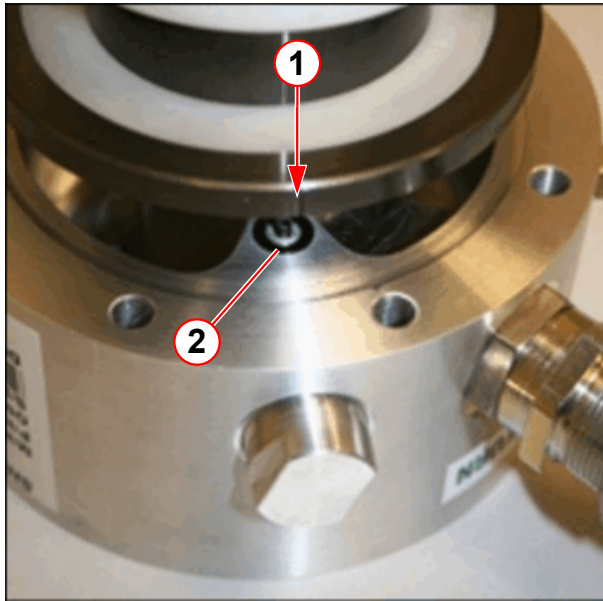
(1) O-rings
(2) Valve stem mating hole

- Very **lightly grease** the valve stem **O-rings**, and fit them to the **valve stem**.
- Do **not** allow **grease** to come into contact with the **displacer**.



Exercise extreme care when applying Apiezon 'N' grease to the thin O-rings. Due to the thin size of these O-rings any stretching will prevent them seating correctly.

Fig. 113: Valve stem alignment



- (1) Roll pin
- (2) Alignment slot

- **Align** the **roll pin** (in the valve stem housing) with the **mating hole** in the valve stem, by aligning **the slot** (pos 1), on the **outer** diameter of the **valve stem**.

Fig. 114: Displacer in sealed bag



- Place the assembled displacer and valve stem assembly into the **resealable bag** provided.
- Open the **gas regulator** valve and fill the bag with helium gas.
- **Maintain** a gas flow of **4 psi** into the bag.
 - **regulate the flow** at the pressure regulator to avoid any accidental **pressure increase**.
- The assembly is now **prepared for installation**.

3.2.12.3 Heating and cleaning

Fig. 115: Heat gun - set to position 8



For cleaning, **both cold head stages** inside the cylinder must be **heated to 25-30 °C**.

- Set the **heat gun to position 8**.
 - **Begin to heat** the cold head cylinder.

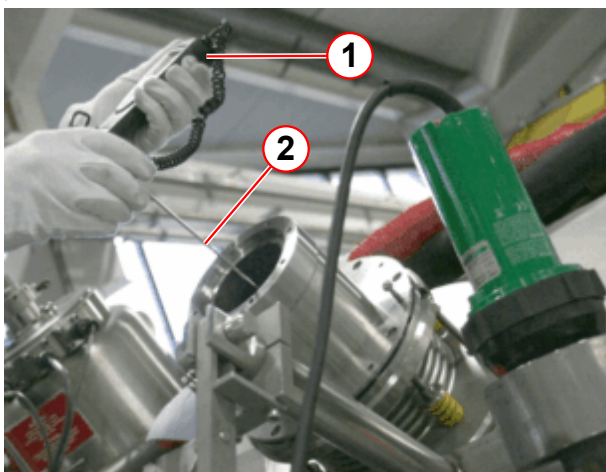
**WARNING**

Damage to internal magnet components.

Overheating of the cold head cylinder can damage the internal shield links and insulation.

- » Do not leave the heat gun in the cold head cylinder un-attended.
- » Continuously monitor the cold head cylinder temperature.

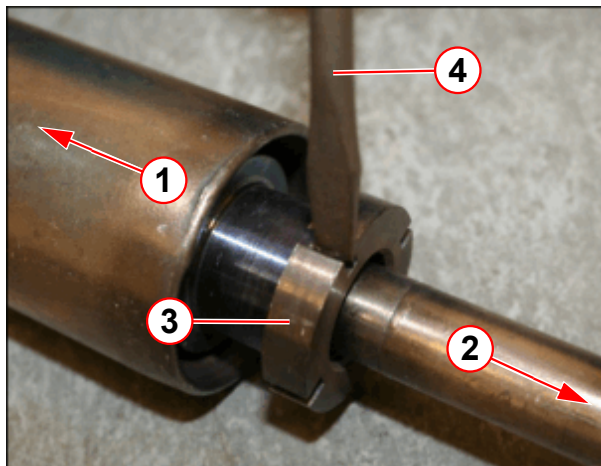
Fig. 116: Check the cylinder temperature with the probe



- (1) Electronic temperature sensor
- (2) Surface probe

- Check the **cylinder temperature** frequently using the **temperature measurement instrument** and **surface probe**.
 - **remove the heat gun**.
 - touch the **surface probe** on each of the stages.

Fig. 117: Adjusting the air flow temperature



- (1) Heat to 20k stage
- (2) Heat to 80k stage
- (3) Air flow adjustment screw
- (4) Screwdriver

- **Maintain an equal temperature at each stage.**
 - **regulate** the hot **air flow** from one stage to the other as required.
 - adjust the **air flow adjustment screw** to adjust the **temperature** from one stage to another (as required).
 - use a **screwdriver** to turn the air flow **adjustment screw**.

**WARNING**

Risk of Hot Burns.

Failure to observe the following safety information may result in physical injury.

- » **Do not touch hot surfaces with bare hands.**
- » **Personnel safety equipment must be worn.**

- It will take approximately **5 min** to raise the cylinder temperature **above a sub zero level**. After this, the temperature should be **checked** every **60-90 seconds**.
- When the **cylinder temperature** has reached **25-30°C**,
 - **wrap** cleaning cloth/lint free paper around the **tweezers**.
 - **moisten** in **cleaning fluid**.
 - **clean** around the upper and lower **cylinder bores** and **copper stages**.



Take great care not to scratch the cylinder and copper stages!!



This cleaning must be done QUICKLY. The cold head cylinder temperature will return to a sub zero value and cause the cleaning material to become frozen and stick to the cylinder.

Approximately 10 seconds cleaning time is recommended.

- If cleaning materials **freeze to the cold head displacer bore**:
 - » **Withdraw** the tweezers and **remove** as much of the cloth/lint free paper as possible.
 - » Use the **heat gun** and continue heating until the cloth **defrosts** and can be removed, or has burned away.
- Continue to **clean the cylinder** (as many times as required), until it is as clean as possible, keeping the **temperature within** the **25-30 °C** range.

Heating the cold head

- When the cylinder is clean, **continue to apply heat**. Monitor until the **critical temperature of 55 °C** is reached.
- During the heating process, **prepare necessary tooling/spare parts** required for the next steps.
 - ensure they are **positioned safely** and **ready for use**.

3.2.12.4 Installation of the new displacer



Wear cotton gloves when handling the displacer.

From **55 °C to 60 °C** you will have **approximately 2 minutes** to prepare the new displacer for installation.

During this time the next steps can be carried out. **Continued monitoring** of the cold head temperatures **will not be required** if this process is **followed without problems**.

- When the final **55 °C measurement** has been made, reinstall the **heat gun**.
- **Remove** the prepared displacer from the **helium filled bag**.
- **Check all seals are compressed** using the compression tools and **then remove them**.
- Move the displacer to the magnet.



The following steps must be completed within 90 SECONDS.
The 90 seconds will start when the heat gun is removed.

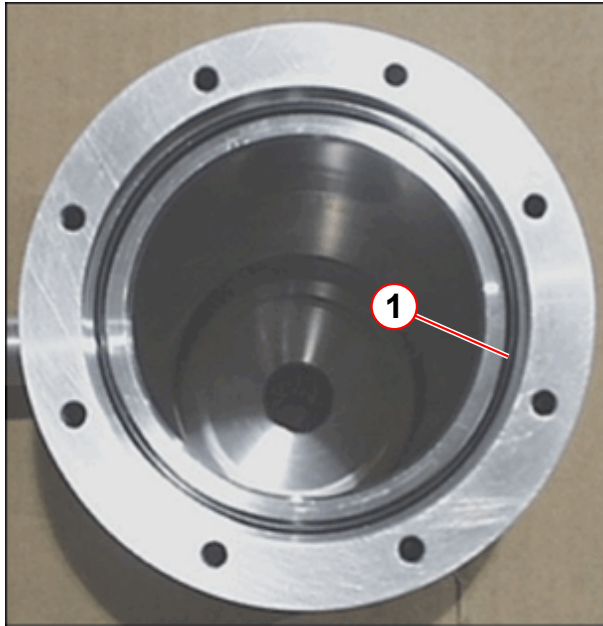


If there is any reason to abort the installation of the displacer, abort within the first 40 seconds to prevent moisture contamination destroying the new displacer.
Reinsert the displacer into the plastic bag and refill with helium gas.



As long as no moisture contamination has occurred on the new displacer, the process can be recovered and another attempt made when the cold head cylinder and displacer is prepared once again.

Fig. 118: Cold head cylinder O-ring



(1) O-ring

- Fit a **new O-ring** to the **cold head cylinder**.
- Check that the **O-ring** is **seated correctly**.



The O-ring can expand due to the temperature of the cold head body. Fit the second new O-ring if required.



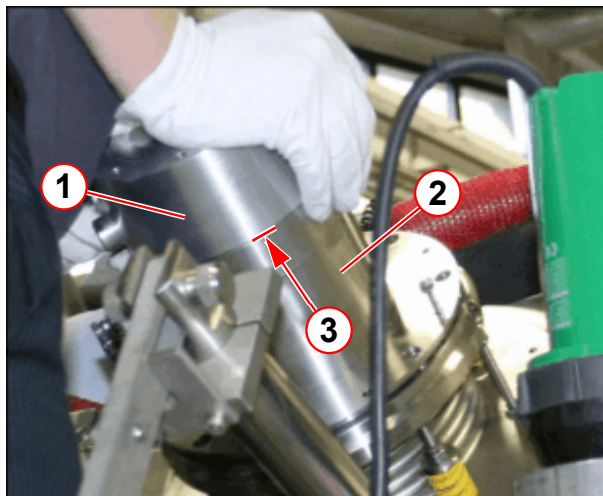
Take great care when inserting the displacer. This operation must be performed quickly.

Fig. 119: Insert the displacer



- **Carefully insert** the displacer into the **cold head**.
 - use a **twisting motion** to ensure the second stage displacer **locates** into its **cylinder**.
 - Ensure the **seals** do not become **unseated** or dislodged during installation.
 - There will be a continuous low **flow of helium gas** throughout this procedure.

Fig. 120: Push down until the valve stem housing is seated on the O-ring



- (1) Valve stem housing
- (2) Cold head cylinder
- (3) O-ring position

- Continue to **push the displacer down** into the cold head, until the **valve stem housing** seats onto the cold head **O-ring seal** (pos 3).



If a slight resistance is felt while inserting the displacer, stop and check if the displacer seals have become unseated.

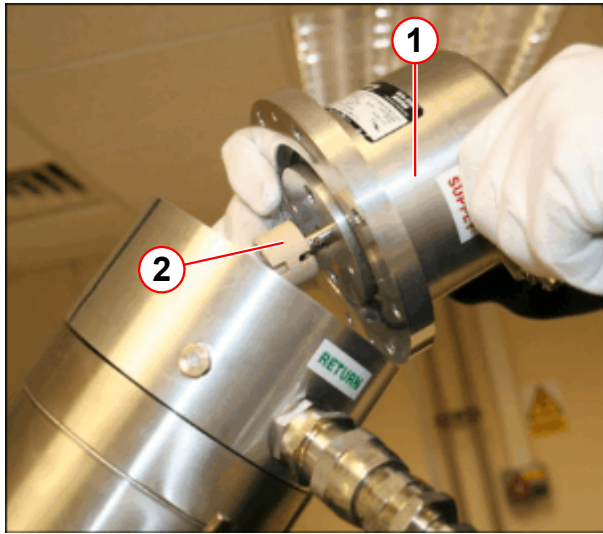
Fig. 121: Fit a new O-ring to the valve stem housing



- (1) O-ring

- Fit a **new O-ring** to the top of the **valve stem housing**.

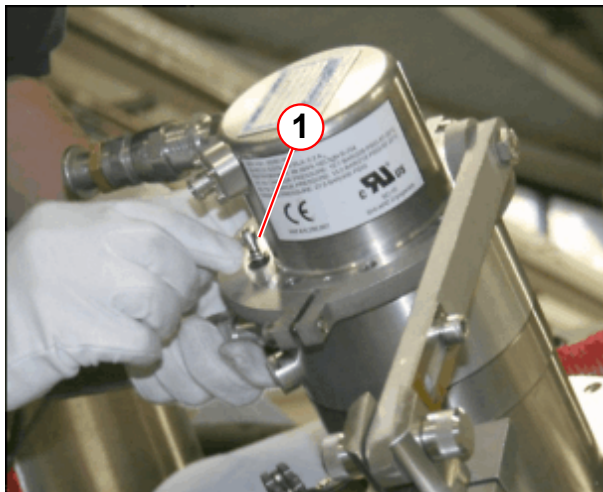
Fig. 122: Fitment of the motor housing



- (1) Motor
- (2) Rotary valve

- Fit the **motor**, (complete with rotary valve fitted), into the **valve shaft**.
 - Hold the rotary valve **in position** with the end of your finger, while the motor is placed onto the valve stem housing.

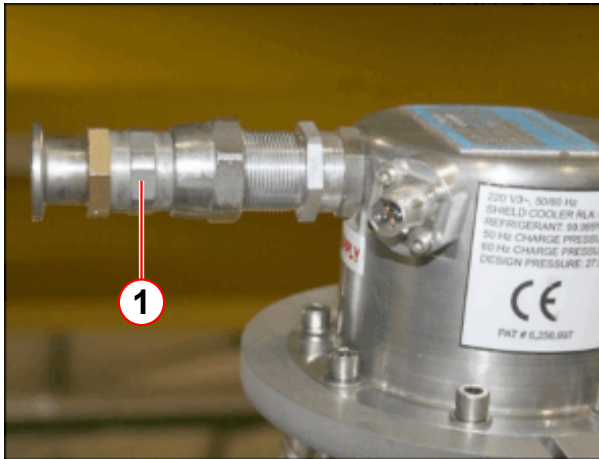
Fig. 123: Secure the cold head motor



- (1) M5 x 90 Allen bolts (8 off)

- Press and **hold the motor down** on to the **valve stem housing**.
 - ensure the **motor** is kept **in place** to **avoid** the **O-rings** moving out of position.

Fig. 124: Purging adaptor



(1) Female aeroquip purging adaptor

- A gentle stream of **gas** should now be flowing from the **female** Aeroquip **purging adaptor**.
- Allow the gas flow to continue whilst **securing** the **motor** to the **cold head cylinder**

- Ensure the motor is **aligned** with the **orientation marks**.
- **Secure** the cold head **motor** using a minimum of 3 off **bolts**.
 - **tighten** the bolts in a **diametric pattern** to spread the load on the O-ring.
 - **at this stage, low pressure gas is flowing**.
- When the **motor** is secured, **remove** the **purging adaptor** from the **Supply coupling**.



When the flow of gas through the purge adaptor has stopped the critical 90 seconds procedure is complete.

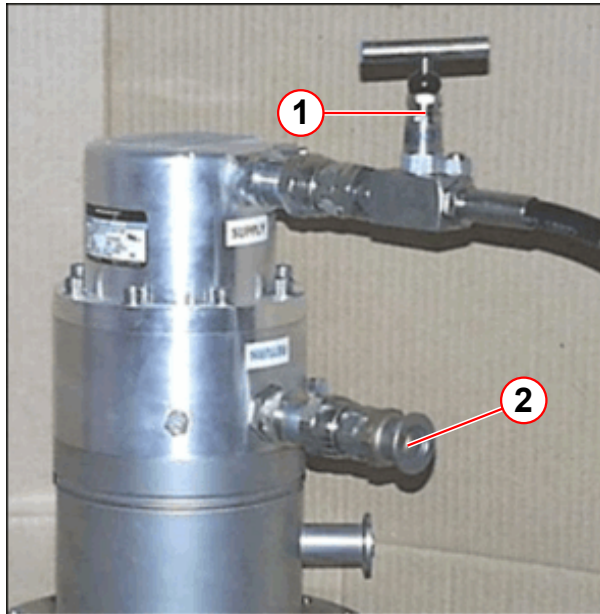
- **Fit** the remaining **motor bolts** and **torque all bolts to 15 Nm**.



Ensure all motor bolts are installed and are secure before commencing the high pressure purging of the cold head.

3.2.12.5 Purging the Cold head with Helium Gas

Fig. 125: Purge adaptor fitted



- (1) He gas supply valve
(2) Purge adaptor

- **Disconnect** the **gas supply** and purge adaptor from the **Return Aeroquip** couplings.
- **Connect** the gas supply to the **cold head Supply Aeroquip** coupling.
- Loosely screw the **purging adapter** into the **Return Aeroquip** coupling (by hand).
- To prevent **air ingress** into the cold head, **do not** screw the **purge adaptor** fully onto the **return** coupling.

- **Reconnect** the **power cable** to the cold head **motor**.
- **Open** the gas supply valve and **set** the regulator to **approximately 5 bar**.
- **Switch on** the motor **power supply**.
- **Screw** the **purging adapter** fully onto the gas **Return Aeroquip** coupling.
 - » The helium **gas** will **discharge** in **surges** from the **Return side** of the cold head.
- After **5 surges**, **close** the **gas supply shut off valve**.
- With the motor **still running**, allow the **pressure to discharge** from the cold head, then quickly **OPEN** the **gas supply valve**.
 - **Do not** allow **air ingress** via the purge adapter
- **Repeat** this sequence **5 times**.
- Continue to purge **continuously** (for at least 15 minutes) or until approximately **50 bar** is left in the gas cylinder.
- **This action:**
 - **purges** the cold head
 - **reduces heat input** from the heat gun, to the **cylinder**



The cold head motor tooling may be removed during purging.

Final steps of purging

- **Increase** the pressure regulator to **16 bar**.
- **Remove** the **purge adaptor** from the **Return** Aeroquip coupling.
 - Ensure the **gas pressure** is **trapped** within the **cold head**.
- **Switch off** the cold head motor **power supply**.
- **Close** the gas supply **shut-off valve**.
- **Disconnect** the **gas supply** from the **cold head Supply** Aeroquip coupling.

3.2.13 Reconnection of compressor and components

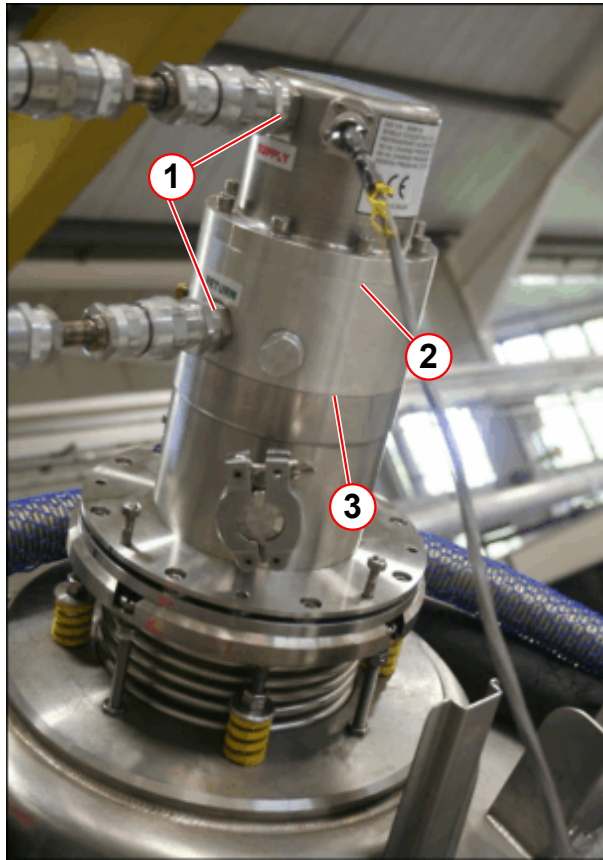
- **Connect** the **cold head/compressor** electrical connections.
- **Connect** the **Supply** and **Return** Aeroquip couplings.
- **Check** and record the compressor **static pressure**.
- **Check** all cold head **electrical connections**.
- **Turn on** the **compressor**.
- **Refit** the **he syphon** onto the bracket. Ensure the **syphon end cap** is **fitted**.
- **Refit covers** and any system **equipment** removed **during disassembly**.



Ensure no tooling or equipment has been left on or in the vicinity of the magnet.

3.2.14 Final steps

Fig. 126: Leak check points



- (1) Aeroquip fittings
- (2) O-ring seals
- (3) Orifice seal

Use a suitable **helium leak detector** capable of detecting leaks **down to 1×10^{-4} mbar l/s**.

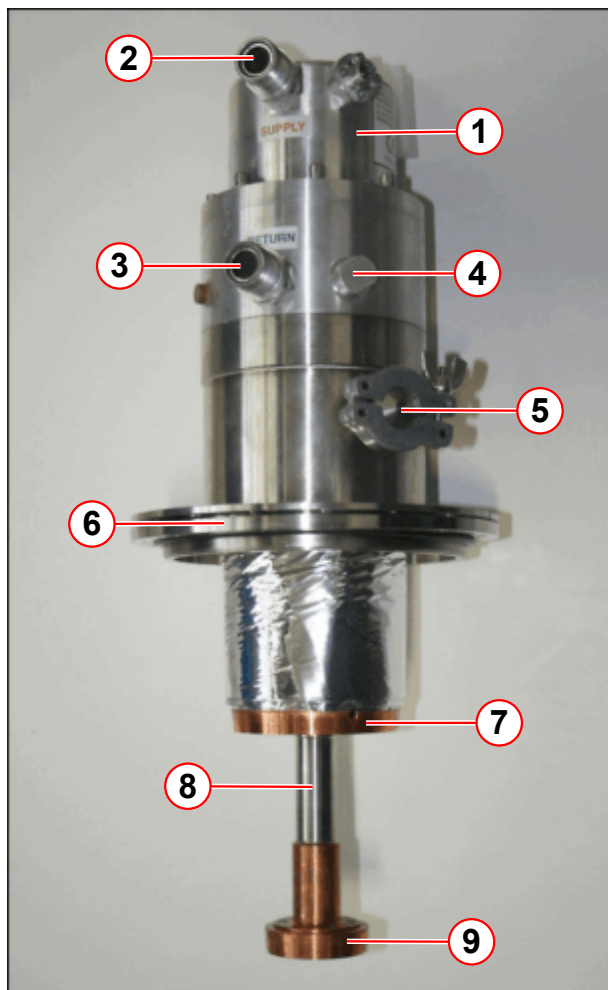
- Record the **shield temperatures**.
- After **2 hours**:
 - » Check and record the **compressor pressure**.
 - » Check and record the **shield temperatures**.

3.2.15 Magnet ramping

If the magnet was **ramped down** prior to changing the displacer, ramp the magnet back to field. Follow the standard ramping procedures.

3.3 Replacing the 10K Cold Head

Fig. 127: 10k Cold head



- (1) Motor housing
- (2) Aeroquip coupling SUPPLY
- (3) Aeroquip coupling RETURN
- (4) Oriface cover (do NOT remove)
- (5) NW16 gas lock port
- (6) O-ring clamp plate
- (7) 80k copper stage
- (8) Thin wall tube
- (9) 20k copper stage

3.3.1 General

This procedure describes the replacement of the **APD 10K Cold Head** for **10K magnet systems**.



Read the procedure and check serviceability of replacement parts before carrying out the replacement process.



Ensure that all tooling and spare components are available before starting the replacement procedure.



This procedure must only be performed by Certified Engineers, who have completed the relevant training course(s).

3.3.2 Safety



Ramp the magnet to ZERO.

This procedure **MUST NOT** be performed with the magnet AT FIELD.



CAUTION

Pressurized helium vessel.

Failure to observe safety precautions may result in serious bodily injury or property damage.

- » The magnet must be depressurized before attempting this procedure.



CAUTION

Risk of asphyxiation.

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogenics.

- » Do not vent helium gas into the magnet room.
- » Use an oxygen monitor.



WARNING

Risk of cold burns.

Failure to observe the following may result in personnel injury.

- » Do not touch cold surfaces with bare hands.

**CAUTION**

Risk of Hot Burns.

Failure to observe the following safety information may result in physical injury.

- » Do not touch hot surfaces with bare hands.
- » Personnel safety equipment must be worn.

3.3.3 Spare parts

- 10K cold head assembly (order no **07461598**)

Lich 2 Conversion to APD

- Cold head power lead (order no **10099938**)
- Compressor adaptor lead (order no **10100427**)

3.3.4 Required time and tools

3.3.4.1 Time

160 min (not including ramping time or removing the magnet cover).

3.3.4.2 Tools

- Standard tool kit.
- Cold head exchange kit (**10119989**).
- 3600 Magnet Power Supply kit (**08395613**).
- Apiezon N grease (**10099740**).
- Torque wrench(5 - 20 Nm)
- 10, 13, 19 mm wrench/spanner.
- 5 and 6 mm hexagon Allen key (5 mm hexagon ball driver).
- Flat blade screwdrivers.
- Bottle 99.999% (5.0) helium gas and high pressure regulator.
- Cotton gloves.
- De-greasing fluid (e.g. petroleum ether - sourced locally).
 - or cleaning fluid which leaves no residue.
- Protective clothing (gloves, safety glasses or face shield, lab coat, safety shoes).
- **Lint free** paper towelling.
- **O₂ monitor** (room or personal). (Personal monitor part no **10142116**).

3.3.5 Prerequisites



This procedure **MUST NOT** be performed with the magnet AT FIELD.

Complete the **Cold Head Exchange Report 10K Units** with the pre-exchange information. The form is supplied with the cold head and is also available to download.

- **Inspect** the new cold head:
 - » Check the **part number** is correct.
 - » Check for **visible damage**.
 - » If **faulty**, return and order a **replacement**.
 - » If correct continue with procedure.
- **Remove** the appropriate **magnet covers**.
- Ensure there is **adequate service height** above the cold head **before removal**.

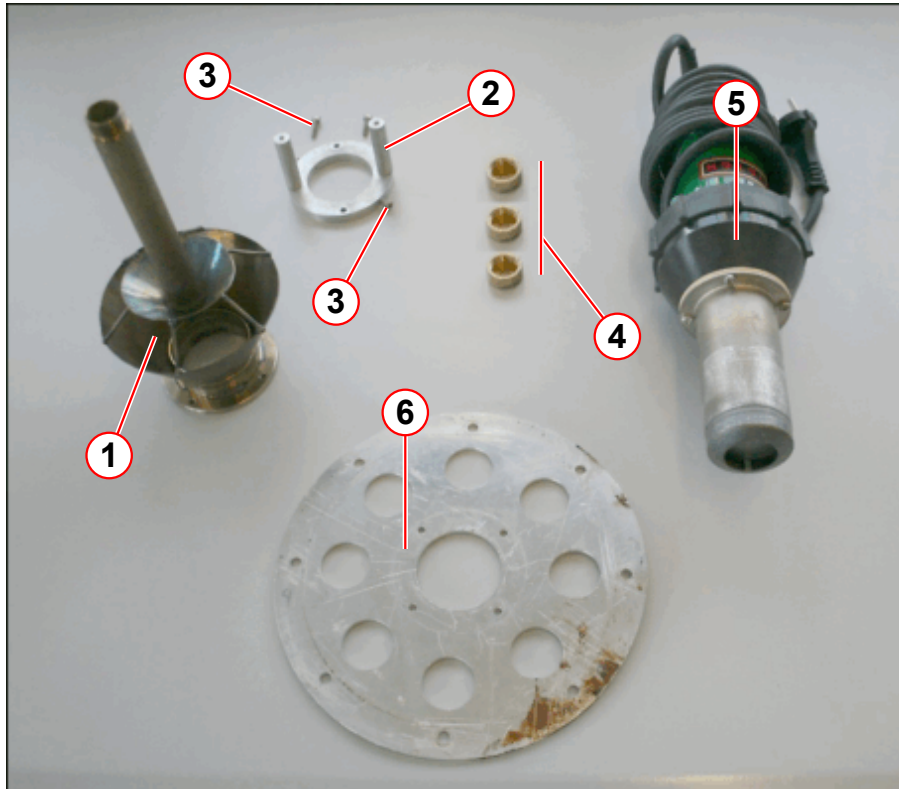


Before removing the cold head check that the area directly above the cold head is clear of cables and venting.

- Note the **field current**.
- **Ramp** the magnet to **zero field**.
- Ensure an **oxygen monitor** is activated in the work area.
- Secure the **service platform** to the system using the straps.

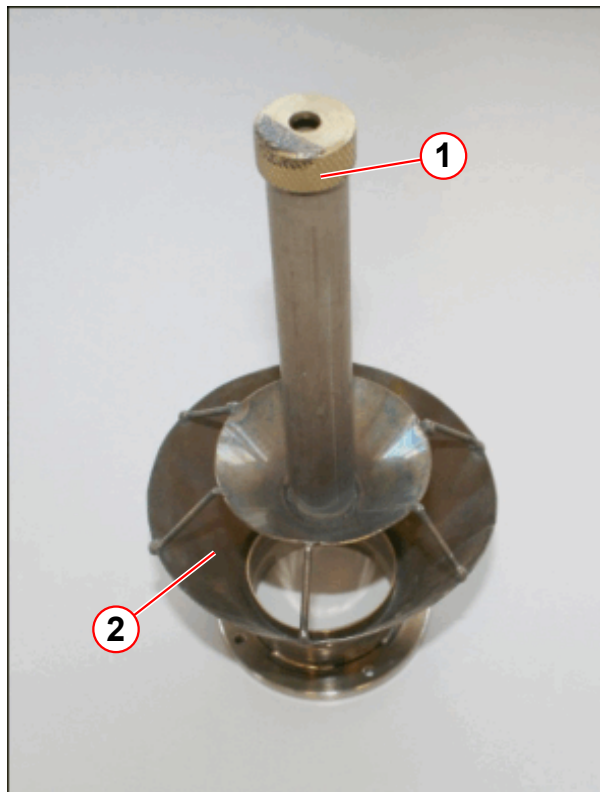
3.3.5.1 Assemble the heat gun and hot window assembly

Fig. 128: Heat gun and hot window assembly



- (1) Tube assembly
- (2) Spacer
- (3) 3mm Allen screws (4 off)
- (4) Brass temperature regulator nozzles (3 off)
- (5) Heat gun
- (6) Hot window

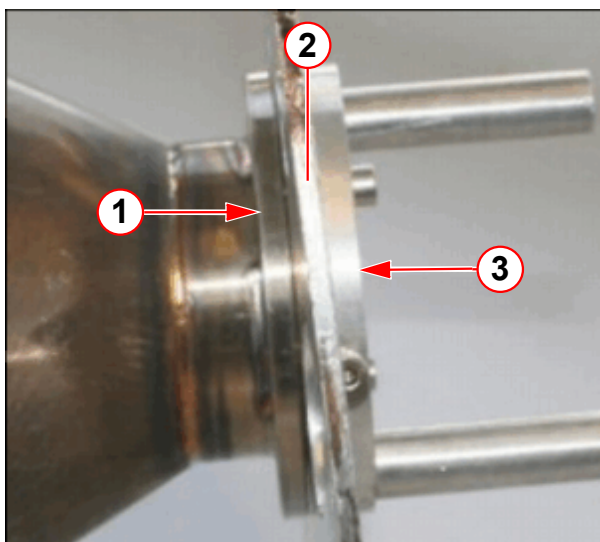
Fig. 129: Fit the nozzle to the tube assembly



- (1) Brass temperature regulator nozzle
- (2) Tube assembly

- Screw the **brass nozzle** (clockwise) to the end of the **tube assembly**.

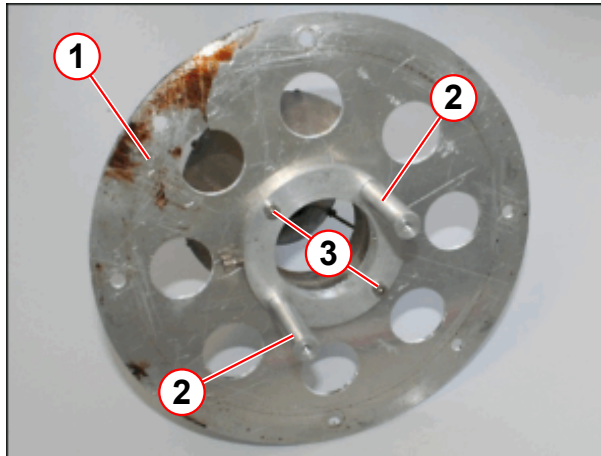
Fig. 130: Fit the hot window



- (1) Tube assembly
- (2) Hot window
- (3) Spacer

- Fit the hot window assembly:
 - place the hot window between the **tube assembly** (pos 1), and the **spacer**.

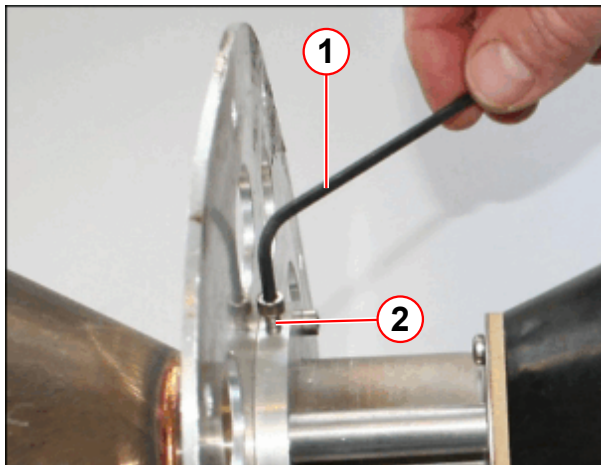
Fig. 131: Hot window assembly



- (1) Hot window
- (2) Spacer
- (3) 3 mm allen screws

- Align the 4 mm holes.

Fig. 132: Secure the hot window assembly



- (1) Allen key
- (2) 3 mm allen screws (2 off)

- Secure the 3 mm screws with an allen key.

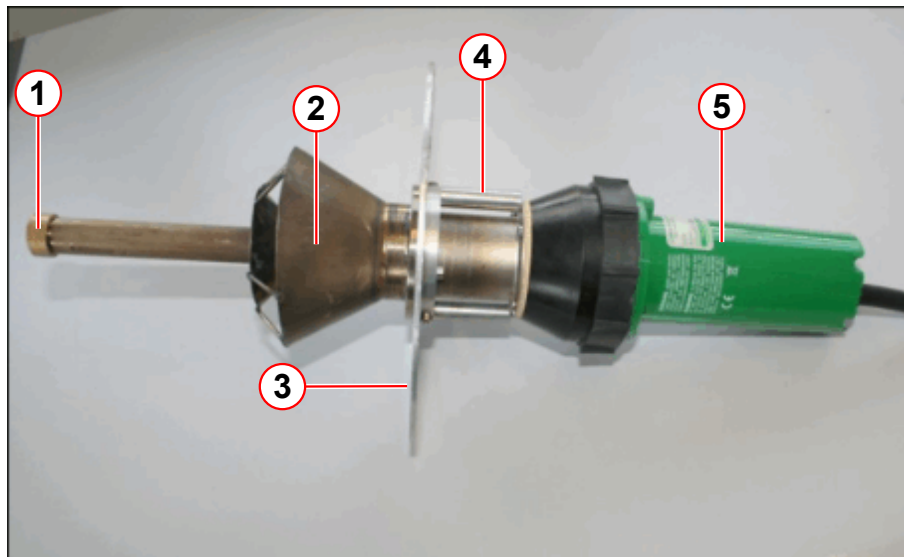
Fig. 133: Fit the hot window assembly to the heat gun



- (1) Heat gun nozzle
- (2) Hot window assembly

- Place the **hot window assembly** over the end of the **heat gun nozzle**.
- **Push down** the hot window assembly until the spacers meet the shoulder of the heat gun.

Fig. 134: Heat gun assembly complete

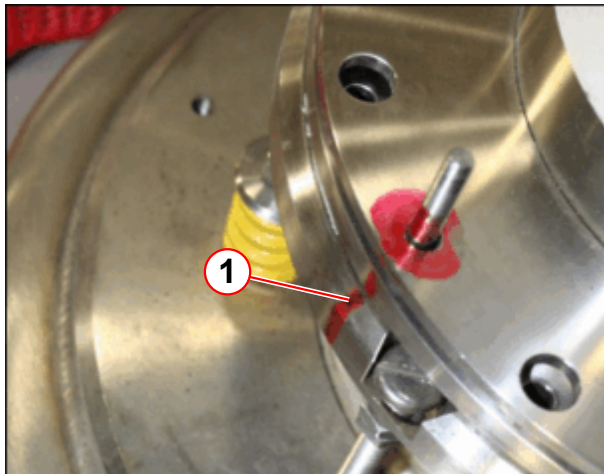


- (1) Brass temperature regulator nozzle
- (2) Tube assembly
- (3) Hot window
- (4) Spacer
- (5) Heat gun

3.3.6 Replacement procedure

- Switch on the **mass spectrometer** to allow the motor to **warm up** to its operating condition.
- Check the **shield temperatures**. Record the values.
- **Turn off** the **compressor and cold head**.
- **Disconnect** the **power cable** from the cold head.
- **Leybold only:** Remove the **iron shield** (part no **10102649**) to allow access to the cold head.

Fig. 135: Unisock flange orientation mark



(1) Orientation mark

- Make a **mark** on the **unisock flange**
- This will ensure **correct orientation** and **alignment** when fitting the new cold head.



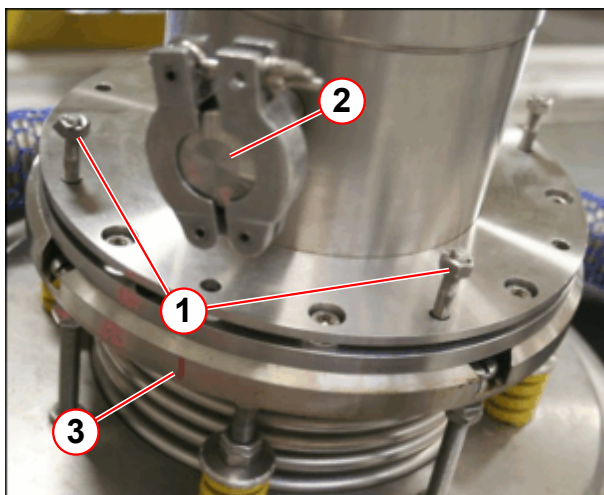
Never lift or support the cold head on the thin walled tube. Damage to the tube could prevent the cold head from working.



To avoid contamination do not touch any of the copper parts or the super-insulation with bare hands. Contamination reduces the efficiency. Clean all surfaces with de-greasing fluid if contaminated.

- **Disconnect the Supply and Return** gas lines from the cold head.

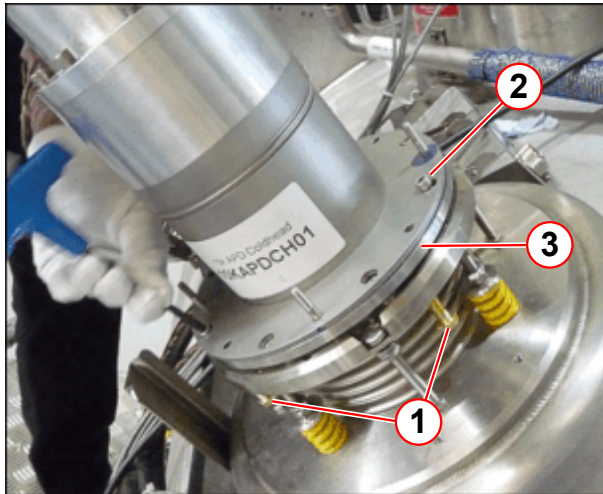
Fig. 136: Cold head fixtures



(1) M6 locking nuts (4 off)
 (2) KF16 blank, clamp and seal
 (3) Orientation mark

- **Remove the 4 x M6 locking nuts** (4 off) from each stud of the unisock.
 - **Grip** the base of the 6 mm stud with a **pair of grips** to stop the stud being screwed out.
- **Remove the KF16 clamp, blank flange and seal** from the cold head pump port (pos 2).

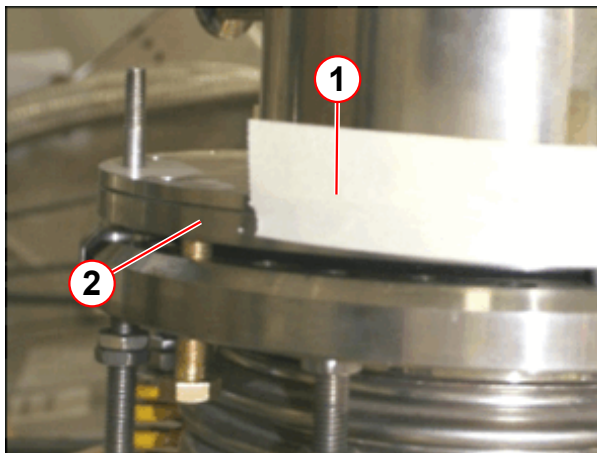
Fig. 137: Disengage the 10k Cold head



- (1) M8 x 40 mm brass release screws (4 off)
- (2) M6 x 8 mm socket screws (8 off)
- (3) Clamp ring

- Insert 4 off **M8 x 40 mm** brass hex head **release screws** to the tapped holes from the **underside** of AV Mount.
 - turn these screws (pos 1), until they make **contact** with the **underside** of the **cold head flange**.

Fig. 138: Secure the clamp ring

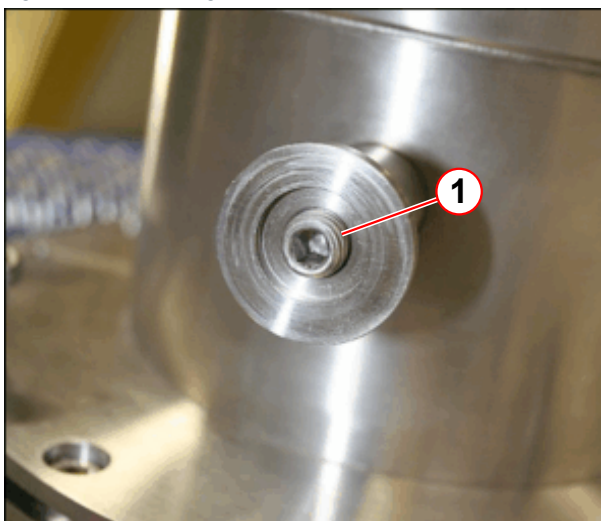


- (1) Masking tape
- (2) Clamp ring

- Secure the **clamp ring**:
 - Use masking **tape** to prevent the **clamp ring** from falling off.

- Using a **5 mm** hexagon (Allen) key, **alternately** remove **4** of the **M6 x 8 mm** socket screws that secure the **clamp ring** to the **unisock**.
- **Check** that the remaining **4 x M6** socket screws can be removed but **leave them in place** holding the O-ring seal.

Fig. 139: Removing the M5 hex socket screw



(1) M5 hex socket screw

- Using a **5 mm** hexagon (Allen) key, undo the vacuum valve **socket screw** inside the **pump port**.
- A sound will now be heard as **air** is **sucked** into the unisock.
- **Wait** until the **pressure** inside the unisock is **equal to atmosphere**, (approximately 2 minutes).

- **Remove** the remaining **4 x M6** socket screws from the **clamp ring**.
- Jack the **cold head** out of the **unisock** until both of the cold head tapered interfaces have **disengaged** from the **copper stages**.

Fig. 140: Removing the cold head from the unisock



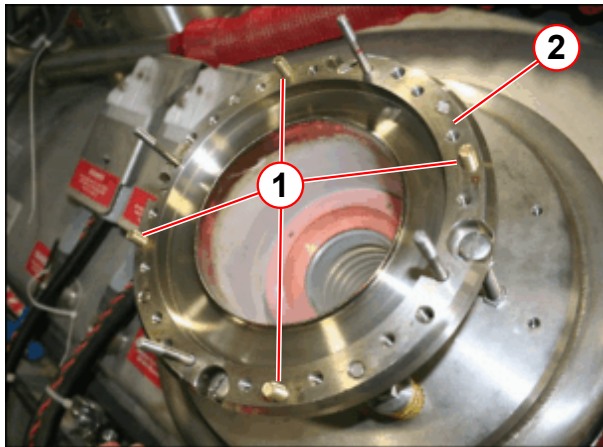
- **Remove** the cold head from the unisock lifting in a **vertical upward action**.



It may be necessary to screw the entire length of the M8 x 40mm release screw into the AV mount to fully disengage the cold head from the unisock.

- **Carefully** place the cold head in a safe place to **warm up**.

Fig. 141: Remove the release screws



- (1) Release screws
- (2) Unisock flange

- Remove the brass **release screws** from the **unisock flange**.
- **Before** fitting the heat gun, wipe the O-ring faces to remove **grease**.

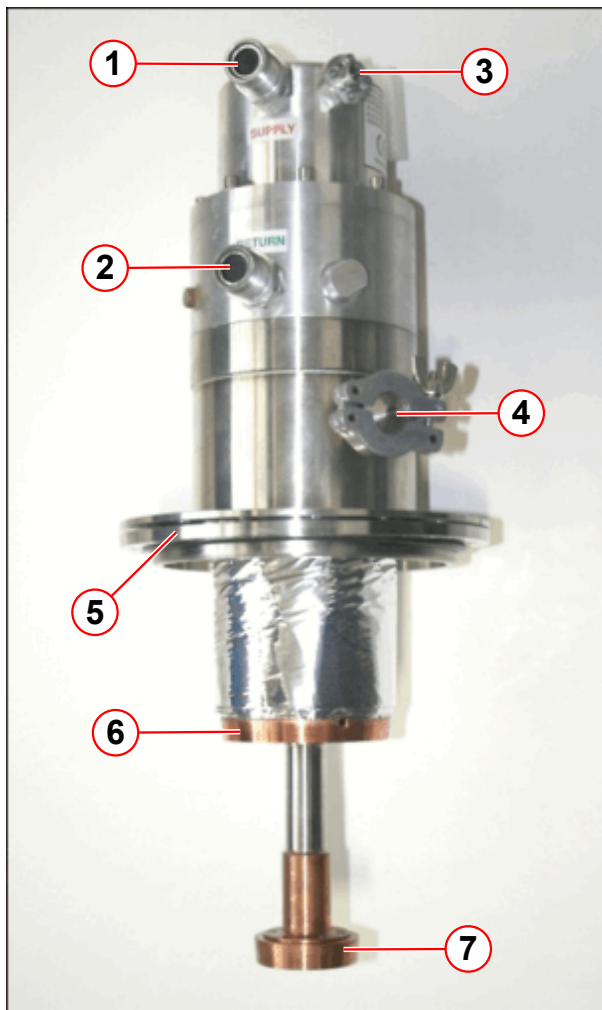
Fig. 142: Heat gun



- **Insert** the **heat gun**, and switch on.
- Set to **position 8**.

3.3.7 Preparing the new cold head

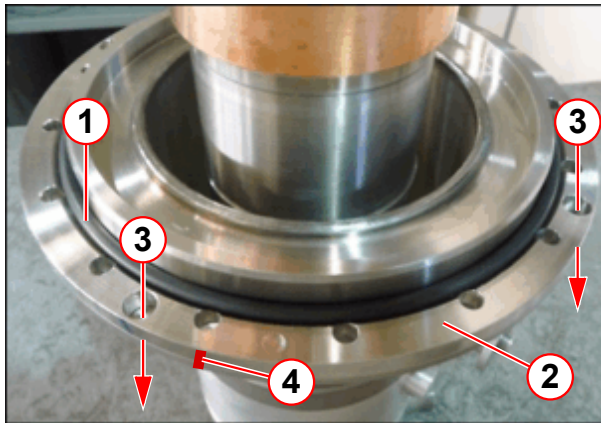
Fig. 143: 10K Cold head



The **preparation** of the new cold head can be completed during the **ramp down** of the magnet.

- (1) Supply Aeroquip he line connection
- (2) Return Aeroquip he line connection
- (3) Electrical power cable connection
- (4) Pump port (KF16 clamp and seal)
- (5) Cold head flange and O-ring seal
- (6) 80K copper stage
- (7) 20K copper stage

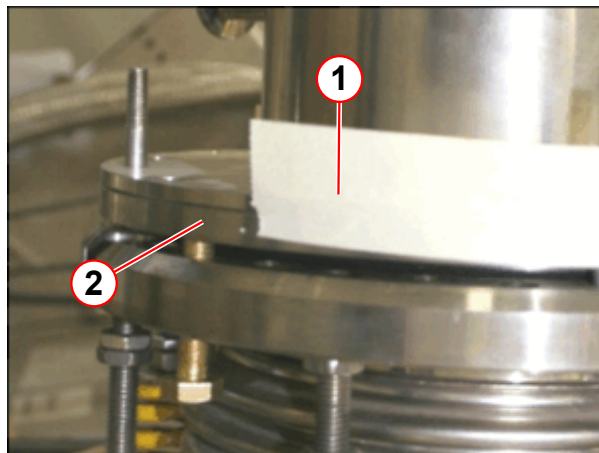
Fig. 144: Fitting the clamp ring and O-ring



- (1) O-ring
- (2) Clamp ring
- (3) M8 release screw holes (4 off)
- (4) Orientation mark

- **Clean** the cold head **clamp ring** and place over the cold head (pos 2).
- **Clean** and lightly **grease** the **O-ring**.
- **Ensure** that the **orientation** of the **compressor lines** is correct during this step **before** fitment.
- Place the **O-ring** into the **chamfer** of the cold head **flange** (pos 1).
- **Ensure** the **M8 release screw** clearance holes are aligned (pos 3).
 - » The **clamp ring** socket screw holes **must not be obstructed** by the cold head **flange**.
 - » The 4 off large holes on the **clamp ring** (pos 3) allow the **M8 release screws** to pass through and impact on the cold head **flange**.

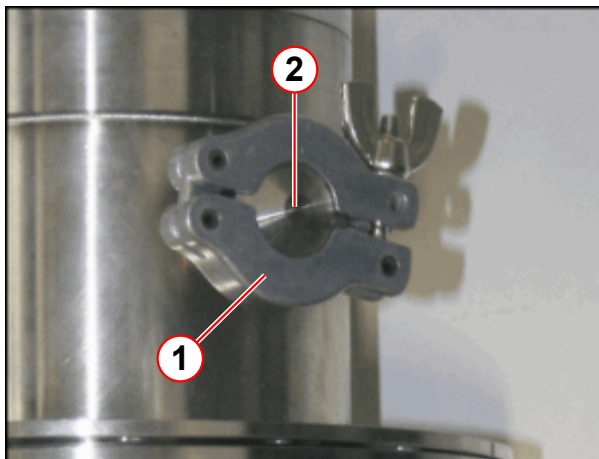
Fig. 145: Secure the clamp ring



- (1) Masking tape
- (2) Clamp ring

- **Secure the clamp ring:**
 - Use masking **tape** to hold the clamp ring in place.

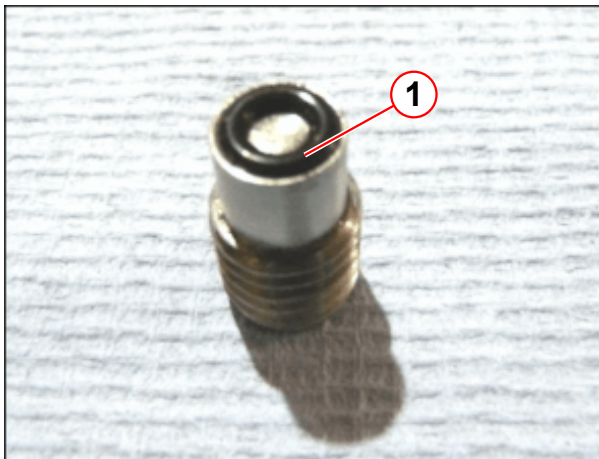
Fig. 146: Remove the KF16 clamp



- (1) KF16 clamp
- (2) Blank and seal

- Remove the **KF16 clamp, blank and seal** from the pump port.

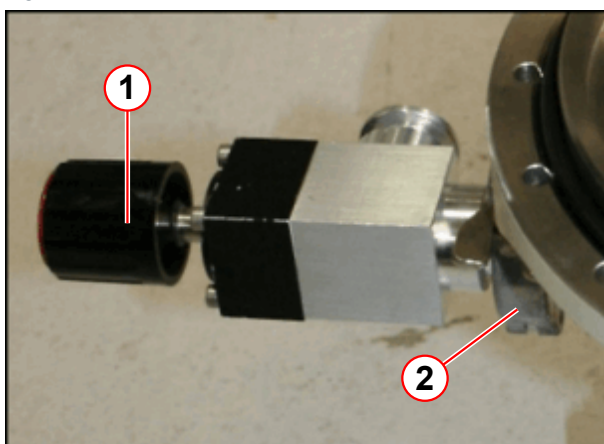
Fig. 147: Vacuum valve screw



- (1) O-ring

- Use a **5 mm allen key** to **remove** the **vacuum valve socket screw** from the **pump port**.
- **Clean** and lightly **grease** the **O-ring** on the end of the **socket screw** and **refit** it into the **pump port**.

Fig. 148: Vacuum lock valve fitted



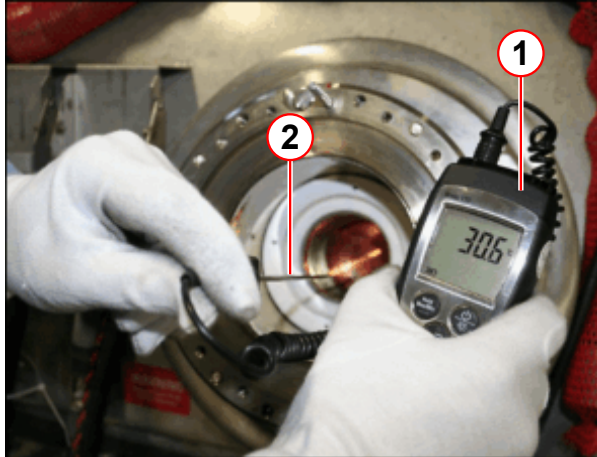
- (1) Valve closed
- (2) Clamp

- Fit the **vacuum lock valve** to the **KF16 port**.
 - locate the 5 mm hexagon shaft into the end of the **vacuum valve screw**.
- **Position** the lock facing **downwards** to enable connection to the mass spectrometer.
- **Do not** open the valve at this point.

- **Cover the cold head** with the **polythene bag** (from the original packing) to **keep clean**, until ready to fit.

3.3.8 Heating the Unisock

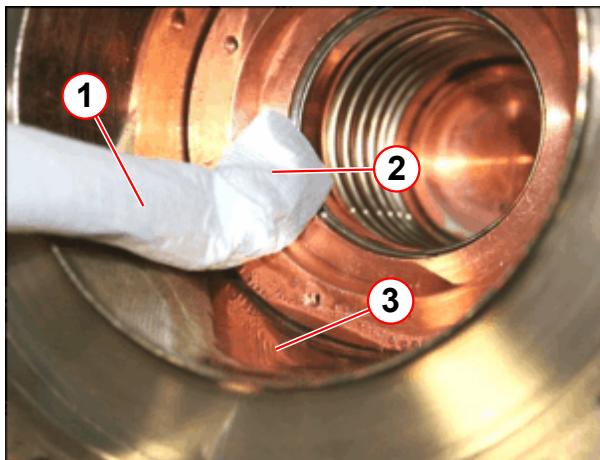
Fig. 149: Checking the the temperature



- (1) Digital thermometer
- (2) Probe

- **Heat the unisock** making regular checks of the **temperature** using the **digital thermometer**.

Fig. 150: Remove any water and moisture



- (1) Tweezers
- (2) Paper towelling
- (3) Water

- When a **temperature** of approximately **25-35C** has been reached:
 - use the **paper towelling** and tweezers to **remove any water and moisture** from the unisock.



Paper towelling will freeze onto the unisock during cleaning. Ensure that no moisture or paper toweling remains before fitment.

Fig. 151: Temperature regulator nozzles



- Change the brass **nozzle** to the smallest diameter.
(→ 1/ Fig. 129 Page 99)
- Ensure that **all moisture** has been **removed** from the unisock and that both copper **stages** are **clean and dry**.

- **Continue** to heat the unisock, regulating the **output temperature** of the heat gun.
- The **temperature** of both the copper stages **must rise** to a **maximum** of **50°C**.
- **Alternate** the heat gun **nozzles** during heating to **regulate the heat** at each stage if necessary.
- **Prepare** the location of the new cold head. Ensure all necessary **tooling is available**.



It is important NOT to heat the unisock above 50°C. Shield temperatures may not recover to their optimal temperature range following the cold head exchange.



Warming up takes 60 - 120 minutes.

- When the **temperature** reaches **50°C** repeat the **unisock measurements**.
 - **record** measurements on the **10K Cold Head Exchange Report Form**.
- **Check** the **unisock condition** and that the **temperature is heated to 50°C**.

3.3.9 Fitting the new cold head

Fig. 152: Heat gun

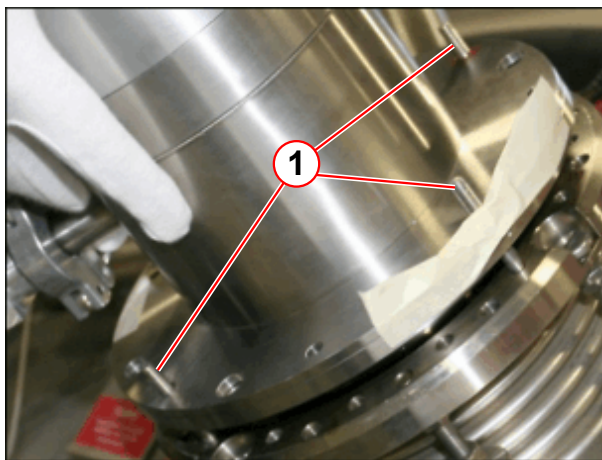


- Set the heat gun to zero.
- Switch off the heat gun, remove, and place in a safe place to cool down.



The next step must be completed within 2 minutes of removing the heat gun.

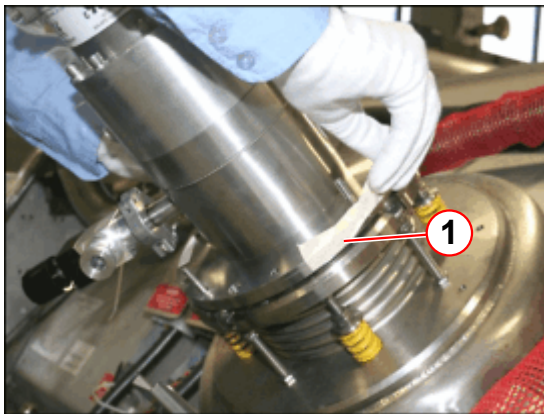
Fig. 153: Locate the cold head over the studs



(1) M6 studs (4 off)

- Ensure fitment is in the **same orientation** as the original cold head.
- Lift the cold head by the **flange plate**, holding the **O-ring** in position with your fingers.
- Insert the new cold head **quickly** into the **uniso**ck.
 - Locate over the **4 x M6 studs**.
- **Avoid damage** to the **O-ring**.
 - **Do not** let the cold head **drop** into the **uniso**ck.
 - **Remove fingers** at the last moment to **avoid trapping**.
- With the cold head in place, **clamp the seal ring** to the **Uniso**ck using 4 of the **8 x M6 Allen head screws**.

Fig. 154: Remove the tape from the flange

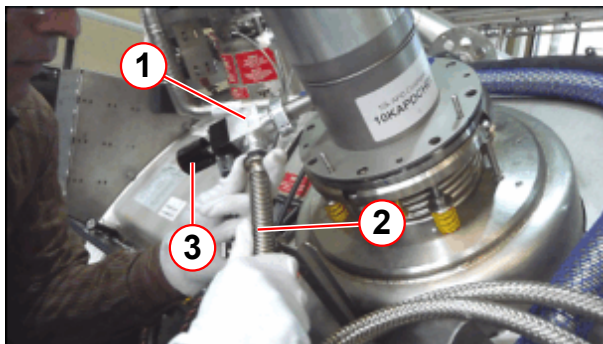


(1) Tape

- Remove the **tape** from the flange.
- With the cold head in place:
 - **clamp** the **seal ring** to the Uni-sock using 4 of the **8 x M6 screws**.
 - tighten the screws in a **diametric pattern**.
 - Torque to **10Nm**.

3.3.10 Vacuum pump and leak check

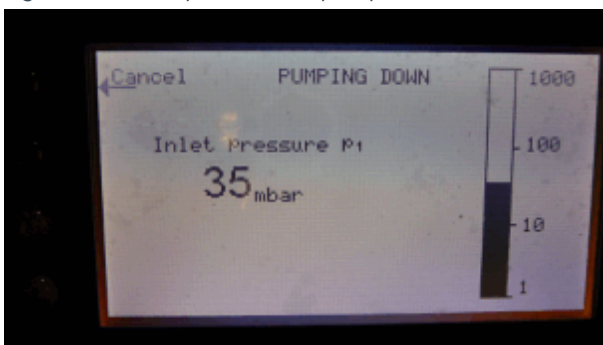
Fig. 155: Connect the mass spectrometer



(1) Vacuum lock valve
 (2) Mass spectrometer flex line
 (3) Handle

- **Secure** the flexible line from the **mass spectrometer**, to the **vacuum lock valve**.
- **Start** the **mass spectrometer** to pump down the **flex line**.

Fig. 156: Mass spectrometer pump start



- **Pump down pressure** on the mass spectrometer.
 - Wait until the **vacuum** reaches **1×10^{-3} mbar**.

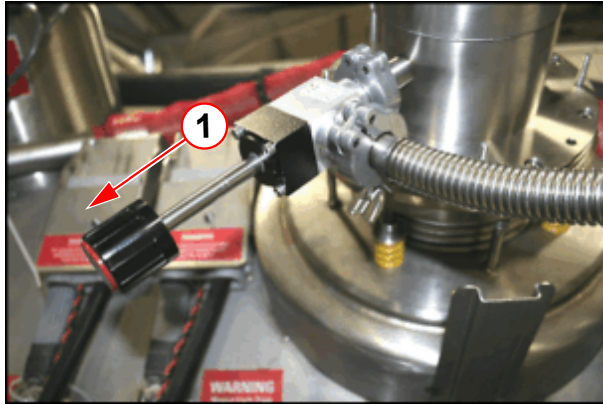
- **Slowly** turn the vacuum lock **valve handle** to **disengage** the socket screw **seal** from the pumping **port**.



Do not turn the handle quickly. it may cause the mass spectrometer to fail due to a large ingress of air into the pump.

If this occurs stop and restart the mass spectrometer.

Fig. 157: Pull the shaft into the valve body



(1) Handle

- When the **screw** feels **free**:
 - draw the **handle** back to **pull the shaft up into the valve body** until resistance is felt (pos 1).
 - slowly **rotate the handle** until it draws back further, then continue to **rotate** a further **half turn**.
- The **handle** is now **locked** in the **open** position.

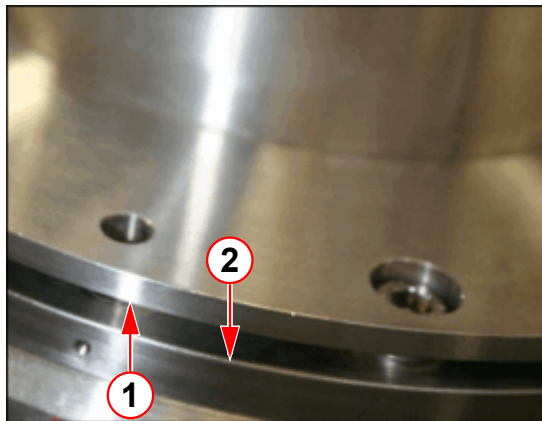
- **Re-fit** the remaining **4 x M6 screws**.
- Tighten the screws in a **diametric pattern**, until the **clamp ring** is secure and **flush** with the **unisock flange**. **Torque to 10Nm**.
- **Wait** for the **vacuum** to reach the **1 x 10⁻³ range mbar**.



The pressure differential created across the cold head as a result of evacuating the unisock causes the cold head to be drawn onto the unisock by about 3 mm.

It is possible that the cold head can come to rest very slightly out of line to the unisock.

Fig. 158: Measuring the gap - nominal 3 - 5 mm



(1) Underside of the cold head flange
(2) Top face of the clamp ring

- **Measure** the **gap** between:
 - **underside** of the **cold head flange** (pos 1).
 - **top** face of the **clamp ring** (pos 2).
- The nominal distance of **3-5 mm** must be **consistent all the way round**.

- **Refit** the **4 x M6 locking nuts** onto the unisock studs.

- Carry out an **internal** cold head vacuum leak check by **blowing helium gas** around the external **seals** and **components**
 - All the **cold head seals** and **connections**.
 - The **O-ring joint** between the **clamp ring** and **cold head flange**.
 - The **vacuum pipe connections** from the mass spectrometer.



There must be no increase in helium background above a leak rate in the range of 1×10^{-6} mbar litre/second.

If an increase is observed, it will be necessary to remove the cold head and repeat the entire procedure.

- **Continue** to pump down the vacuum **for at least 60 minutes** after the leak check has been successfully completed.
- **During** the 60 minute pumping procedure you can complete the following steps:
 - **Clean** the Aeroquip **couplings**.
 - **Check** that the **gasket seals** are **clean** and fitted correctly.
 - **Reconnect** the supply and return **cold head lines** to the **cold head**. Check the **seals**.
 - **Note** the **static pressure** of the **compressor**. **Top up** the gas pressure if required.
 - **Reconnect** the cold head **power cable**.
 - (**Leybold only**) **Refit** the **iron shield** around the cold head using the original fittings.
- At the **end of the 60 mins**, ensure the **vacuum** is **less than 1×10^{-3} mbar** and the helium **background reading** is less than 1×10^{-6} mbar litre/sec.
- **CLOSE** the vacuum lock **valve**.
- **Turn** the **handle** of the vacuum lock **valve** to overcome the **valve locking device**.
- **Push** the **handle inwards** until it stops.
- **Turn** the handle **clockwise** to **relocate the socket screw** into the cold head pumping port.
- **Continue** to turn **clockwise** until the socket screw seal is **fully inserted** and **tight**.
- **Pull back** on the valve **handle** to **release** it from the socket screw.
- **Switch off** the vacuum pump and remove the **KF16 valve** from the cold head pumping port.
- **Refit** the **KF16 blank**, **O-ring** and **clamp** to the cold head pumping port.
- **Refit** the KF-16 pumping port **seal** and **blank**.
- **Start** the **cold head**:
 - Switch on the **compressor** and perform an **external leak check** of the **cold head** and **compressor lines**.

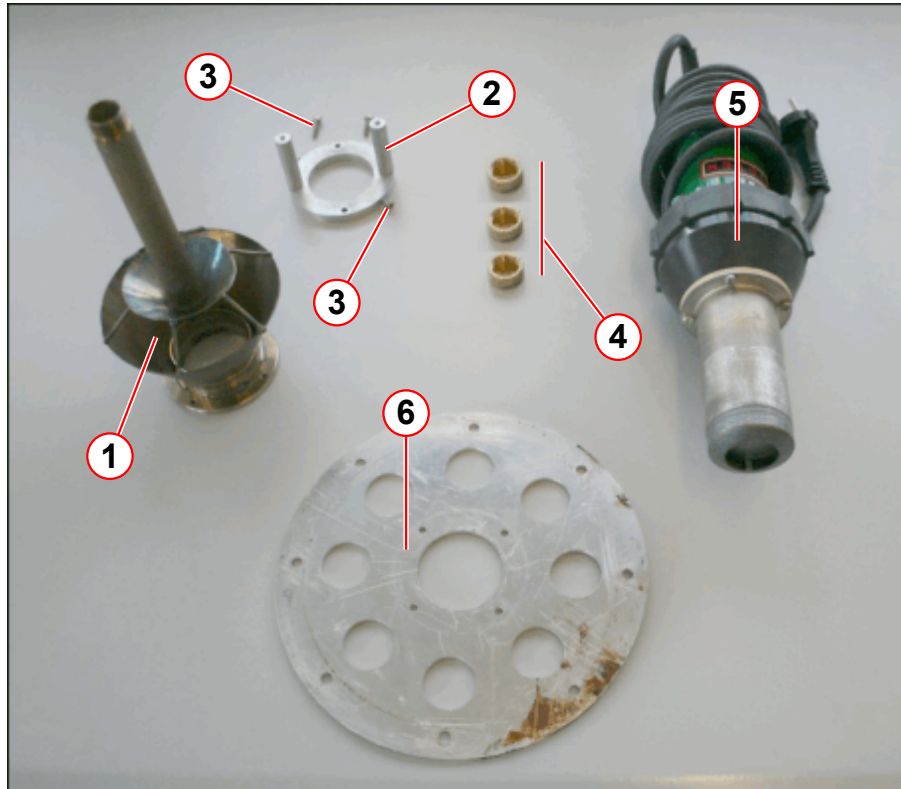
3.3.10.1 Final tasks

- **Clear** all **tools and test equipment** from the magnet area.
- **Re-energise** the magnet following the **ramping procedure**.

- **Pack** the original cold head using the box and packing material from the replacement cold head.
- **Return** the cold head.

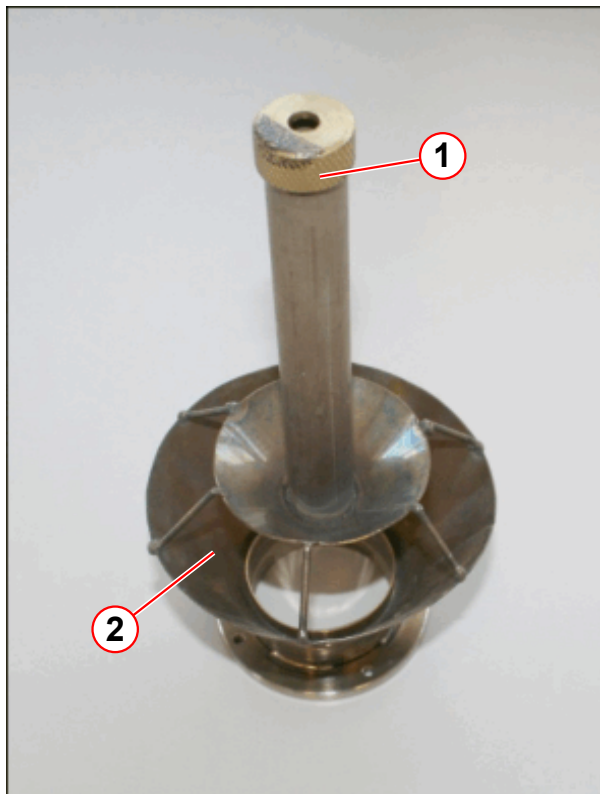
3.3.11 Assemble the heat gun and hot window assembly

Fig. 159: Heat gun and hot window assembly



- (1) Tube assembly
- (2) Spacer
- (3) 3mm Allen screws (4 off)
- (4) Brass temperature regulator nozzles (3 off)
- (5) Heat gun
- (6) Hot window

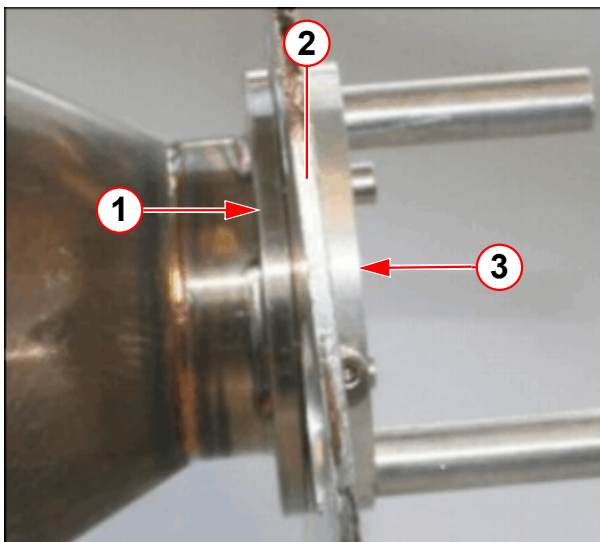
Fig. 160: Fit the nozzle to the tube assembly



- (1) Brass temperature regulator nozzle
(2) Tube assembly

- Screw the **brass nozzle** (clockwise) to the end of the **tube assembly**.

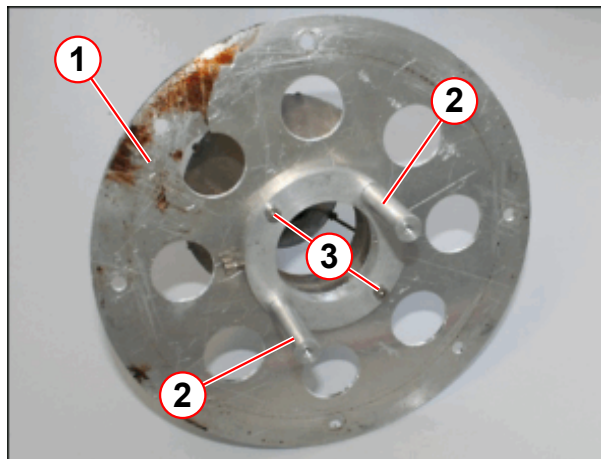
Fig. 161: Fit the hot window



- (1) Tube assembly
(2) Hot window
(3) Spacer

- Fit the hot window assembly:
 - place the hot window between the **tube assembly** (pos 1), and the **spacer**.

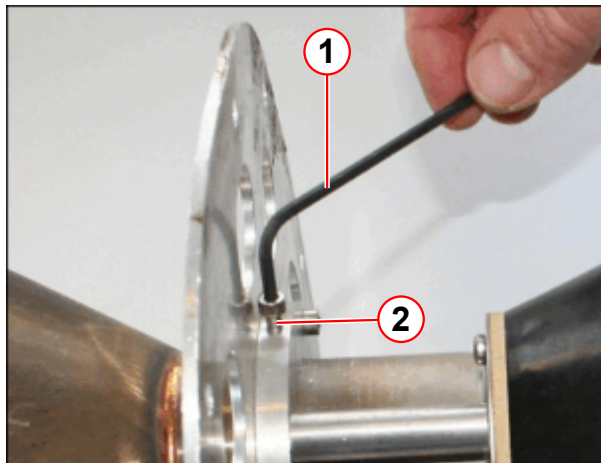
Fig. 162: Hot window assembly



- (1) Hot window
- (2) Spacer
- (3) 3 mm allen screws

- **Align the 4 mm holes.**

Fig. 163: Secure the hot window assembly



- (1) Allen key
- (2) 3 mm allen screws (2 off)

- **Secure the 3 mm screws with an allen key.**

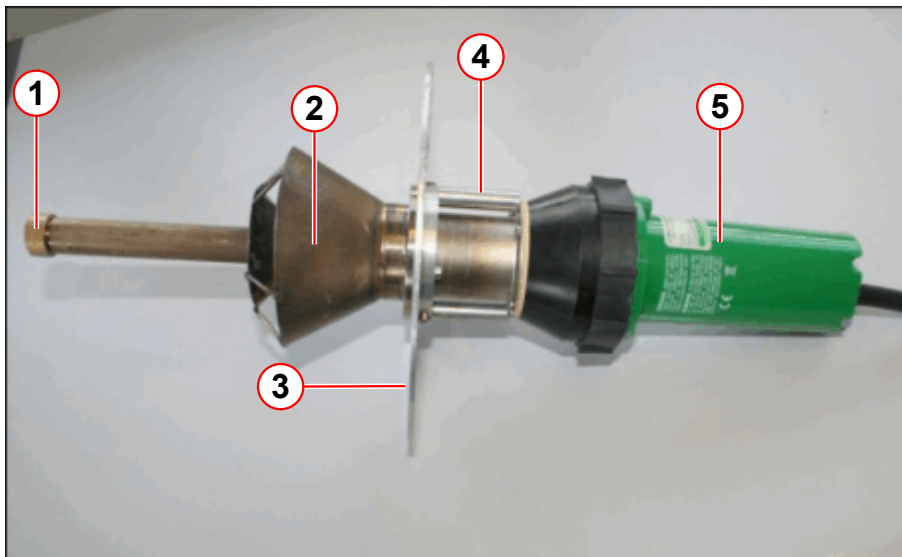
Fig. 164: Fit the hot window assembly to the heat gun



- (1) Heat gun nozzle
- (2) Hot window assembly

- Place the **hot window assembly** over the end of the **heat gun nozzle**.
- **Push down** the hot window assembly until the spacers meet the shoulder of the heat gun.

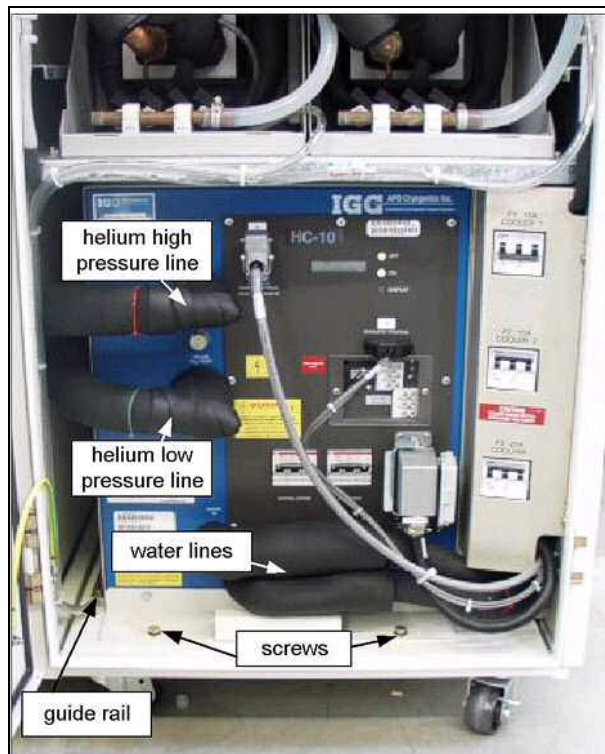
Fig. 165: Heat gun assembly complete



- (1) Brass temperature regulator nozzle
- (2) Tube assembly
- (3) Hot window
- (4) Spacer
- (5) Heat gun

3.4 Replacement of the Compressor (APD)

Fig. 166: APD compressor



3.4.1 Removing the APD compressor

Step	Instructions	Required Tools
1	Switch off compressor Switch off F3	
2	Unplug the power cable Unplug the shield collar cable Unplug the Diagnostic cable	
3	Disconnect the high and low pressure helium flexlines Using two wrenches	Wrench 1 3/16" and 1"
4	Unscrew compressor from the base plate Removing the compressor can be facilitated by removing the guide rail screw first	
5	Carefully pull out the compressor. The compressor should be pulled onto a sturdy support (e.g., wood, pallet) of approx. 8-10 cm.	

Fig. 167: APD compressor



6	Pull the compressor on a pallet	
7	Close the ball valve of primary water, supply and return	Wrench
8	Disconnect the water lines from the compressor	



The weight of the compressor is approximately 95 kg

Fig. 168: Water lines



Fig. 169: Water lines



Drain the water from the pipes before returning the replaced unit

3.4.2 Installing the new compressor

1	Connect the water lines	
---	-------------------------	--

2	Slide compressor into the cabinet	Wrench
3	Tighten the compressor on the base plate	
4	Connect the high and low helium flexlines 1. High-pressure (red) 2. Low-pressure (green)	
5	Connect the electrical cable	
6	Check the helium pressure on the pressure gauge Operation (compressor off) at: 50 Hz: 15.6 - 15.9 bar (226 - 230 psig) 60 Hz: 14.9 - 15.2 bar (216 - 220 psig)	
7	Open the ball valve of the water supply	



To run the compressor with the auto restart module, the compressor must be started from the remote switch.

DO NOT switch the compressor ON using the on/off button on the front panel of the compressor

8	Check if remote switch is in OFF position Switch fuse F3 ON Close MAIN POWER AND CONTROL SYSTEM circuit breaker on the compressor CHECK that the display is ON but DO NOT switch the compressor ON using the on/off button on the front panel of the compressor	
9	Switch the compressor ON using the Remote on/off switch Typical operating pressure (compressor on): 50/60 Hz: 20.0 - 22.8 bar (290 - 330 psig)	

3.5 Replacing the F70 Compressor with Tooling kit

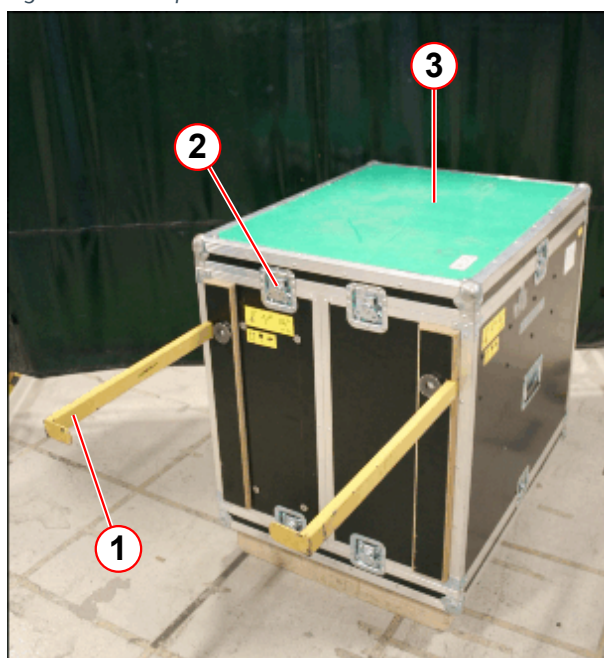
3.5.1 General

This procedure describes the replacement of the compressor unit when using the replacement tooling and transport kit (material number **11060700**)

The replacement compressor unit is shipped in a specially designed wooden transport box. The transport box contains the tooling and accessories required for a safe replacement of the compressor unit:

- It allows for the safe transport and replacement of the compressor unit
- The transport box allows 1 person to pack a damaged compressor unit.
- The shipping crate is suitable for worldwide transport and complies to shipping regulations IEC 60721-3-2.

Fig. 170: Transport box



- (1) Lifting handles (x 2)
- (2) top lid latch (x 6)
- (3) Lid with PTFE sheet covering

The transport box is fitted with 2 x lifting handles to allow movement. Built-in wheels allow movement over a short distance without requiring auxiliary lifting tools.



If necessary, the transport box can also be moved using a pallet truck.

3.5.1.1 Prerequisites

The document refers to compressor units identified by material no:

- 10121435 - Sumitomo F70
- 07461119 - compressor unit M/N HC-10
- 07758191 - SHI Compact compressor unit CSW71TW

- 07014827 - Robust Coolpack
- 04296518 - RCA - compressor unit Coolpak 6000 (MKI)
- 07120913 - Helium compressor unit. Coolpack. MKII

3.5.2 Safety Information

Carefully read the Operator Manual supplied in the transport box prior to using the tooling for the first time!

- The Operator Manual contains additional Safety Information. As well as general safety, transport, installation, operation, and maintenance.
- Also observe the special Safety Information and Notes included in the chapters!

NOTICE

Tripping hazard

Tripping hazard caused by edges, thresholds, cables and lines when moving the compressor unit on the casters.

- » Check for hazards before moving the compressor unit.



CAUTION

Unsafe lifting and movement of the tooling

Personal injury or equipment damage if safety procedures are not followed.

- » It is not permitted to lift or lower loads on inclines!
- » Do not park the shipping crate or tooling on inclines.
- » Only lift compressor units that the tooling is designed for.



CAUTION

Personal injury

When performing the replacement procedure personal injury can occur.

- » Safety shoes must be worn.
- » Wear protective gloves and suitable clothing when handling the equipment.
- » The operator must ensure that non-authorized personnel do not use the tooling.

**CAUTION**

High-pressure helium gas.

Follow the Safety Information provided to avoid injury.

- » Protective clothing, gloves, and eye protection must be worn.

NOTICE

Risk of injury and damage to equipment

The weight of the compressor unit is 100 kg. (220 lbs)

- » Use the supplied lifting equipment provided in the transport box.
- » Assemble the lifting equipment correctly to the instructions provided.

3.5.3 Completion Time and Number of Persons

Tab. 8 Completion time

Work steps	Time	Persons
Preparation	n.a.	1
Replacement	90 min	1
Startup/Tune-up/QA	n.a.	1
Final steps	30 min	1

3.5.4 Tools

- Transportation lifting kit (material number **11060700**, F70 compressor kit).
- 14-mm and 17-mm open-ended wrench.
- 17-mm socket.
- Torque wrench (0–20 Nm).
- 8-mm Allen (Hex) key.

- 7-mm socket and ratchet.

Fig. 171: Edwards Leak detector



- Use the Edwards hand-held leak detector (material no **10614850**).

The Edwards is the **recommended equipment** for leaks **greater than 1×10^{-4} mbar l/s**.

NOTE: The hand-held device is calibrated in **ml/sec**:-

1×10^{-4} mbar l/s = 1×10^{-4} ml/sec at normal pressure and temperature.

3.5.5 Disconnecting and Removing the Compressor Unit



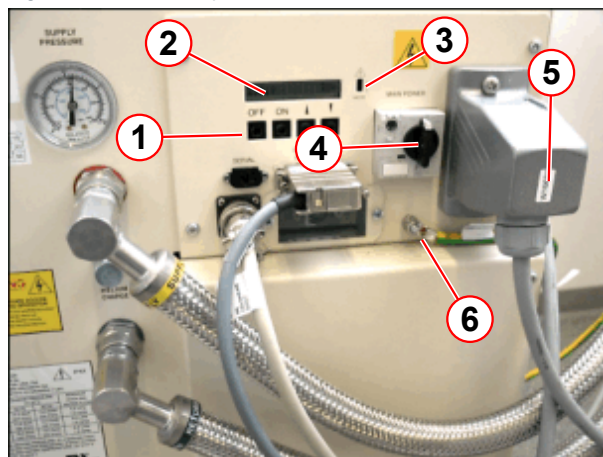
CAUTION

High-pressure helium gas.

Failure to observe the following Safety Information may result in injury.

- » Protective clothing, gloves, and eye protection must be worn.

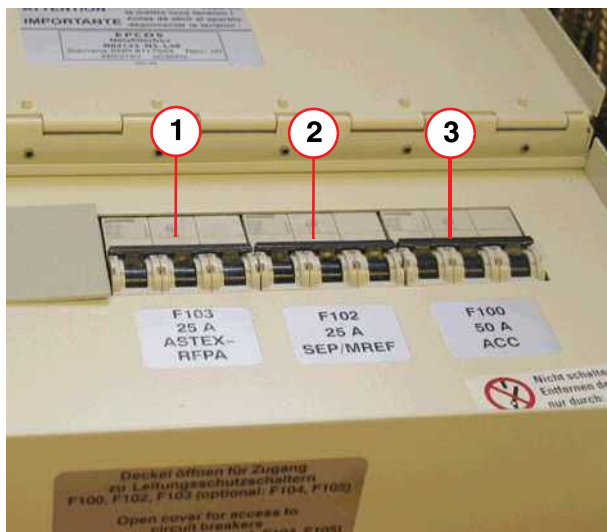
Fig. 172: F70 Compressor control buttons



- (1) Control buttons
- (2) LCD display
- (3) Configuration selector switch
- (4) Main power switch
- (5) Mains supply cable
- (6) Ground wire

- Turn **off** the compressor unit at the **main power switch** (pos 4).
 - The **ON/OFF** buttons are disabled in **mode 2**.
 - Switch off F3

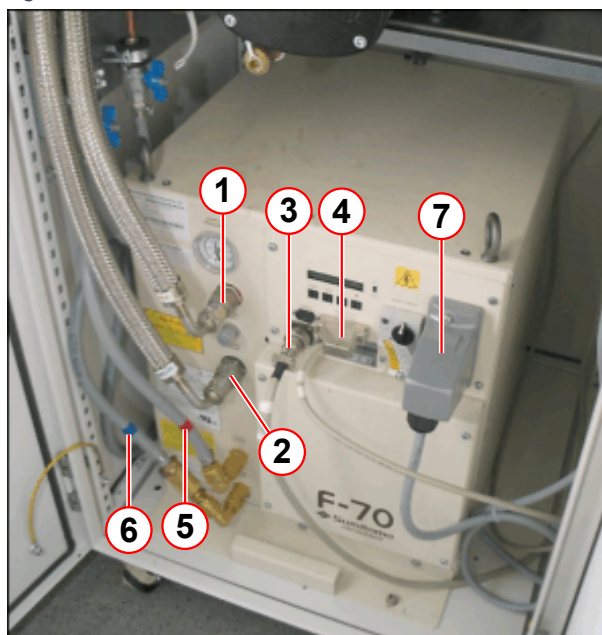
Fig. 173: ACC cabinet - Circuit breakers F100, F102, F103



- (1) F103 - isolates RFPA
- (2) F102 - isolates SEP or IFP (stand-alone compressor)
- (3) F100 - isolates ACC cabinet

- The **alarm box** displays a warning LED, and an audible alarm.
- Press the **AUDIO ALARM OFF** button to cancel the audible alarm.

Fig. 174: F70 installation



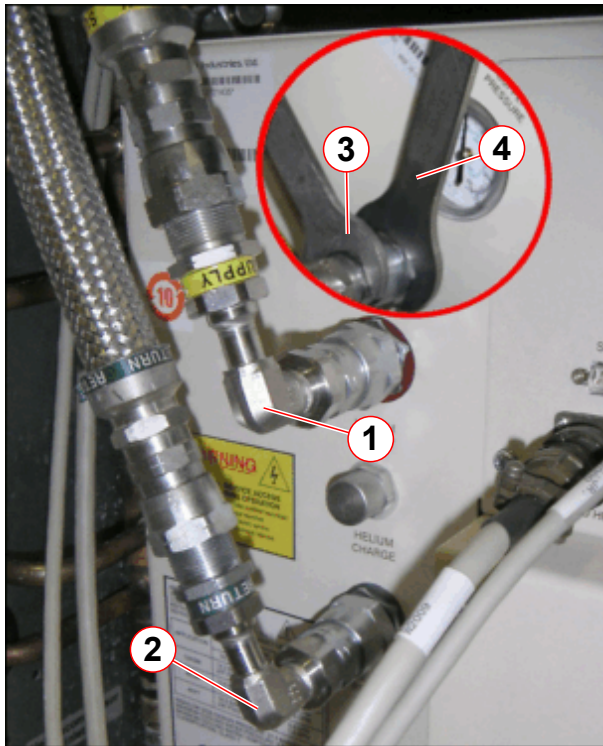
- (1) Helium supply (high pressure)
- (2) Helium return (low pressure)
- (3) Cold head cable (to filter plate)
- (4) Diagnostic interface
- (5) Water line (inlet)
- (6) Water line (outlet)
- (7) Mains power connections

ACC cabinet

- Isolate power to the compressor:
 - Switch **OFF** circuit breaker F102 (2).

- The base of the compressor unit is fitted with 2 nylon slide rails (pre-fitted) to ease movement.

Fig. 175: F70 He compressor connections

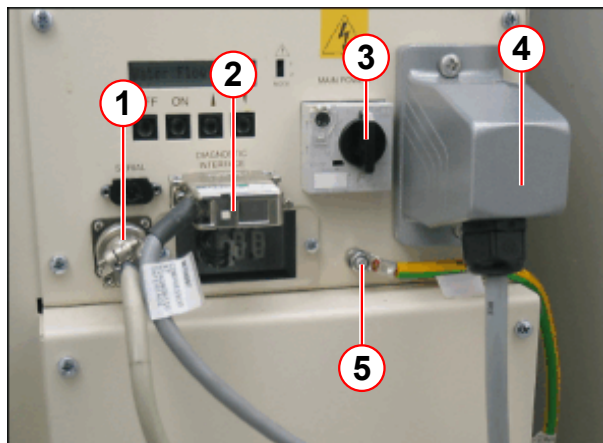


- (1) SUPPLY (high pressure) 90-degree elbow
- (2) RETURN (low pressure) 90-degree elbow
- (3) Hold nut (1" spanner/wrench)
- (4) Turn nut (1 3/16" open end crowfoot spanner/wrench)

Each helium pressure line is connected to the compressor unit via a **90 degree elbow**.

- **Remove** the self-sealing helium pressure lines in the following order:
 - **Return** (low pressure).
 - **Supply** (high pressure).
- **Hold** nut (3) and **turn** nut (4).

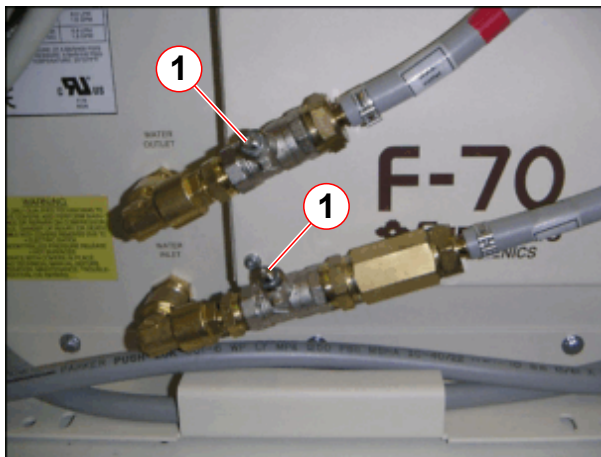
Fig. 176: F70 Electrical cable connections



- (1) Cold head cable
- (2) MSUP DB25 Diagnostic interface cable
- (3) Main power switch
- (4) Mains supply cable
- (5) Ground wire

- Disconnect the **mains voltage**, **cold head motor**, and **MSUP** (DB25 diagnostic interface) cable.
- Disconnect the **ground wire** (5).

Fig. 177: F70 Compressor water inlet/outlet shutoff valves



(1) Shutoff valve

- Close the **shut-off valves** on the **inlet** and **outlet** water lines.
- Disconnect the **water lines** from the compressor.



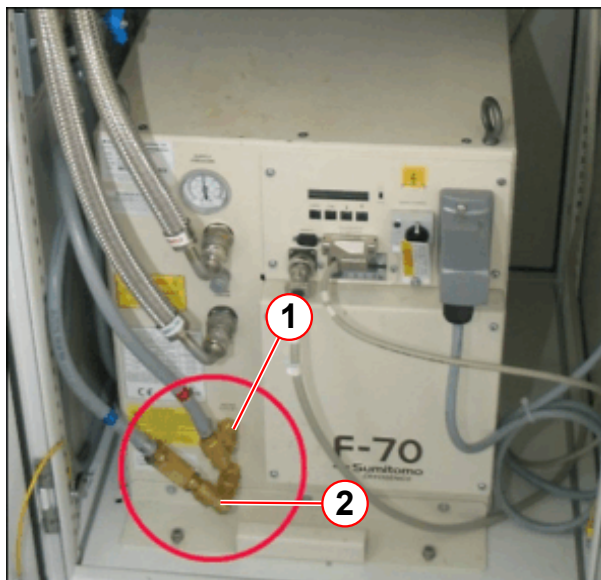
CAUTION

Chemical hazard

If the chilled water contains ethylene glycol, it must be collected in a suitable container while draining.

- » Return the collected water back to the water circuit after the new unit is installed.
- » If necessary, dispose of in line with local regulations for hazardous material.

Fig. 178: F70 connect the water lines



(1) Water line (inlet)
(2) Water line (outlet)

- Disconnect the **water pipes** from the compressor unit.



There is the potential for irreversible damage to the unit if the internal water pipes freeze during shipping.

Before returning a faulty compressor unit, use a gas line on either the inlet and outlet pipes to expel any water.

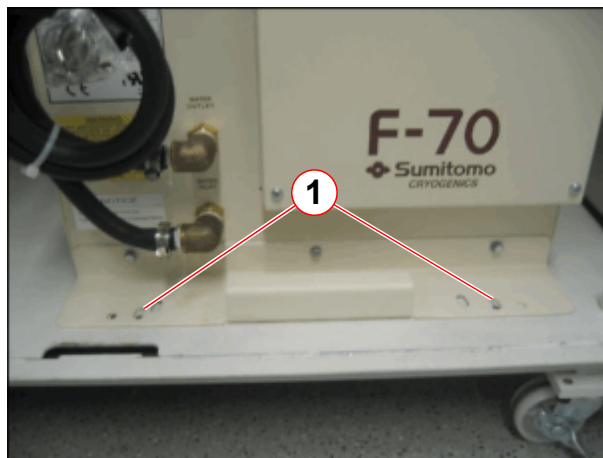
Fig. 179: F70 compressor - draining cooling water notice



Always **drain the coolant** from the cooling circuit when the compressor unit is shipped (or stored).

Refer to the disposal instructions for further information.

Fig. 180: F70 Compressor fixing plate



(1) M8 Hex hd screws (2 off)

RCA cabinet

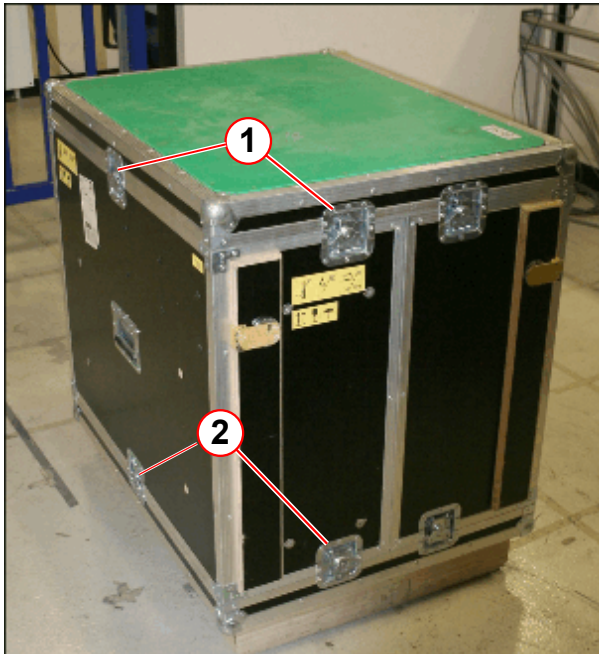
- **Remove** the 2 x M8 hex head screws securing the **fixing plate** to the cabinet.
- Leave the compressor unit in the cabinet until the replacement compressor unit is ready to install.

3.5.6 Unpacking and Pre-installation Checks

Before installation, check the compressor transport box for any transport damage or mishandling.

- Check the shock indicators when the transport box covers are removed.

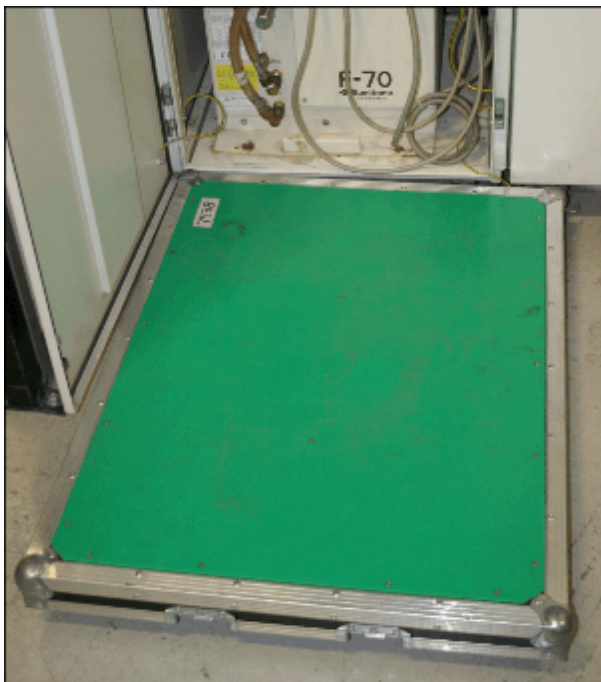
Fig. 181: Transport box



- (1) Top retaining latch x 6
- (2) Bottom retaining latch x 6

- Unclip the 6 x latches securing the top cover.
- Lift the top cover away from the transport box.

Fig. 182: Cover lid of transport crate



- The transport box cover has x 4 casters fitted to the underside. The cover lid is used as a support ramp when removing the defective F70 compressor unit.
 - » The upper side includes a PTFE plate with gliding ability.

Fig. 183: Transport lid castors



When not being used stand the transport lid upright in a safe area against a wall.

- » Do not place the cover on the floor! Danger of rolling away.
- » Do not stand on the cover!



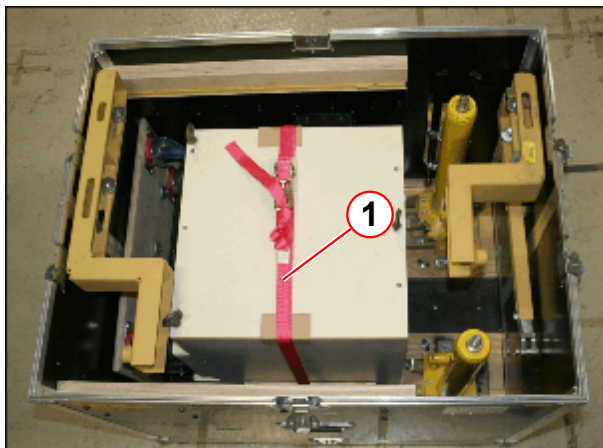
CAUTION

Storing the cover:

Risk of accident! Do not step on the cover!

- » When not in use, place cover upright in a safe area.

Fig. 184: Top view of compressor kit

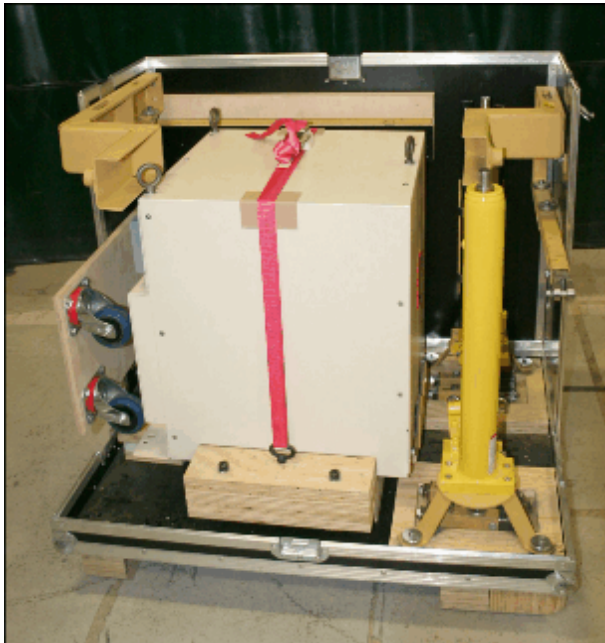


(1) Securing strap

Top view of the transport box.

- Check that the compressor unit has remained strapped securely to the transport base.

Fig. 185: Right side panel removed



All tooling components remain attached in position inside the transport box when removing the right side cover.

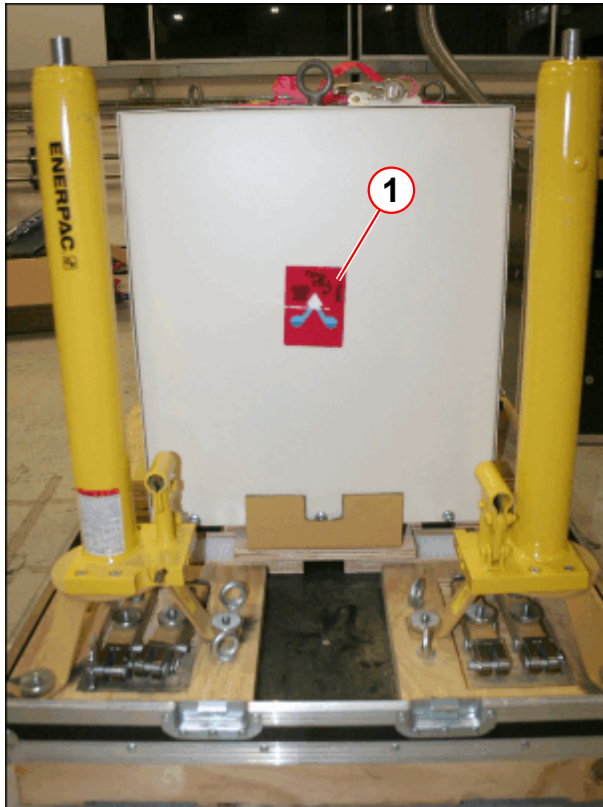
- Unclip the 3 latches and remove the right side cover of the transport box.
- Place the cover in a safe location to the side.



Do NOT tip the compressor unit more than 10 degrees from horizontal to avoid contamination.

DO NOT connect gas lines to the compressor unit if Tip-N-Tell indicator shows that mishandling has occurred.

Fig. 186: Crate sides removed



(1) Shock indicator

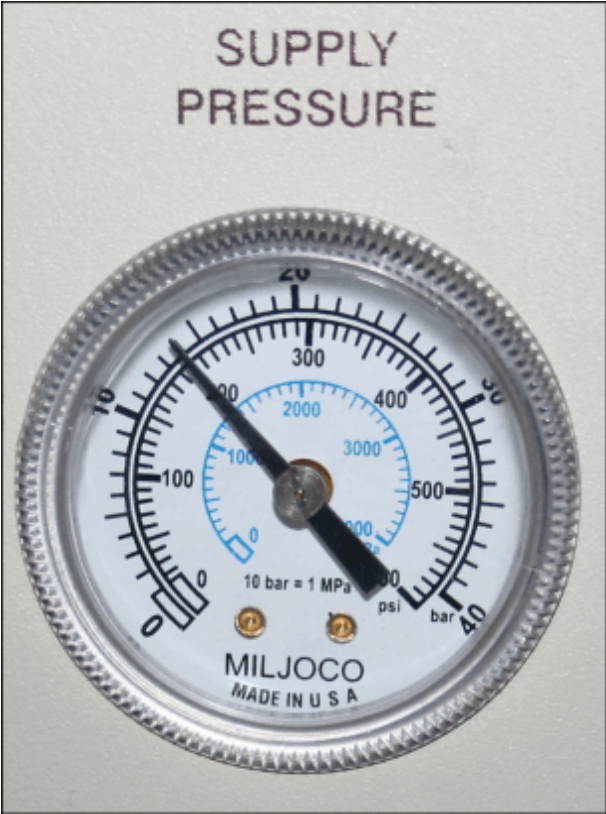
- Check the Tip-N-Tell indicator on the compressor unit.
- The sensor is blue (above the line) when the unit has been tipped or mishandled during transport.

Fig. 187: F70 Compressor Tip-N-Tell indicators



- If the sensor has been activated (the arrow point is blue), and the pressure is within specifications, allow the compressor unit to stand upright for **2 hours**. Refer to (→ Testing the Compressor Unit after Mishandling / Page 155)
- If the sensor has **not** been activated, continue with the replacement procedure.

Fig. 188: F70 Static pressure reading



- Check that the **static pressure** is **positive**.
 - If the gauge indicates **zero pressure** the compressor unit **cannot** be used.
 - Arrange a replacement compressor unit.

Tab. 9 F70 static pressure on arrival

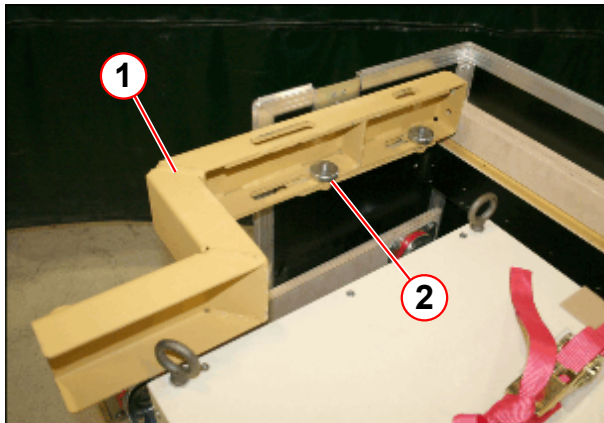
Pressure	Ambient temperature
207 - 212 psi (G)	20°C (68°F)
1430 - 1460 kPa	
13.5 - 14.0 bar	

Continue with the replacement procedure if:

- The sensor has **not** been activated
- The static compressor pressure is positive.

Remove the tooling components from the transport kit and assemble.

Fig. 189: Long Cross beam



- (1) Cross beam
(2) Knurled head fastener

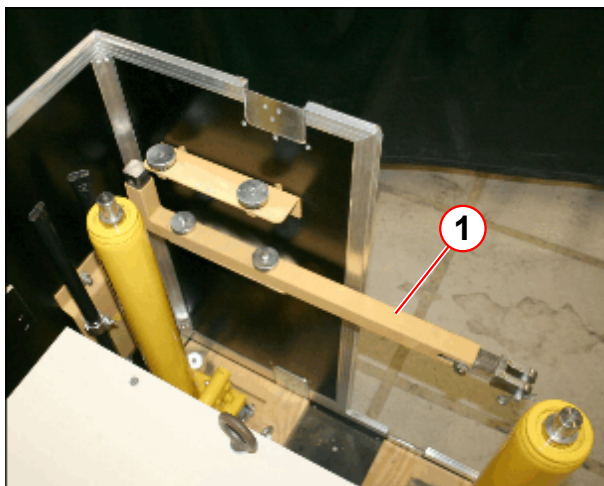
- Unscrew the 2 x knurled head fasteners securing the long 'cross beam'.
- Refit the fasteners to the case.

Fig. 190: Short cross beam



- Unscrew the 2 knurled head fasteners and remove the short cross beam.
- Refit the fasteners to the case.

Fig. 191: Vertical lifting beam

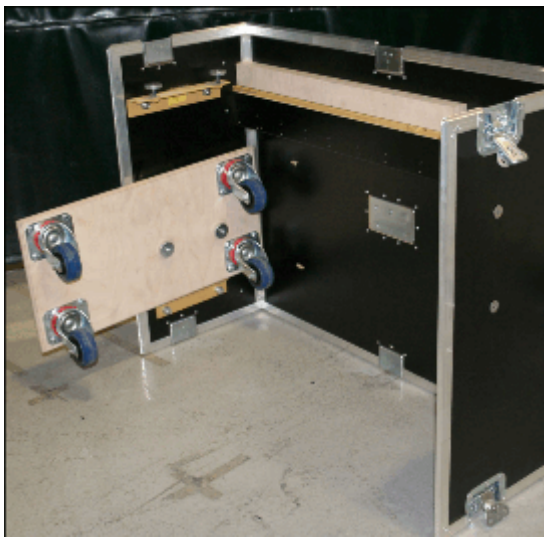


- (1) Vertical lifting beam tool

If there are only 2 x lifting eyes available for lifting the compressor unit:

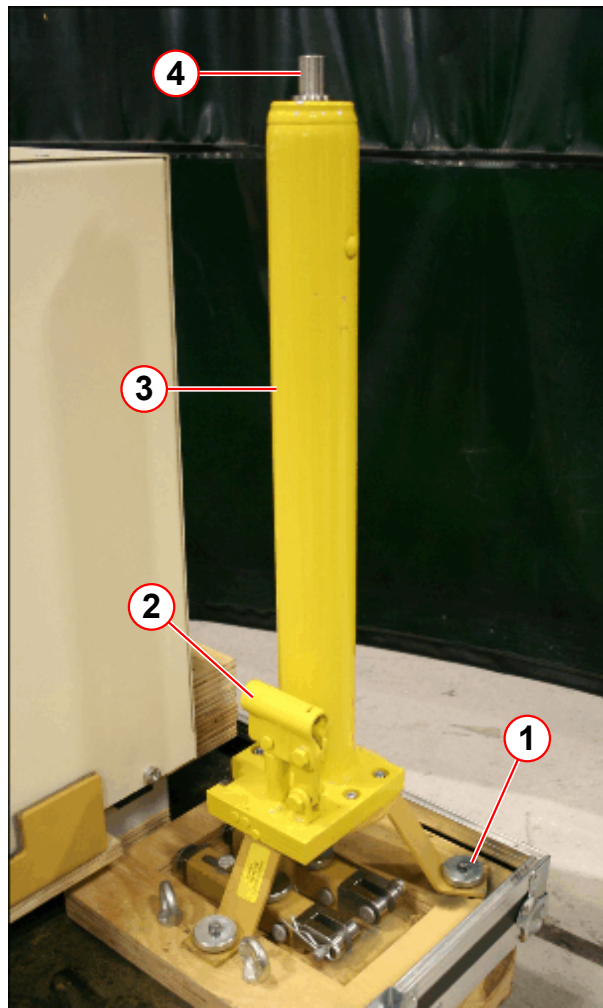
- Unscrew the 2 knurled head fasteners and remove the vertical support arm.
- Replace the fasteners.

Fig. 192: Board with castors



- Remove the knurled head fasteners and detach the moveable board with castors from the transport box side.
 - » Do not leave the board unattended.
 - » When not in use, place it upright in a safe area.

Fig. 193: Lifting Jack x 2



- Remove the knurled head fasteners x 3 from each of the 2 jacks.
- Remove the 2 lifting jacks from the base of the transport box.

- (1) Knurled fastner x 3
- (2) Hydraulic pump handle attachment
- (3) Cylinder body
- (4) Hydraulic lifter

**CAUTION****Hydraulic steel jack**

Non-observance of these warning notices may lead to damage or error of the steel jacks. The results are property damage and serious injuries.

- » Immediately support the compressor unit.
- » Working on a compressor unit when supported by the 2 x steel jacks is not permitted
- » Use only a hard, even floor area with sufficient load bearing capacity for the lifting device.
- » Prevent excess weight. The load bearing capacity cannot be exceeded!
- » Do not open the oil filler rubber plug under any circumstances. Exception: the jack has been fully lowered.
- » The operator has to monitor all movements of the load and the configuration.

Fig. 194: Support arms x 4



- Below the 2 jacks there are 4 x horizontal support arms.

The quantity required will correspond to the number of lifting eyes fitted to the compressor unit.

Fig. 195: Removing straps

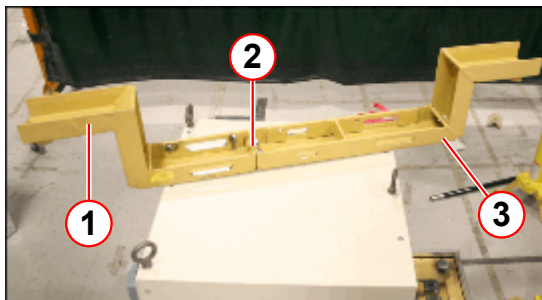


- Carefully open the restraining strap securing the compressor unit.
- Remove the edge protection pieces and the strap.



During assembly of the lifting frame place some cloth / cardboard on top to avoid paint damage to the paint.

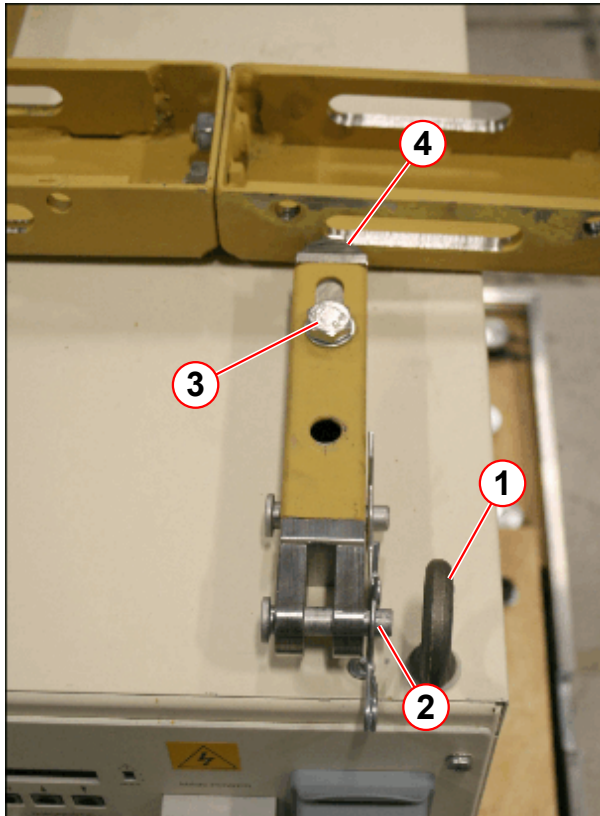
Fig. 196: Assemble lifting beams



- (1) Short beam
- (2) Bolts x 2
- (3) Long beam

- Screw both cross beams together with M10 nuts and washers.
- Torque screws to 23 Nm (17 Ft/lbs).

Fig. 197: Fitting the Support arms (x3)



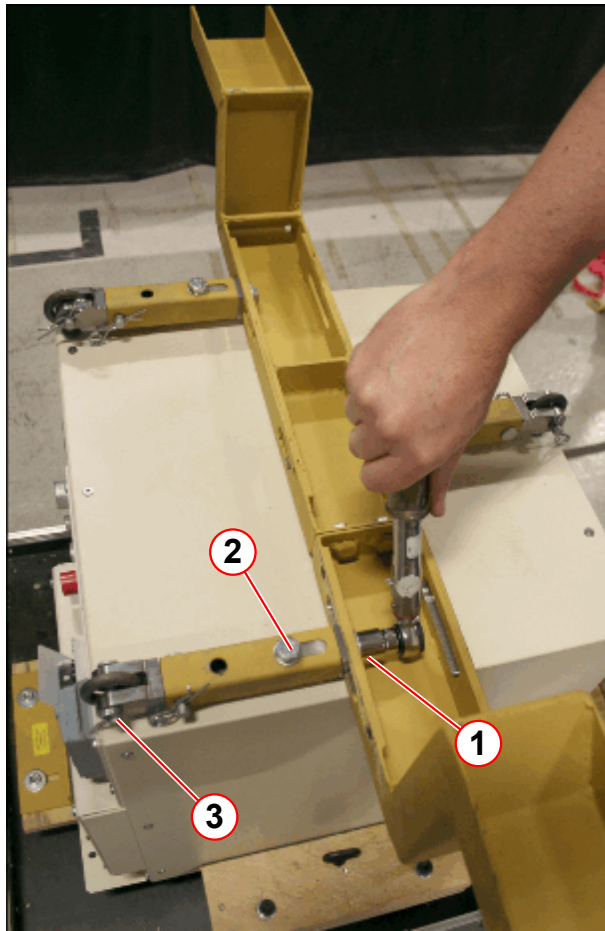
- (1) Lifting eye
- (2) Bolt and Split pin
- (3) Adjustment screw M10
- (4) Cross beam securing screw M10

Before attaching the support arms check that the lifting eyes are orientated at 90 degrees to fit into the lifting arms.

Attach the support arms:

- If there are no lifting eyes fitted on compressor unit, fit the lifting eyes supplied with the tooling.
- Insert the bolt and secure with the split pin, (pos: 2).
- For fine movement undo the adjustment screw, (pos: 3).
- Remove the cross beam securing bolt, (pos: 4).

Fig. 198: Secure support arms



- (1) Support arm & Lifting beam screw
- (2) Adjustment screw
- (3) Split pins

- Attach the support arms to the cross beams M10 x 20 screws and washers.
- Torque screws to 23 Nm (17 Ft/lbs).
- Center the complete cross beam assembly and tighten all screws with the 17-mm wrench.
- Torque the M10 adjustment screws to 23 Nm (17 Ft/lbs).
- Check that the split pins are all attached.

The cross beam lifting assembly is ready for the hydraulic jacks to be attached.

Fig. 199: Alternative lifting method



(1) Lifting eye

Alternative Lifting method

- If only 2 x lifting eyes are available on the upper side of the compressor unit:
- Attach the additional vertical support arm with the supplied crane lifting eye to the back of the compressor unit.

NOTE: The horizontal support arm seen in the figure is omitted here.

Fig. 200: Front F70 fixing adjustable plate



- Loosen the 2 x screws at the front angle adjustable plate.
- Move the angle plate away from the F70 cabinet housing.

3.5.7 Installing the Compressor Unit

i

Prevent stress situations caused by shocks generated by abruptly opening or closing the actuating valve while lowering the piston.

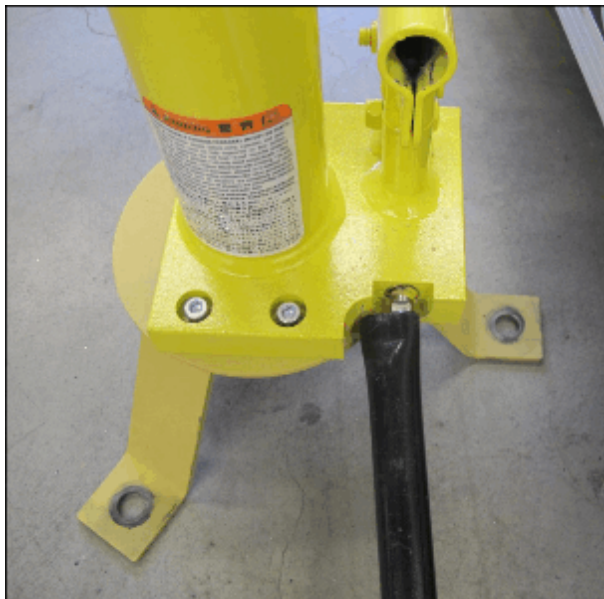
Shock loads can lead to hydraulic circuit overloads that damage the jack.

When the jack is not in use, the piston must be in the lowest position.

i

Before using, check the jack and the cross beam tooling for damage and oil leaks. If faulty, do not use the tooling.!

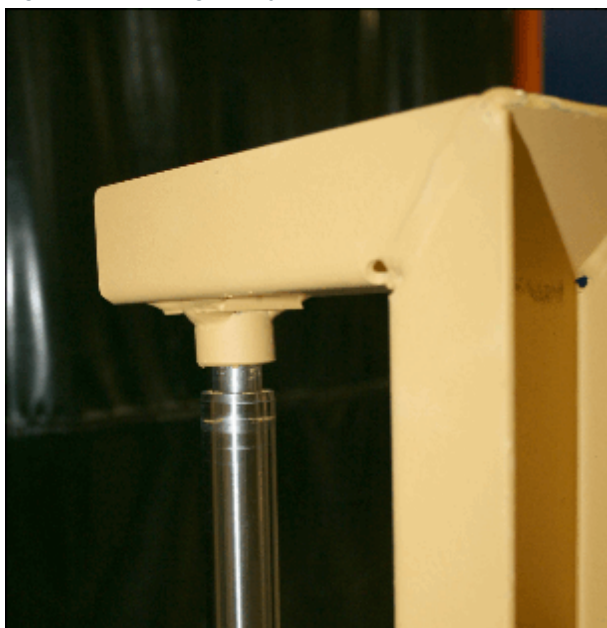
Fig. 201: Pressure relief valve



Lifting operation

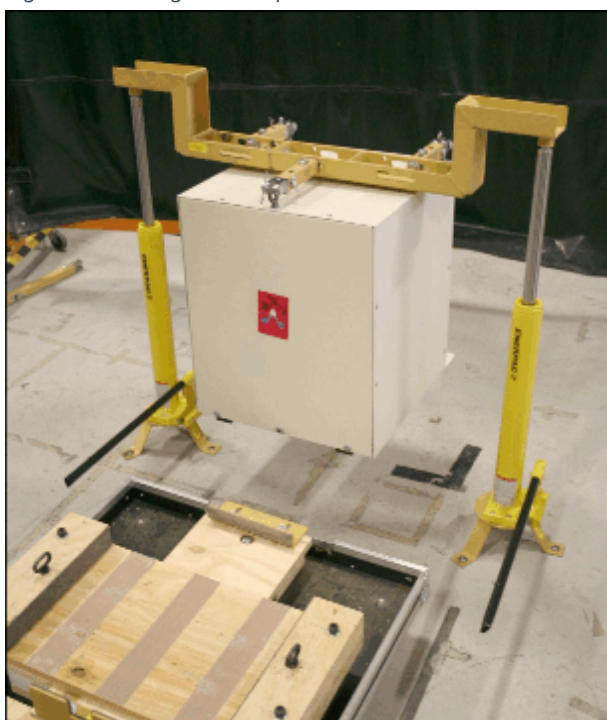
- Fit the rod onto the pressure relief valve screw.
- Turn it clockwise until the valve is firmly closed.
- Remove the rod and insert into the hydraulic pump bracket.
- Repeat this operation for the 2nd hydraulic jack.

Fig. 202: Locating the Hydraulic arms



- Slide hydraulic supports underneath the each lifting arm.
- Using the rod, pump each hydraulic arm up to the socket on the lifting beam.
- Check that the hydraulic arm is located into the socket on the lifting beam before taking the full weight of the compressor unit.

Fig. 203: Lifting the compressor



- Continue pumping the hydraulic jack rods **at the same time** to ensure a uniform lift of the compressor unit.
 - » Keep the compressor unit in a horizontal position. It must not be tipped at more than a 10 degree angle.
- Move the lower assembly of the transport box away from the compressor unit.

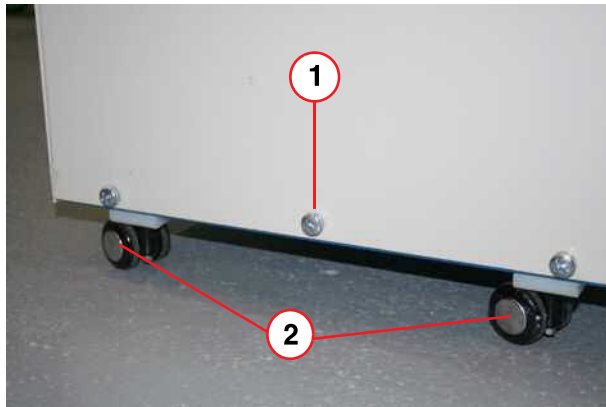


The new compressor unit is supplied fitted with 4 x caster wheels. When the compressor unit is held off the ground on the hydraulic jacks remove the caster wheels.

Compressor caster Removal

- The casters are to be removed for fitting the compressor into a SEP cabinet.
- For Standalone and IFP systems the caster wheels can remain fitted.

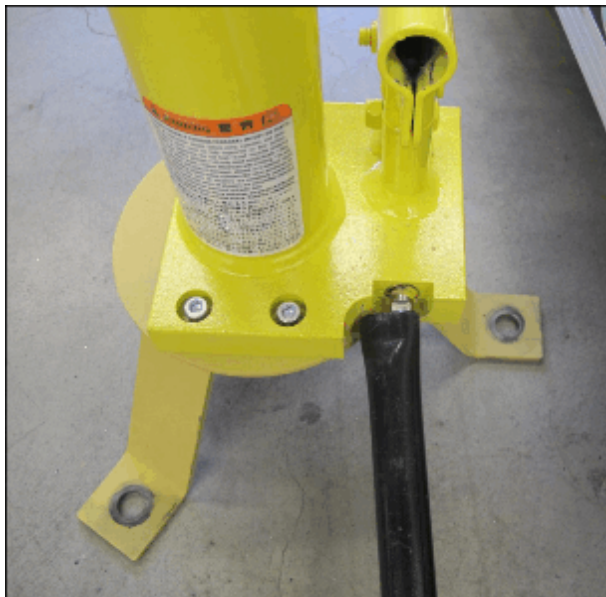
Fig. 204: F70 Back panel view



- (1) Center screw is removed for SEP cabinet installation
 (2) Casters x 4 (stand-alone units only)

- To remove the 4 casters from the compressor unit.
 - Use a 7 mm ratchet and socket and a 14 mm open ended wrench.
 - **Remove** the center screw (pos 1). (The screw hole aligns with the **guide pin** at the back of the SEP cabinet).

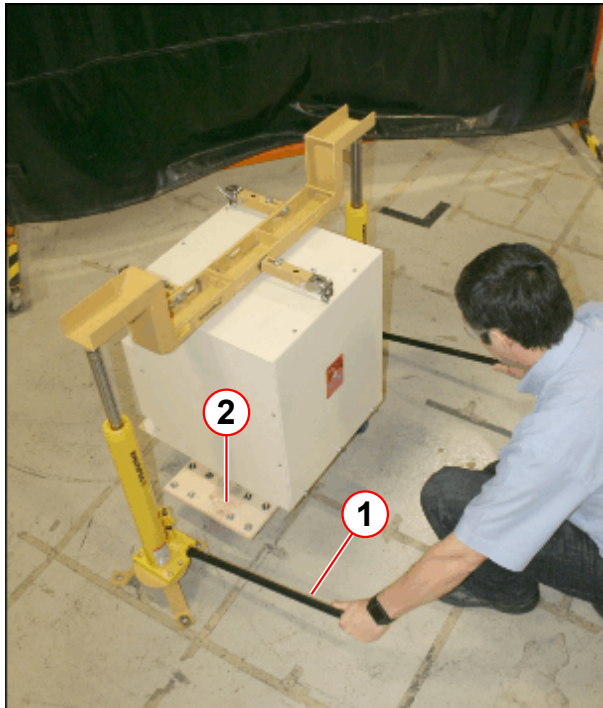
Fig. 205: Pressure relief screw



Lowering operation

- Remove the 2 x rods from the hydraulic pump bracket.
- Fit the rods over each pressure relief valve screws
- Slowly turn it counter-clockwise by not more than 1 full turn.
- Turn it slowly to be able to check the lowering speed.

Fig. 206: Lowering compressor



- (1) Hydraulic jack rods x2
- (2) Moveable board

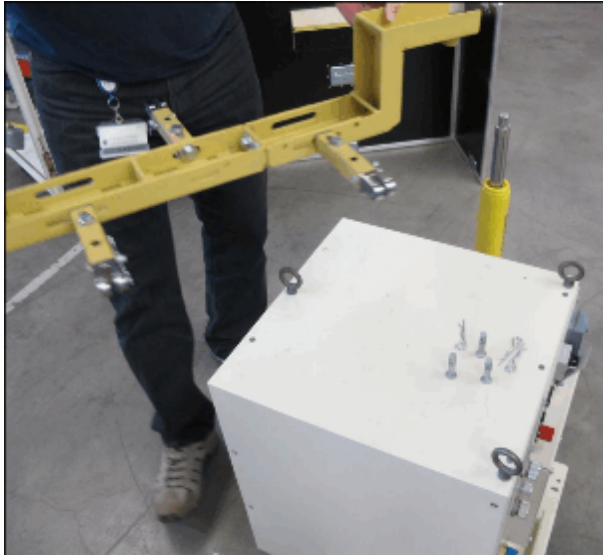
- Place the moveable board below the compressor unit.
- Turn both pressure relief valves slowly counter-clockwise to lower the compressor unit onto the moveable board.

Fig. 207: Support arm removal



- Remove the splints and bolts from the support arms fitted to the new F70 compressor unit.

Fig. 208: Removing lifting beam



- Remove the cross beam assembly.
- Carefully move the compressor unit to the final position near to the cabinet.



The moveable board casters are fixed. Movement will only be in a straight line. Check that the assembly tool is in the direct vicinity of the cabinet.



Uneven flooring leads to the hazard of tipping over!

Fig. 209: Removing the defective compressor



- Position the top cover board of the transport box in front of the cabinet.
- Carefully slide the defective compressor unit out of the cabinet onto the board.

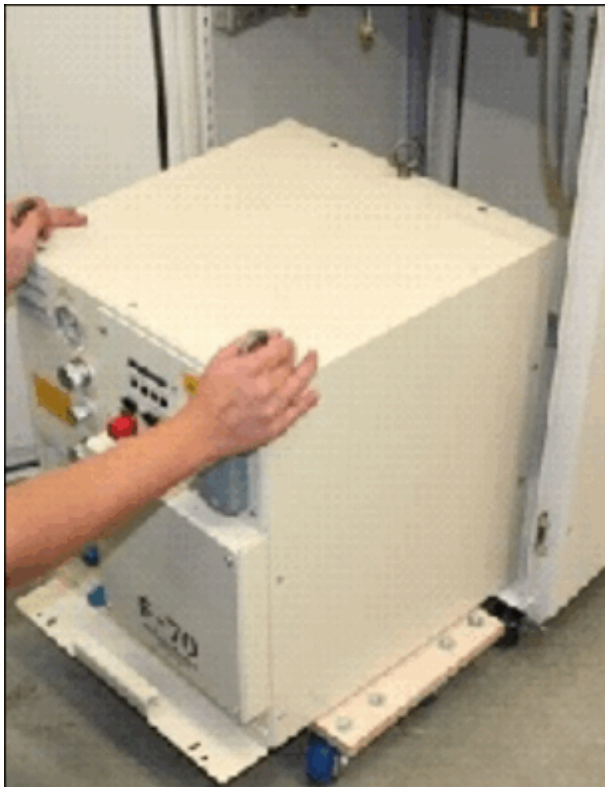
Fig. 210: Remove defective compressor (2)



When the defective compressor unit has been removed, move it away from the cabinet.

The cross beam and support arm assembly can be fitted later. The hydraulic jacks will be used to lift the defective compressor unit onto the transport box.

Fig. 211: Moving in the compressor

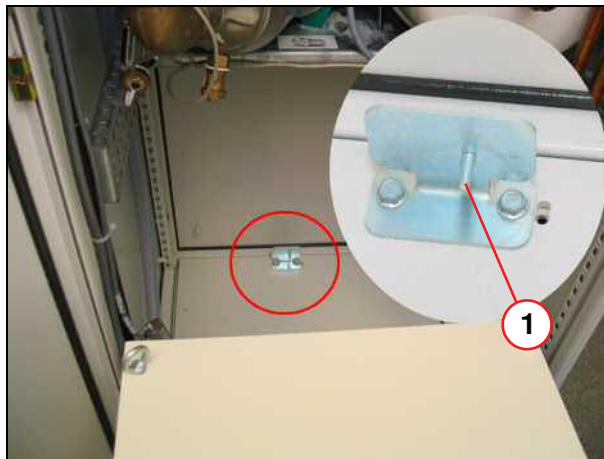


- When fitting the new compressor unit:

- » Check that the compressor unit is directly in line to the front of the cabinet.

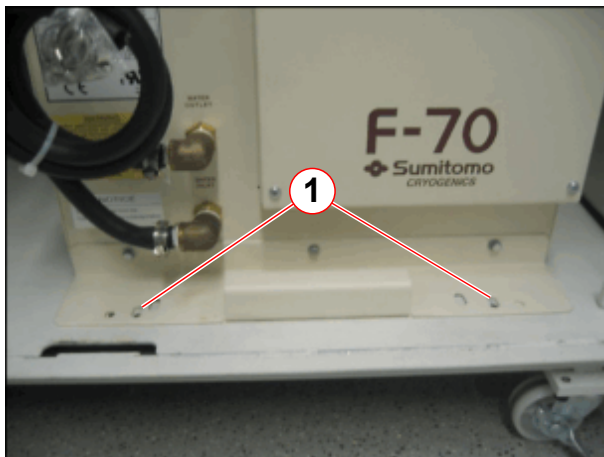
- Slide the new compressor unit into the cabinet.

Fig. 212: SEP - Compressor guide pin



- The (center) screw hole on the rear panel of the compressor unit should align with the **guide pin** at the back of the cabinet.

Fig. 213: F70 Compressor fixing plate



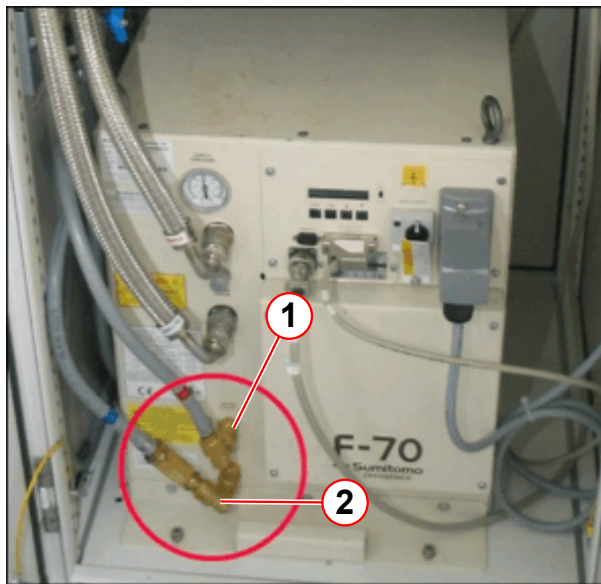
- Secure the **fixing plate** to the cabinet using the 2 x M8 hex head screws.

(1) M8 Hex hd screws (2 off)

3.5.8 Compressor Unit Configuration and Connections

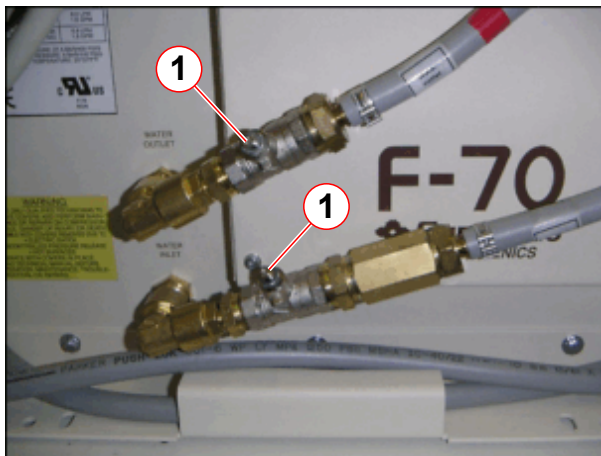
The new compressor unit must be operated for **10 minutes** to clean up any oil contamination incurred during transportation. Run the compressor unit **before** the **helium pressure lines** and **cold head** are connected. For this test, only the **water** and **power** connections are required.

Fig. 214: F70 connect the water lines



- (1) Water line (inlet)
- (2) Water line (outlet)

Fig. 215: F70 Compressor water inlet/outlet shutoff valves

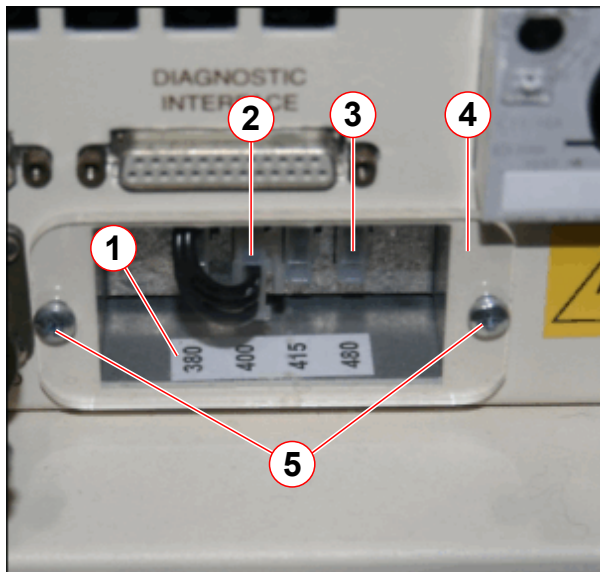


- (1) Shutoff valve

Connecting the water:

- Remove the protective caps from the water in and water out fixings.
- **Connect the water lines** to the compressor.
 - Ensure the **water in** is connected to the **water supply line**, and the water is turned **on**.
 - **Open the shut-off valves**, turn on the coolant supply, and check for leaks.

Fig. 216: F70 Compressor voltage selection access panel



- (1) Voltage label
- (2) Voltage selection connectors
- (3) Voltage tap
- (4) Clear plastic cover
- (5) Screws (2 off)

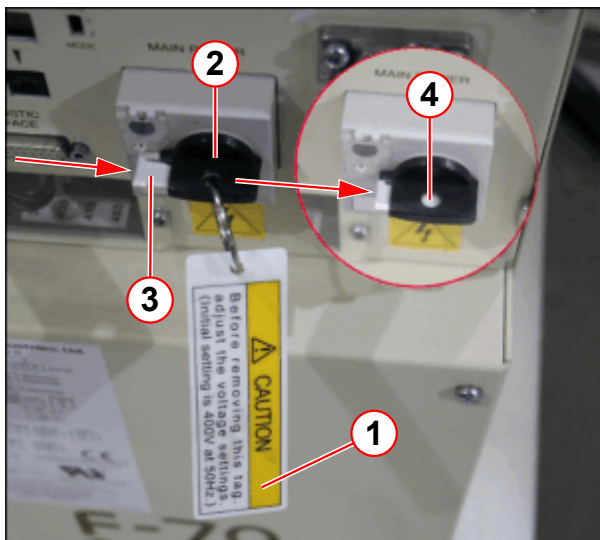
- Check that the **transformer voltage taps** are correctly selected to match the supply voltage.
- If necessary, remove the plastic cover (3), reposition the tap, and replace the cover.



The unit is configured in the factory to 400 V/3~50 Hz.

If the compressor unit is not wired correctly, connecting to a higher voltage may damage the control circuit.

Fig. 217: F70 Main power switch - locking device



- (1) Voltage caution label
- (2) Main power switch
- (3) White locking device
- (4) Device fully engaged (view through hole)

- **Remove** the voltage caution label (pos. 1) (fitted through the main circuit breaker).
- Engage the white switch locking device fully **before** attempting to turn on.
- **Push in** the white locking device fully.
 - » The device should be visible through the blocking hole (pos. 4).



Ensure that the white switch locking device is **FULLY** engaged. **Do not use too much force on the switch.** A new compressor unit will be required if damaged.

- Connect the mains voltage cable.

Fig. 218: F70 Main power supply connection

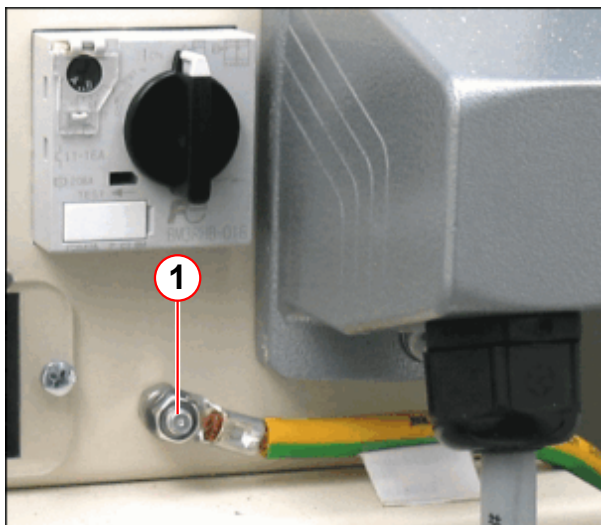


- (1) Pull cover to remove
- (2) Connect power supply cable
- (3) Fasten the cover by attaching the screws

3.5.9 Compressor Unit (Ground) Earth Connection

A protective ground connection test must be performed as a final step **before** switching on the compressor unit.

Fig. 219: Compressor ground wire connection



- (1) Ground wire connection

- Connect the ground wire to the helium compressor terminal.

Fig. 220: RCA



(1) Protective conductor

- Connect the protective conductor W0821 from the RCA to the ACC.

Test the protective earth connection

- Measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent). Check between the protective earth connector on the compressor unit, and the protective earth wire connected to the top of the RCA

3.5.9.1 Compressor Unit Run Mode



Compressor unit is supplied in configuration **mode 1**.

Before the compressor unit is switched on, the **mode selector switch** must be set to either mode **1** or **2**.

Tab. 10 Configuration mode switch

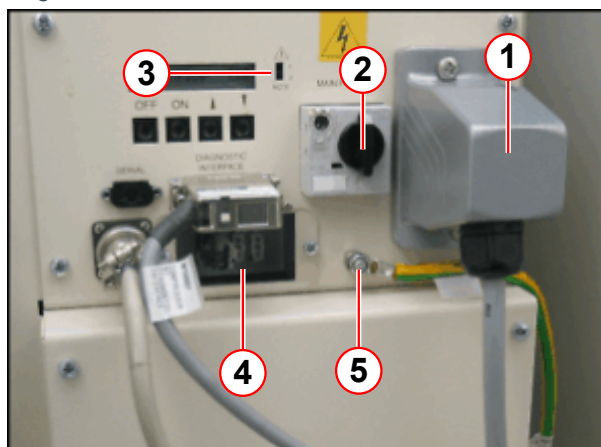
Configuration	Switch position	Magnet system type	ON/OFF
Mode 1	UP	10-K shield cooling	Enabled
Mode 1	UP	F70 Startup checks	Enabled
Mode 2	DOWN	4-K recondensing	Disabled



DO NOT change the configuration-mode switch position **AFTER** the main power has been switched **ON**. Loss of control and incorrect errors will result.

3.5.9.2 Testing the Compressor Unit on Startup

Fig. 221: F70 Compressor - connections for test running before connection



- (1) Power supply
- (2) Main power switch
- (3) Configuration mode switch
- (4) Voltage selector
- (5) Ground wire connection

- Ensure that the **main power** switch is **OFF** before adjusting the selector switch.
- Check that the configuration mode switch is set to 1.

Use a flat-blade screwdriver to move the switch up and down.

- » The **ON/OFF** buttons only work in configuration mode 1.
- » The compressor unit must **NOT** be run in stand-by in mode 2.

In **mode 2** both **ON** and **OFF** buttons are disabled. Operation is controlled remotely via the DB25 diagnostic interface.

- Switch on the circuit breaker F102 (GPA cabinet).
- Turn the main power switch.
- Press the ON button.
- Check the temperature of the water supply (water in) and return (water out) lines:-
 - » The **return** (water out) pipe should be **warmer** than the **supply** (water in).
 - » **Stop** the compressor unit and reverse water pipes if the **return** line is **cooler**.
- Run the compressor unit for **10 minutes** and then press the **OFF** button.
- Turn **OFF** the main power switch.
- The test run is now complete.



DO NOT change the configuration-mode switch position **AFTER** the main power has been switched **ON**. The result is loss of control and incorrect indications.



The compressor unit is protected from running in reverse.



If the compressor unit starts but does not build pressure, **TURN OFF IMMEDIATELY**. Contact Siemens Magnet HSC.

3.5.10 Testing the Compressor Unit after Mishandling

Follow this procedure if:

1. The Tip-N-Tell indicator has been activated.
2. The compressor pressure is lower than specification.

Fig. 222: F70 Compressor Tip-N-Tell indicators



Pressure within specification:

- If the Tip-N-Tell has been activated but the pressure is in specification:
 - Stand compressor unit upright for **2 hours**.
 - Perform the test run.

Pressure outside specification:

- Notify the delivering carrier of the damage if:
 - The compressor unit has been delivered on its side or upside down.
 - There is physical damage to the unit.

NOTICE

If the compressor unit arrives at zero pressure - **DO NOT CONNECT THE HE PRESSURE LINES or COLD HEAD POWER CABLE** to the compressor unit.

Connecting the lines to the compressor unit at zero pressure may cause contamination of the whole refrigeration system.

- » Check the compressor pressure before attempting to connect the helium pressure lines.
- » Contact Siemens Magnet HSC for advice.

- **Remove** the compressor unit from the **transport box**.
 - **Retain** all packaging materials for reuse if returning the compressor unit for replacement or repair.

Install the replacement compressor unit into the SEP cabinet before performing the test.

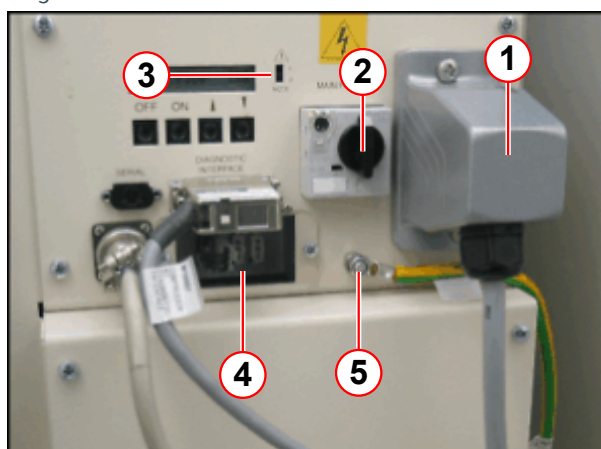
For the test, only compressor water and power connections are required.

Ensure all other lines and cable connections are disconnected from the compressor unit.



DO NOT change the configuration-mode switch position **AFTER** the main power has been switched **ON**. Loss of control and incorrect indications will result.

Fig. 223: F70 Compressor - connections for test running before connection



- (1) Power supply
- (2) Main power switch
- (3) Configuration mode switch
- (4) Voltage selector
- (5) Ground wire connection

- Check that the configuration mode selector switch is set to **1**.
Use a flat-blade screwdriver to move the switch up and down.
 - » The **ON/OFF** buttons only work in configuration mode **1**.
 - » The compressor unit must **NOT** be run in stand-by in mode **2**.
In configuration **2**, both **ON** and **OFF** buttons are **disabled**. Operation is controlled remotely via the **DB25 diagnostic interface**.
- Switch **ON** the circuit breaker F102 (EPC).
- Turn on the main power switch.
- Press the **ON** button to start the compressor unit. The LED displays **RUN**.
- Check the water supply and return line temperatures.
 - » The **return** line should be **warmer** than the **supply**.
- **Stop** the compressor unit and reverse the water pipes if the **return** line is **cooler**.
- Run the compressor unit for a **MINIMUM of 2 HOURS**.



If the compressor unit shuts down during the test run, contact Siemens Magnet HSC.

- If there are no problems during the test run, stop the compressor unit.
 - **Press** the **OFF** button.
 - **Turn OFF** the **main power** switch.
 - The test run is complete.

- Reset the configuration mode switch to 1.



The compressor unit is protected from running in reverse.



If the compressor unit starts but does not build pressure, **TURN OFF IMMEDIATELY**. Contact Siemens Magnet HSC.

3.5.11 Cold Head and Compressor Unit Start up



CAUTION

High-pressure helium gas

Failure to observe the following Safety Information may result in injury.

» Protective clothing, gloves, and eye protection must be worn.

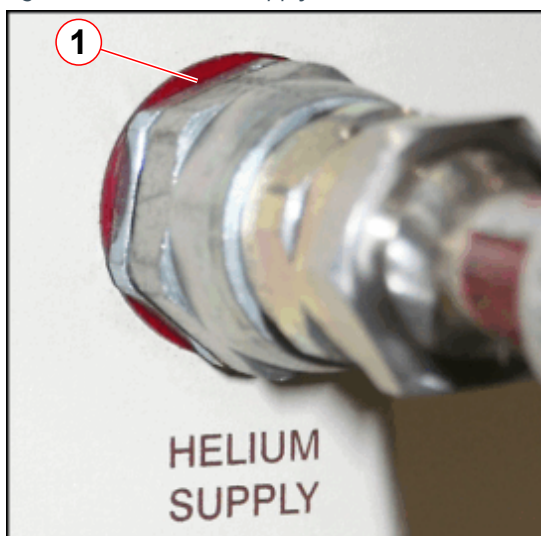
Helium gas lines

Ensure that the **Supply** (HP) and **Return** (LP) lines are connected correctly. The 4K helium pressure lines are colour coded as shown in the table below.

Tab. 11 Gas line connections

System	Gas line color	F70 He connection
4K supply (HP)	Yellow	Red
4K return (LP)	Green	Green

Fig. 224: F70 Helium supply connections - color coding



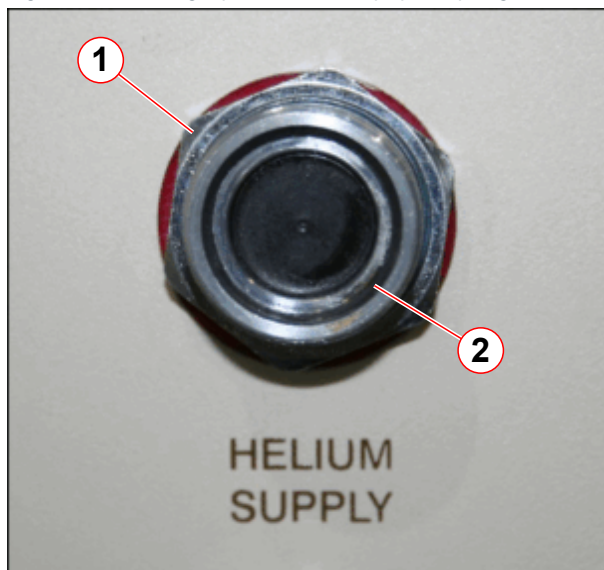
(1) Helium supply = RED

Fig. 225: F70 Helium supply connections - color coding



(1) Helium return = GREEN

Fig. 226: F70 High-pressure Aeroquip coupling

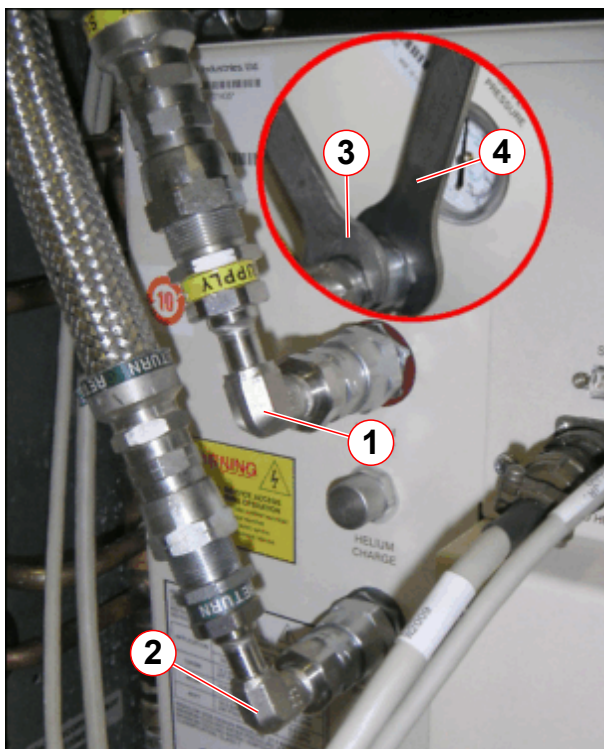


- (1) Locknut
- (2) Flat rubber gasket

- Remove the protective transport covers from the return and supply connections.
- Check the **rubber gasket** seal for dirt or dust and check that it is correctly fitted.

NOTE: An incorrect fit may cause a helium gas leak.

Fig. 227: F70 He compressor connections



- (1) SUPPLY (high pressure) 90-degree elbow
- (2) RETURN (low pressure) 90-degree elbow
- (3) Hold nut (1" spanner/wrench)
- (4) Turn nut (1 3/16" open end crowfoot spanner/wrench)

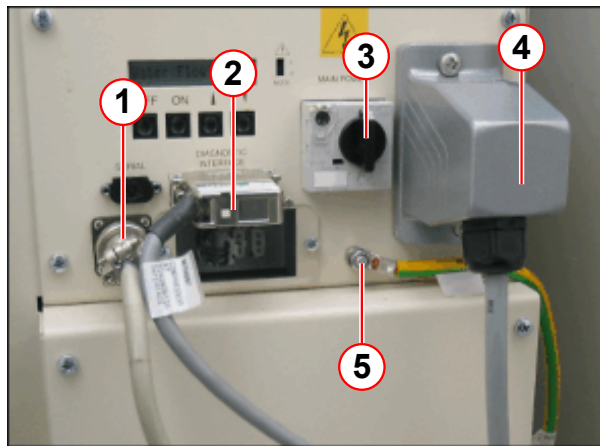
Each **helium pressure line** is connected to the compressor unit via a **90 degree elbow**.

- **Fit** the self-sealing helium pressure lines in the following order:
 - **Return** (low pressure).
 - **Supply** (high pressure).
- **Hold** nut (3) and **turn** nut (4).
- **Tighten down** all couplings as far as possible, and then turn them back by 1/4 turn to **relieve strain**.



When connecting the Helium pressure lines, **DO NOT OVER-TIGHTEN** the couplings.

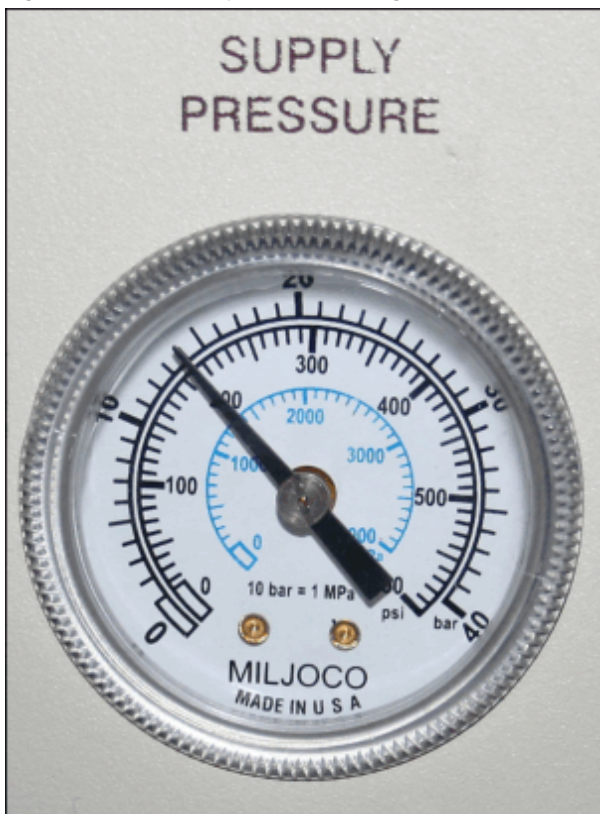
Fig. 228: F70 Electrical cable connections



- (1) Cold head cable
- (2) MSUP DB25 Diagnostic interface cable
- (3) Main power switch
- (4) Mains supply cable
- (5) Ground wire

- Connect the **cold head cable** (1) and **MSUP cable** (DB25 diagnostic interface) (2).

Fig. 229: F70 Static pressure reading



- Check the **static helium pressure** on the pressure gauge.
- Charge with helium gas if pressure is lower.

Equalization (static) pressure at 20⁰ C (68⁰ F) for 12 m to 20 m long gas lines:

Tab. 12 Helium gas pressures

Cold head	Static pressure	Dynamic pressure
4 K - 50/60 Hz	13.5 - 14.0 bar (13.6 nominal) 196 - 203 psi (G) 1350 - 1400 kPa	21 - 22 bar 304 - 319 psi (G) 2100 - 2200 kPa

- Check that the **configuration switch** is set to **mode 1**.
- Check for **water flow** to the compressor unit.
- Switch **ON** the **main power** switch.
- Check the **alarm box** status and clear any faults.

3.5.12 Final Checks

- Wait **20 minutes** and check the operating (**Dynamic**) helium gas pressure on the pressure gauge.
 - **Supply: 21-22 bar (304-319 psi (G)).**



If the dynamic pressure is outside of this range, refill.

- **Check** the helium pressure line and connectors for **leaks**.
- **Check** for leakage at the water connections.
- Document the measurement in the **service report**. The (measured value, and measured equipment) **is required by IEC 62353**.

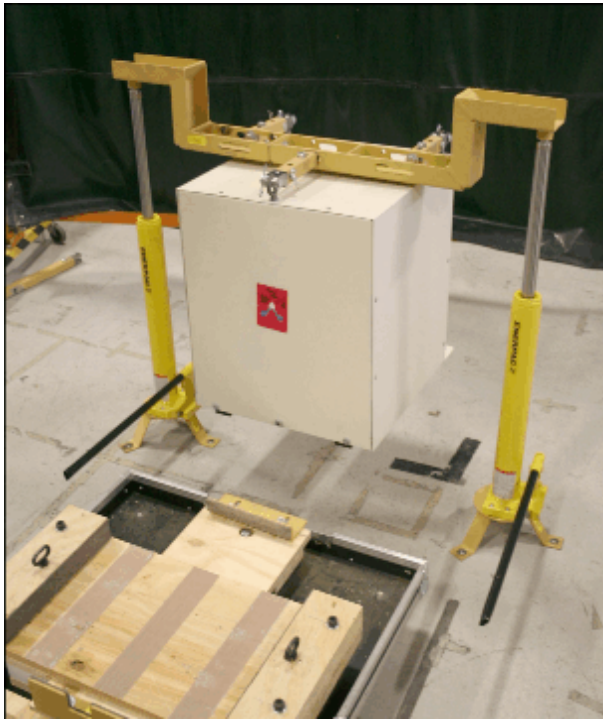
3.5.13 Packing the Compressor Unit

Fig. 230:



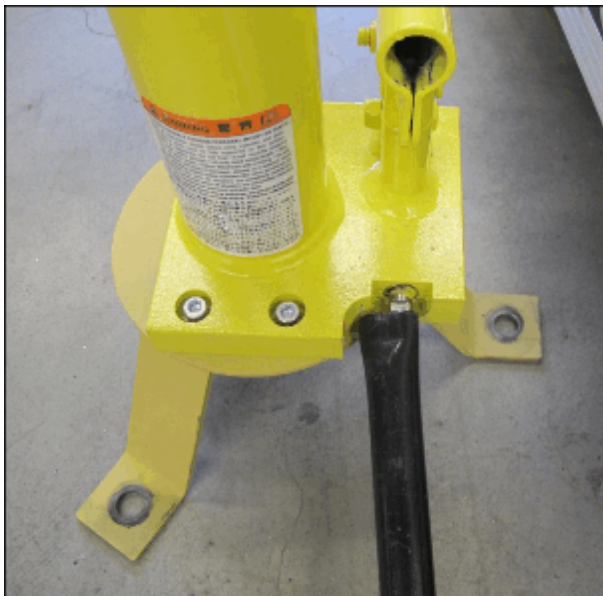
- Move the failed compressor unit on the moveable board next to the lifting jacks.
- Fit the lifting beam assembly.
- Pumping the 2 x hydraulic jacks at the same time lift the compressor unit.
- Remove the moveable board.

Fig. 231: Lifting the compressor



- Keep the compressor unit in a horizontal position
- Move the lower assembly of the transport box under the compressor unit.

Fig. 232: Pressure relief screw



Lowering operation

- Remove the 2 x rods from the hydraulic pump bracket.
- Fit the rods over each pressure relief valve screws
- Slowly turn it counter-clockwise by not more than 1 full turn.
- Turn it slowly to be able to check the lowering speed.

Fig. 233: Back F70 fixing plate



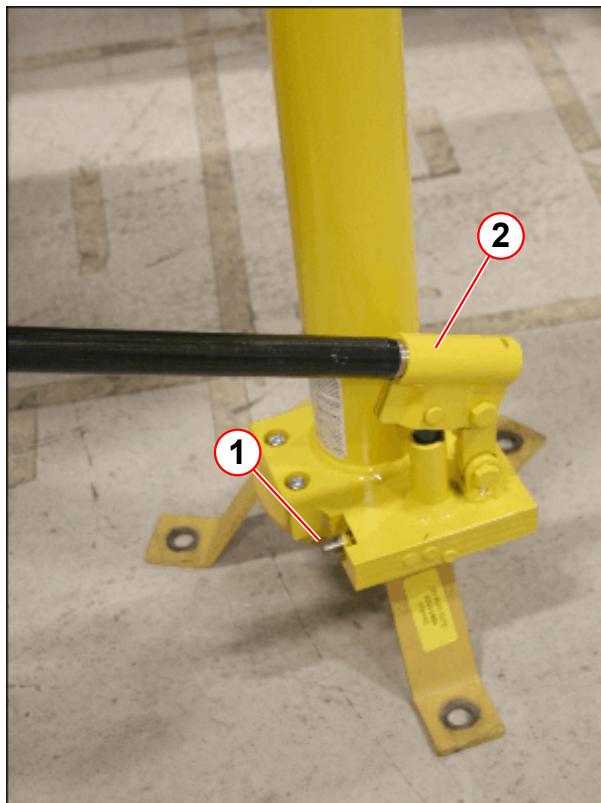
- Lower and place the defective F70 compressor unit onto the transport box in the location provided.
- Check that the F70 cabinet fits flush with the back angle fixing plate.

Fig. 234: Front F70 fixing adjustable plate



- Loosen the 2 x screws at the front angle adjustable plate.
- Move the angle flush and tight with the F70 cabinet housing and retighten the 2 x screws.

Fig. 235: Packing the hydraulic jacks



- (1) Pressure relief valve screw
(2) Hydraulic pump bracket

To removing air from the hydraulic jack

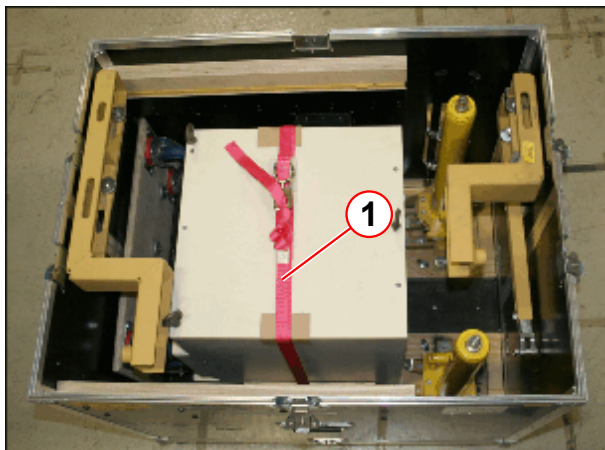
- Place the rod onto the pressure relief valve screw.
- Turn it counterclockwise by not more than 3 turns.
- Insert rod into the neck of the rod.
- Perform between 6 to 8 complete pump strokes.
- Repeat if necessary.

Fig. 236: Manual press piston



- If the piston has not fully lowered, manually press down.
- Repeat for the 2nd jack.

Fig. 237: Top view of compressor kit



(1) Securing strap

- For the return shipment, pack all the tooling parts back into the transport box in the correct locations and tighten all screws.

Fig. 238: Transport crate



- (1) Lifting handles (x 2)
- (2) top lid latch (x 6)
- (3) Lid with PTFE sheet covering

- Replace the top lid of the transport box.
- Return the compressor transport box to CSML.

4.1 Replacing the Magnet Supervisory Unit (MSUP)

4.1.1 Tools and time required

4.1.1.1 Time

60 min

4.1.1.2 Tools

- ERDU test load
- Non-magnetic tool kit

4.1.2 Preliminary

- Download the magnet-specific values from the MSUP to the HOST, if possible. Open SeSo / Magnet&Cooling / Initialization
- Note the following magnet-specific parameters:
 - Resistance of probe 1 and probe 2.
 - Pulse length and pulse height of probe 1 and probe 2.
 - Pressure heater resistance.



CAUTION

The replacement procedure disables the magnet emergency shut-down system at the MSU.

Failure to adhere to safety precautions may result in serious bodily injury or property damage.

- » Connect the ERDU Test Load.
- » The magnet may only be quenched using the MAGNET STOP button on the ERDU Test Load
- » Exclude all non-service personnel and all magnetic materials / tools from the magnet area.
- » Close the door of the RF room

Connecting the ERDU Test Load (ETL):

**CAUTION**

The ERDU test load is slightly ferromagnetic.

The ERDU Test Load is attracted by the magnetic force and projected into the magnet bore, resulting in injury or damage.

- » Keep the ERDU test load as far from the magnet bore as possible.

1. Connect the ETL PSU to a convenient AC outlet, ensuring that the PSU is as far from the magnet as possible.
2. Connect the PSU to the ETL X4 using the extension cable.
 - » Check that the SUPPLY and SWITCH OK indicators glow on the ETL.

**CAUTION**

The magnet cannot be quenched while disconnecting and reconnecting the turret connector.

- » Connect the turret connector immediately.

3. Connect the "<MSUP-ETL>" cable to the 25-pin connector X1 on the ERDU Test Load.
4. Connect the <41-pin to 10-pin adapter> to the 41-pin connector on the ERDU Test Load.
5. Ensure that the Magnet STOP button on the ETL is not activated.

**WARNING**

Magnet quench!

A magnet quench is caused by pressing the ETL Magnet Stop button.

- » Ensure that the MAGNET STOP button on the ETL is NOT OPERATED!
- » DO NOT operate the Magnet Stop button on the ERDU Test Load unless an emergency magnet quench is required.

6. Set the mode switch to OR64 / 70
7. Set all HIGH R / LOW R switches on the ETL to NORMAL.

**CAUTION**

It is not possible to quench the magnet while disconnecting and reconnecting the 10-pin turret connector "B".

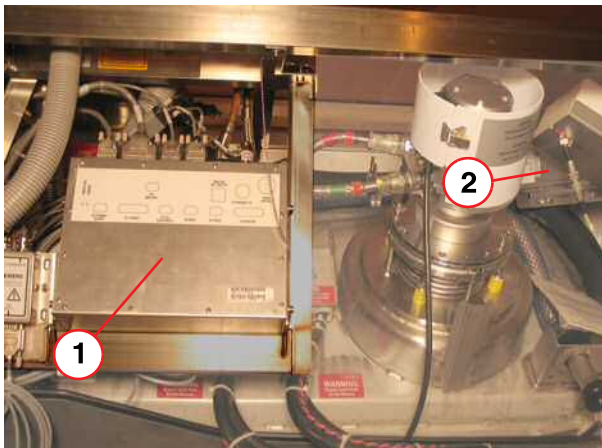
- » Connect the 10-pin turret connector "B" without delay.
- » In case of an emergency, immediately connect the 10-pin turret connector "B".

- ### 4.1.3 Replacement procedure

- 

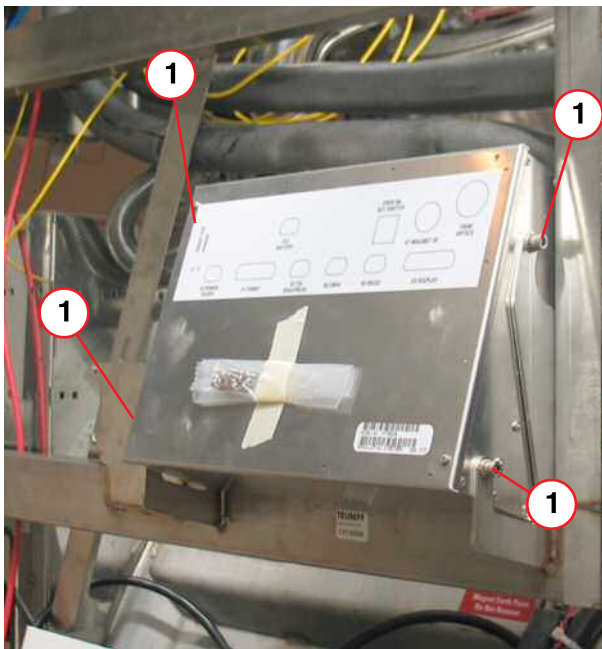
- Disconnect the turret cable from X1 on the MSU.

Fig. 240: MSUP - ERDU battery



- (1) MSUP
- (2) ERDU (MSUP) battery

Fig. 241: MSUP

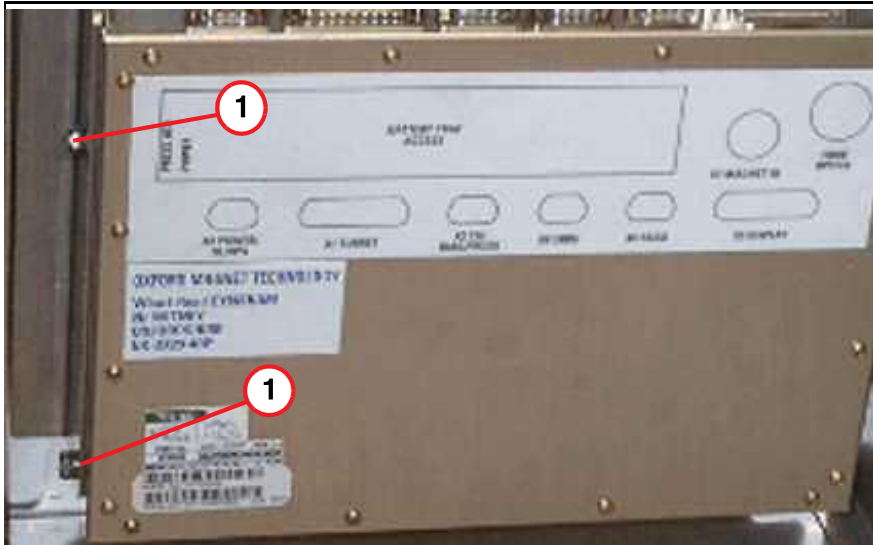


- (1) Mounting bolt

- Disconnect power cable X3 on the MSUP.
- Disconnect battery cable X12.
- Disconnect the remaining electrical connections.
- Remove the lid of the MSUP.
- Carefully disconnect fiber optic cables U38, U39, U40, U41 from the connectors on the MSUP PCB.
- Withdraw the fiber optic cables through the wave guide.

- Remove the bolts securing the MSUP to the cradle.

Fig. 242: MSU



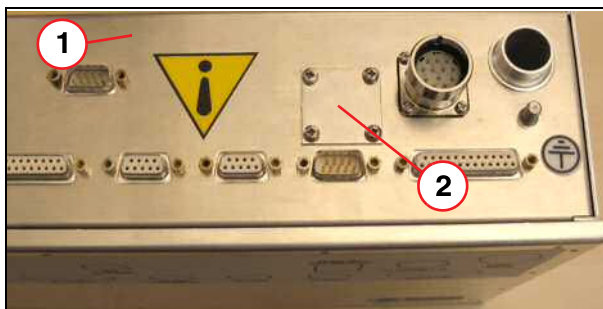
(1) Bolts

4.1.4 Installation of the new MSUP

- Carefully secure the MSUP using the bolts.
- Check the setting of the Magnet Stop switch.

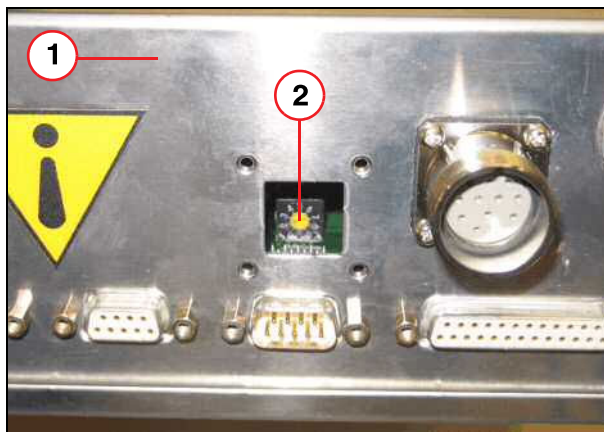
Fig. 243: MSUP (ERDU No SET SWITCH) Cover

- Remove cover (2).



(1) MSUP
(2) Cover

Fig. 244: MSUP (ERDU No SET SWITCH)



(1) MSUP
(2) Switch

- Check switch setting:
 - » Position 2 for 2 ERDU buttons (Alarm box button + ERDU button 1)
 - » Position 3 for 3 ERDU buttons (Alarm box button + ERDU button 1 + ERDU button 2)
- Attach the cover

- Carefully insert fiber optic cables U38, U39, U40, U41 into their correct connectors on the PCB.
- Reinstall the lid of the MSUP.



Do not mix up the lids of the MSUP. Check the serial number labels.

- Connect external connectors X2, X5, X6, to the MSUP.
- Connect the magnet ID plug to X7 taken from the old MSUP.
- Connect battery connector X12.
- Connect power connector X3.
- Insert F3 (ACC).
- Connect the MSUP to the magnet:
 - » Connect the "MSUP to the turret" cable to X1 of the MSU.
 - » Disconnect the ETL from the 41-pin connector on the magnet turret
 - » Immediately connect the "MSUP to the magnet turret" lead at the 41-pin connector on the magnet turret.
- Reboot the MR system.
- Check that all alarm and fault indications show that the system is OK.
-

4.1.5 Concluding checks

- Open SeSo / Magnet&Cooling / Initialization.
 - Press the UPLOAD button to transfer the magnet-specific values from the HOST to the new MSUP.
- Exit the Magnet&Cooling menu.

- Reenter Magnet&Cooling/Initialization menu

Compare the magnet-specific parameters with the parameters written down at the beginning

- Check the helium probe resistance.
- Check the pressure heater resistance.
- Check the pulse length and pulse height (can be ignored with SW version VB11D)



These magnet-specific values can be found in the Magnet "INSTALLATION DATA" sheet.

- Measure the helium level.

4.1.6 Final steps

- Performing an ERDU test using the ERDU test load. Refer to (→ ERDU / M6-015.833)

4.2 Replacing the Magnet Supervisory Unit (MSUP 2657B)

4.2.1 Tools and time required

4.2.1.1 Time

60 min

4.2.1.2 Tools

- ERDU test load kit (part no **03861197**)
- Non-magnetic tool kit

4.2.2 Preliminary

Data from the old MSUP must be transferred to the replacement unit .

- Download the magnet specific values from the MSUP to the HOST (items 1,2 and 3 in table below (→ Tab. 13 Page 173)).
 - Open **SeSO > Magnet&Cooling > Initialisation**.
 - If control and communication of the old MSUP still works, item 1,2 and 3 can be downloaded to the host.
- Note the MSUP parameters from the data locations detailed below.

Tab. 13 MSUP 2657B Configurable parameters

	Parameter	Data Location
1	0% helium level calibration He Probe 1,2	Acceptance document/or MSUP download
2	100% helium level calibration He Probe 1,2	Acceptance document/or MSUP download
3	Pressure heater resistance	Acceptance document/or MSUP download
4	Magnet serial number	Acceptance document and Magnet ID plate
5	Magnet revision	Acceptance document and Magnet ID plate
6	Supervisory serial number	Label on MSUP
7	Supervisory revision	Label on MSUP
8	Operating current	Last ramp current saved on MSUP
9	Date	Current date
10	Time	Current time
11	Link and Bore Shield limits	Link and Bore Shield limits
12	CCR Calibration (as required)	Magnet Acceptance Document, or MSUP download.

**CAUTION**

The replacement procedure disables the magnet emergency shut down system at the MSU.

Failure to adhere to safety precautions may result in serious bodily injury or property damage.

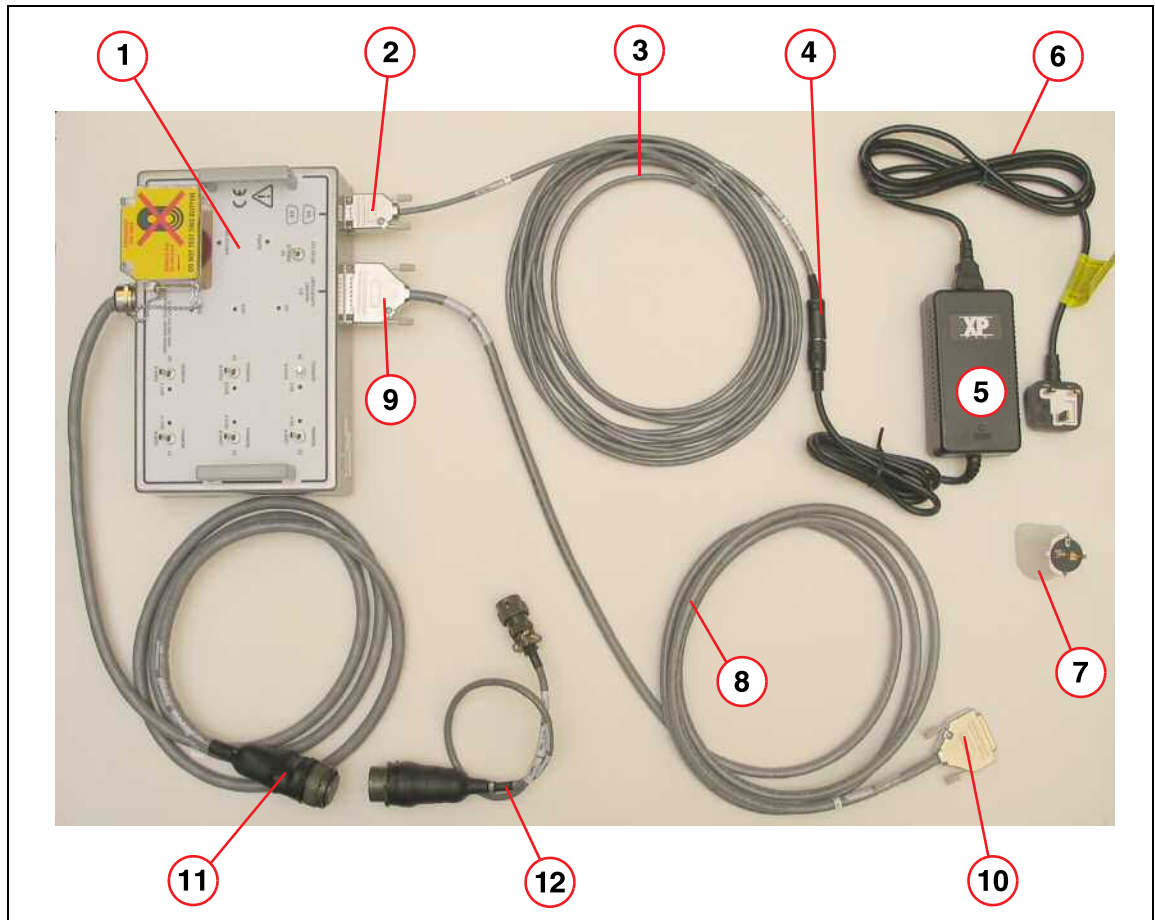
- » Connect the ERDU Test Load.
- » The magnet may only be quenched using the MAGNET STOP button on the ERDU Test Load
- » Exclude all non-service personnel and all magnetic materials / tools from the magnet area.
- » Close the door of the RF room

4.2.3 Prerequisites

1. The ERDU Test Load (ETL) is for use by qualified personnel only.
2. Restrict access to the magnet if it is at field while the ETL is connected to the MSUP.
3. Observe ESD precautions while connecting to or disconnecting from the MSUP.
4. Check that all alarm and fault indications show that the system is OK.

5. Check that the ERDU Test Load Kit is complete with all the listed parts present:

Fig. 245: ERDU Test Load Kit (2661)



- (1) ERDU Test Load 3 (ETL)
 - (2) PSU output extension lead to ETL X3 or X4
 - (3) PSU output extension lead
 - (4) PSU output extension lead connector to PSU
 - (5) Power Supply Unit (PSU)
 - (6) AC supply lead
 - (7) International adapter - AC supply
 - (8) ETL - MSUP lead
 - (9) 25-pin connector to ETL X1
 - (10) 25-pin connector to MSUP X1
 - (11) 41-pin connector, magnet turret
 - (12) Adapter for OR64 / OR70, 41-pin connector to 10-pin magnet turret connector 'B' (optional ETL 3 2661)
6. Remove the service access cover, refer to 'Replacement of Parts'.
7. Remove the turret cover, refer to 'Replacement of Parts'.
8. Connect the Magnet Display Unit to X9 on the MSUP.

4.2.3.1 Magnet display unit (MDU) operations

To initiate a full test sequence, press the **SAMPLE** button on the MDU.

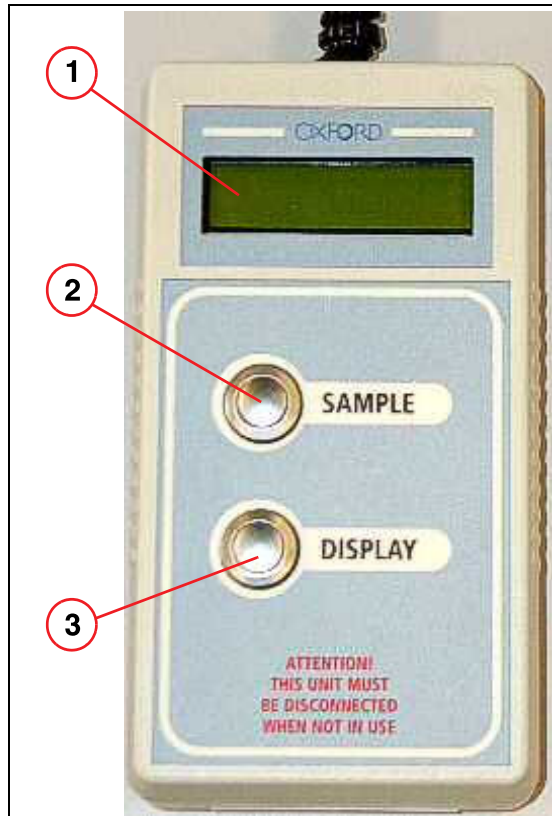
To initiate scrolling functions, press the **DISPLAY** button.

Scroll through the parameters by pressing the **DISPLAY** button, one parameter at a time. The parameters are presented in a set order. If a specific parameter is required, you have to press the **DISPLAY** button repeatedly until that parameter is displayed. The actual level or reading for the parameter is displayed.

If the parameter is incorrect, either (W) or (A) will be displayed next to the parameter reading to indicate either a warning (W) or an alarm (A) situation.

Where a control function is available e.g. EIS, the function may be toggled ON and OFF by pressing the **SAMPLE** button while that function is being displayed.

Fig. 246: Magnet Display Unit



- (1) LCD Display
- (2) Sample (On) button
- (3) Display (Scroll) button

The parameters in order of presentation are:

1. He level (press sample to recheck level)
2. Battery volts
3. 50K link, bore (K, K)
4. Carbon resistor 1, 2 (ohm, ohm)
5. Carbon resistor 3, 4¹ (ohm, ohm)
6. Turret 1, 2 (K, K)
7. Switch heater (Ohm, status)
8. Quench heaters 1, 2 (Ohm)
9. Cold head (K)
10. ERDU buttons (status)
11. Magnet pressure (psiA)
12. Pressure heater (Off/On, %) (press Sample to switch on, 50 min time-out)
13. Pressure heater (average power over last hour of previous measurements)
14. EIS (Off, Sample to switch On, 130 min time-out)
15. Self-test
16. Compressor status (OFF/ON,P,T)
17. LVQD (Waiting trigger/quench location)
 - Time & Date
 - Time and date
 - Magnet type
 - Part no and software version

1. Carbon resistor 4 is shown in Kelvin for CCR type magnets.

4.2.4 Connecting the ERDU Test Load (ETL)



Always use the magnet service platform when working on the service turret.



WARNING

It is not possible to quench the magnet using the normal system **MAGNET STOP** buttons while the ERDU Test Load is connected to the MSUP and magnet.

- » The magnet may be quenched only by using the **MAGNET STOP** button on the ERDU Test Load.
- » Restrict access to the magnet accordingly.



CAUTION

The ERDU Test Load is slightly ferromagnetic.

The ETL is attracted by the magnetic force and projected into the magnet bore, resulting in injury or damage.

- » Keep the ETL as far away as possible from the magnet bore.

1. Connect the ETL PSU to a convenient AC outlet ensuring that the PSU is as far away from the magnet as possible (magnetic field < 50 Gauss).
2. Connect the PSU to the ETL X4 using the extension cable(→ 2/Fig. 245 Page 175) .
 - » Check that the **SUPPLY** and **SWITCH OK** indicators glow on the ETL.
3. Connect the "<MSUP-ETL>" cable to the 25-pin connector X1 **on the ERDU Test Load** (→ 9/Fig. 245 Page 175)
4. Connect the <41-pin to 10-pin adapter> (→ 12/Fig. 245 Page 175)to the 41-pin connector on the ERDU Test Load (→ 11/Fig. 245 Page 175).



WARNING

Magnet quench!

A magnet quench is caused by pressing the ETL Magnet Stop button.

- » Ensure that the **MAGNET STOP** button on the ETL is **NOT OPERATED!**
- » **DO NOT** operate the Magnet Stop button on the ERDU Test Load unless an emergency magnet quench is required.

1. Ensure that the **MAGNET STOP** button on the ETL is **NOT OPERATED!**
2. Set the mode switch to OR64 / OR 70

3. Set all HIGH R / LOW R switches on the ETL to NORMAL.



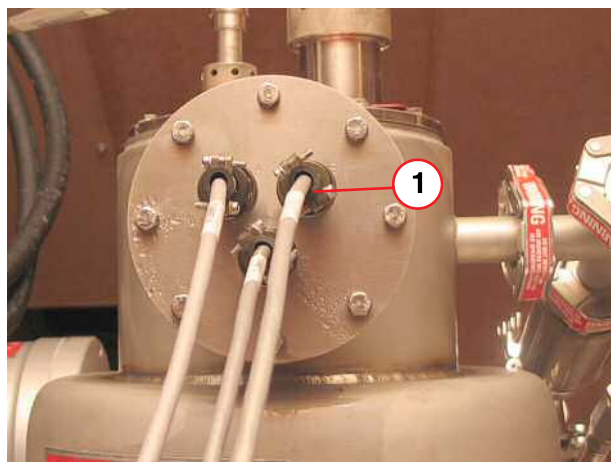
CAUTION

It is not possible to quench the magnet while disconnecting and reconnecting the 10-pin turret connector 'B'.

- » Connect the 10-pin turret connector 'B' without delay.
- » In case of an emergency, immediately connect the 10 -pin turret connector 'B'!

OR 64

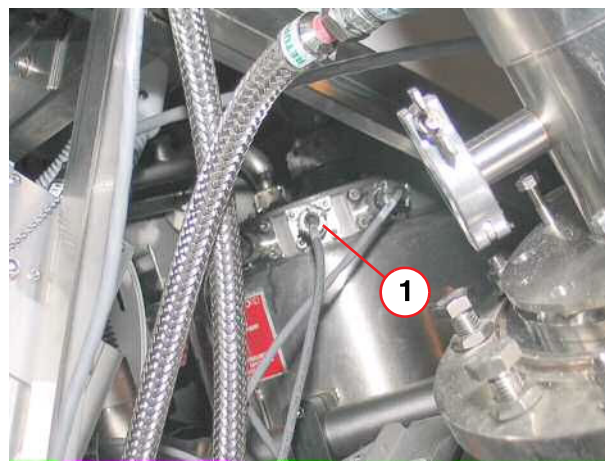
Fig. 247: Magnet turret



(1) Connector 'B'

OR 70

Fig. 248: Magnet turret



(1) Connector 'B'

1. Disconnect connector 'B' on the magnet turret (→ 1/Fig. 247 Page 178) (→ 1/Fig. 248 Page 178) and
2. immediately connect the 10-pin connector (→ 12/Fig. 245 Page 175) from the ETL to the magnet turret connector 'B'. Keep the ETL as far away as possible from the magnet bore.
 - » Check that the alarm sounds and that the MAG STOP indicator lights up on the Alarm Box.
 - » Check that the QH1, QH2, SH and SWITCH OK indicators glow on the ETL.
3. Disconnect the 25-pin connector on the MSUP to the turret lead from X1 on the MSUP and in its place connect the < ETL to the MSUP lead > (→ 10/Fig. 245 Page 175).
 - » Check that the buzzer sounds in the ETL.
4. Acknowledge the alarms at the Alarm Box.

4.2.5 Replacement Procedure

4.2.5.1 LVQD Functionality

To use the MSUP 2657B, the loadware version must be at least **VA07B**. Therefore the minimum **system software** must be **VB13A + SP2**.

Order the correct replacement MSUP according to the spare part list.



MSUP 2657B can be used on any TIM system, as long as the system software has been upgraded to VB13A + SP2 or higher.
X14 should NOT be used on older systems** due to the LVQD circuit invalidating UL approval

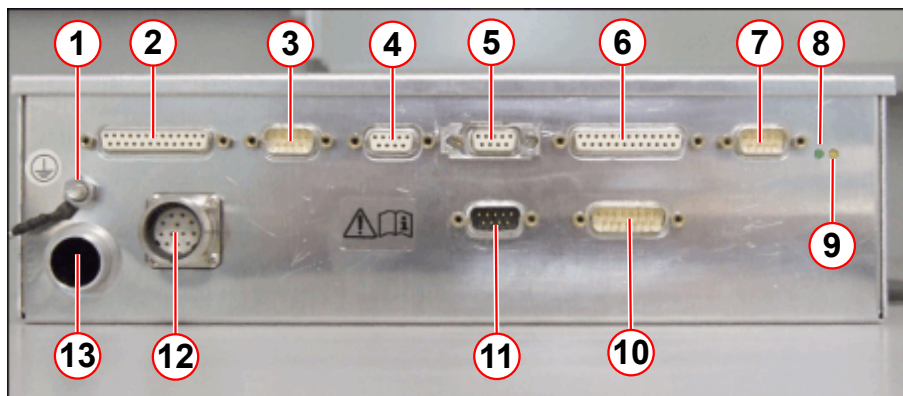
4.2.5.2 Removing the old MSUP

- Remove **fuse F3** (ACC) or turn off **F3** on the ACC.
 - » The power for the MPSU is now **off**.



Use the ESD kit during the following steps.

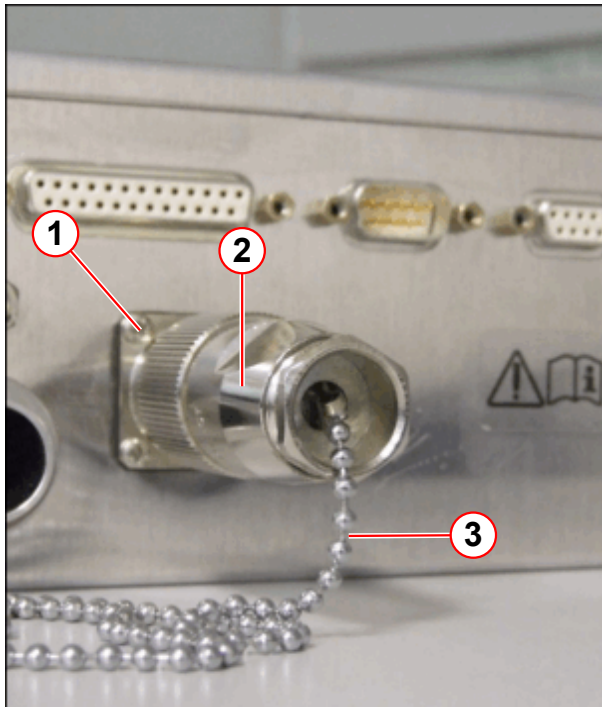
Fig. 249: MSUP 2657B - Indicators and input/output connections



- (1) Earth stud (6mm)
- (2) X9 DISPLAY
- (3) X5 RS232
- (4) X6 ERDU
- (5) X2 CH/DIAG/PRESS
- (6) X1 TURRET
- (7) X3 POWER/SL/UPS
- (8) Power ON LED (green)
- (9) Pressure heater LED (yellow)
- (10) X14 LVQD
- (11) X12 BATTERY
- (12) X7 Magnet ID
- (13) CAN Fiber optics wave guide

- **Disconnect** from the MSUP:-
 - Turret cable **X1**. (→ 6/Fig. 249 Page 179)
 - Power cable **X3**. (→ 7/Fig. 249 Page 179)
 - Battery cable **X12**. (→ 11/Fig. 249 Page 179)

Fig. 250: MSUP 2657B - Magnet ID



- (1) Connector X7
- (2) Magnet ID connector
- (3) Securing chain

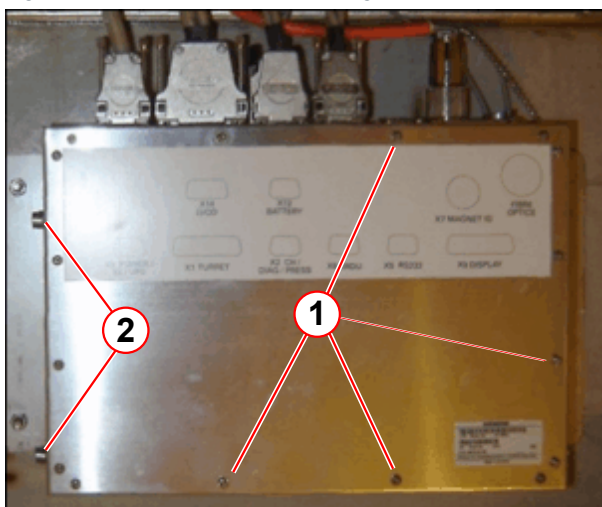
- **Disconnect** the **Magnet ID connector** from **X7** (pos 1).

The **Magnet ID** contains information about this **specific** magnet.

Ensure that the Magnet ID connector is transferred to the replacement MSUP unit.

- **Disconnect** the remaining 'D' type electrical connectors. (→ Fig. 249 Page 179)

Fig. 251: MSUP 2657B - Mounting bolts and screws

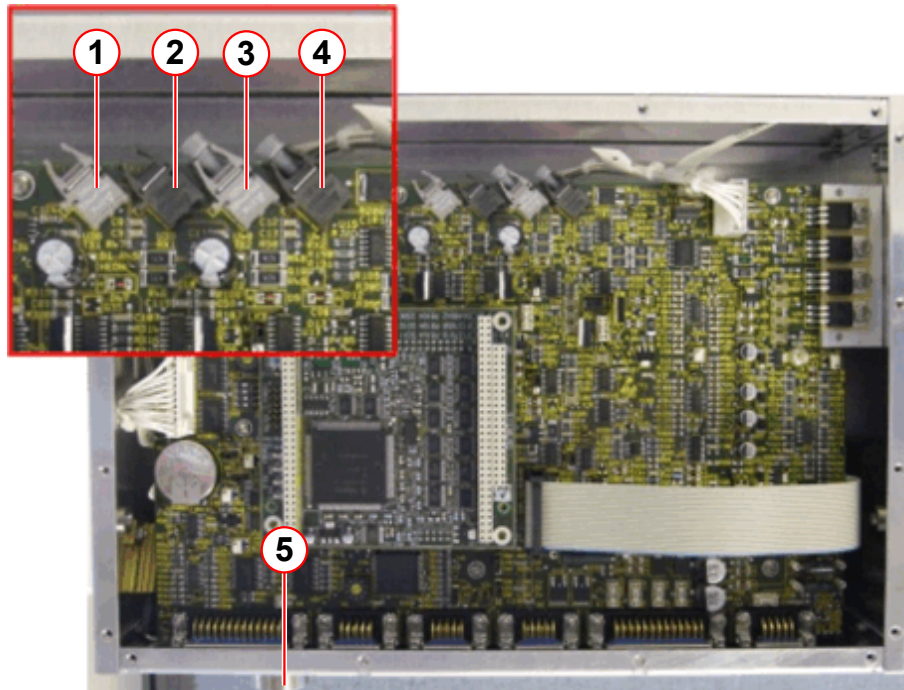


- (1) Retaining screws (16 off)
- (2) Securing bolts and washers (2 off each end)

- **Remove** the lid of the MSUP by removing the screws (16 positions).

- Carefully **disconnect** fiber optic cables **U38, U39, U40, U41** from the connectors on the **MSUP PCB**, and withdraw the **fiber optic cables** through the **wave guide**.
(→ 5/Fig. 252 Page 181)

Fig. 252: MSUP 2657B - Fiber Optic cable connectors



- (1) U38
- (2) U39
- (3) U40
- (4) U41
- (5) CAN Fiber optics wave guide

- **Remove** the (4 off) bolts securing the MSUP to the cradle.
- Carefully lift out the MSUP unit.
- **Re-fit** the lid using the (16 off) screws.
- Return the defective unit to CSML.

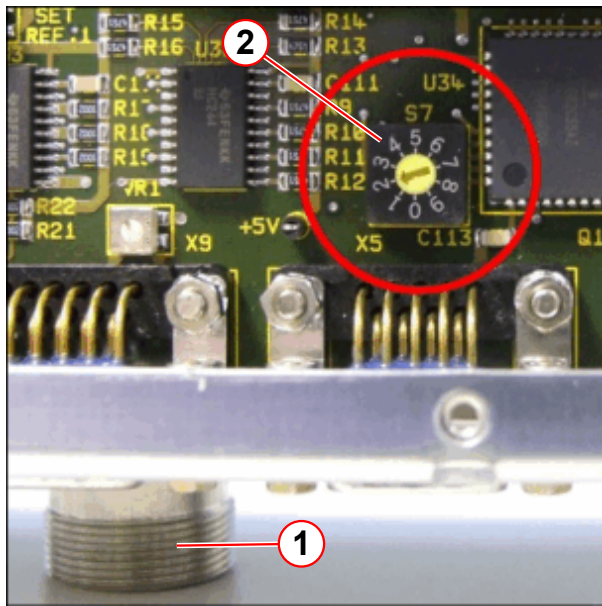
4.2.5.3 Installing the new MSUP 2657B

Tab. 14 ERDU Switch Settings

Number of ERDU Buttons	ERDU Selection Switch
Alarm Box	1
Alarm Box + 1 ERDU Button	2
Alarm Box + 2 ERDU Buttons	3

- **Remove** the lid of the MSUP by removing the screws (16 positions).

Fig. 253: MSUP - 2657B ERDU Selection switch



- (1) Magnet ID connector
(2) ERDU selection switch

- Set the **ERDU switch** as required refer to (→ Tab. 14 Page 181).

- Carefully insert the fiber optic cables **U38, U39, U40, U41** through the **waveguide** and into their correct connectors on the PCB. (→ 5/Fig. 252 Page 181)
- **Secure** the new MSUP to the support cradle using the (4 off) bolts.
- **Refit** the MSUP lid using the (16 off) screws.



Do not mix up the lids of the MSUP. Check the serial number labels.

- **Re-connect to the MSUP:-**
 - External connectors **X2, X6**. (→ Fig. 249 Page 179)
 - **Magnet ID** connector to **X7**. (→ 1/Fig. 250 Page 180)
 - Battery connector **X12**. (→ 11/Fig. 249 Page 179)
 - Power connector **X3**. (→ 7/Fig. 249 Page 179)
- Insert fuse **F3** (ACC), (or switch on F3).
 - » Check that the **green LED** on the MSUP is illuminated. (→ 8/Fig. 249 Page 179)
 - » The Alarm box may sound an **audible alarm** - press the **ACKNOWLEDGE** button to cancel the alarm.



Even though the MSUP is connected to the system, the ERDU buttons will NOT be able to quench the magnet.

- **Re-boot** the scanner.
 - » this will ensure the correct software is installed on the supervisory.
- **Open** SeSo > Magnet&Cooling > Initialization.
 - **Check** the magnet specific values and compare with the **acceptance document**.

- Press the **UPLOAD** button to transfer the magnet-specific values from the **HOST** to the **new MSUP**.
- **Exit** the Magnet&Cooling menu.



If the Host PC is not available, or unable to complete this task then refer to the section 'Calibration of MSUP using a Laptop' (→ Calibration of MSUP using a Laptop / Page 183)

4.2.6 Concluding Checks

- Connect the **ETL-MSUP lead** (→ 8/ Fig. 245 Page 175) to **X1** on the MSUP.
- Perform an **ERDU test** using the ERDU test load - (refer to **Maintenance, Safety-Related Tests- ERDU**).
- **Check** that all alarm and fault indications show that the system is OK.
- **Connect** the MSUP hand-held display (**X9**).
- Perform a **manual sample** (press SAMPLE).
- **Check** that all alarm and fault indications show that the system is OK.
- **Disconnect** the hand-held display from X9.

4.2.7 Calibration of MSUP using a Laptop

4.2.7.1 Direct input to MSUP

If it was not possible to upload the old data to the host, refer to the **acceptance documents**, (refer to (→ Tab. 13 Page 173)) and then do the following:-

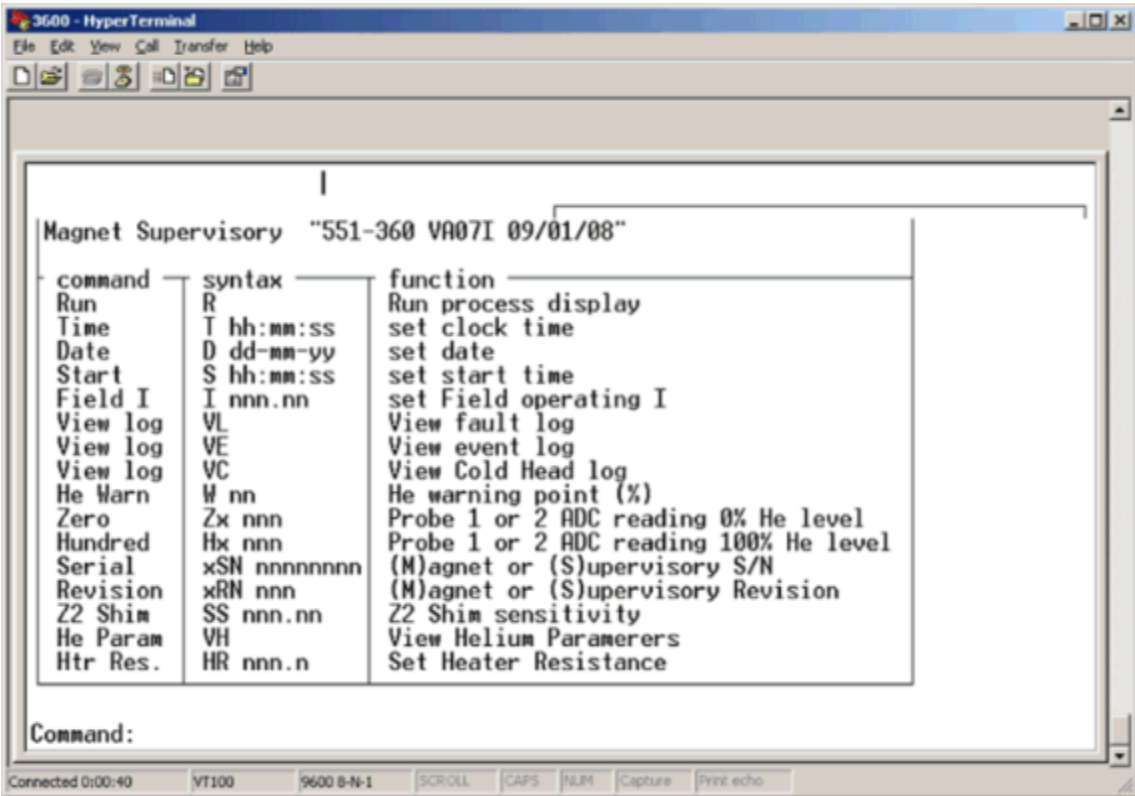
- **Open** the **HyperTerminal** application and select the appropriate **Com Port**.
- Select the following settings:

Baud rate	9600
Bits	8
Parity	None
Stop Bits	1
Flow	None
Terminal Emulation	VT100

- **Connect** the laptop via a **RS232** interface cable to **X5**. (→ 3/ Fig. 249 Page 179)

- Press **Escape** - the **Command Input** screen is displayed.

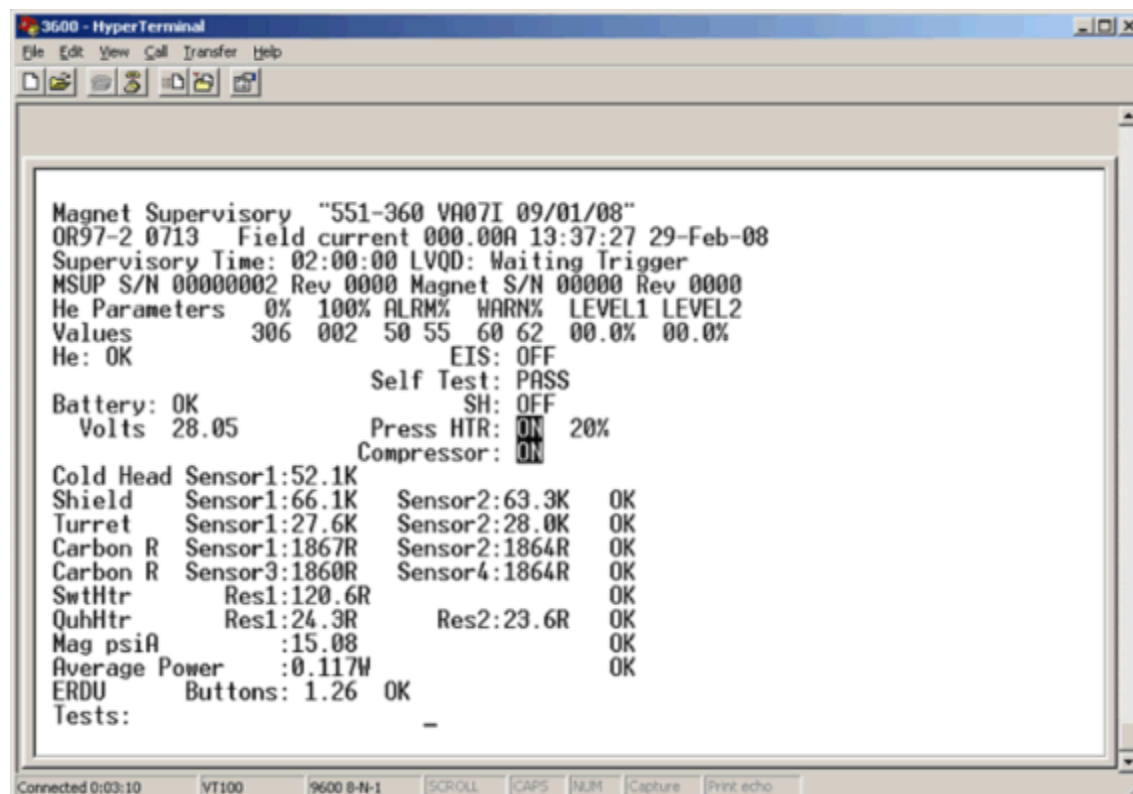
Fig. 254: MSUP 2657B - Command Input screen (typical)



- Enter the required values into the **Command: _** line (refer to (→ Tab. 15 Page 190)).

- Press **R <R>** to run the **Process Display**. The updated data will be displayed.

Fig. 255: MSUP 2657B - Process Display screen (typical)



Calculating Helium Probe Calibration at early sites

The latest version of the acceptance document provided by SMT lists both the resistance and the converted values.

For earlier sites, (where the calculated value may not be available) use the formula below to calculate the calibration values.

$R \times 0.25 \times 36.27 = \text{Input Value}$, where **R = Helium Probe Resistance**.

- Enter the resulting value into the MSUP as a **whole number**. Example:-
 - » Resistance = 88.8 Ohms
 - » $88.8 \times 0.25 \times 36.27 = 805.194$
 - » Value used to calibrate is **805**

All values are entered from the **command menu**. (→ Tab. 15 Page 190)

Calibration of CCRs

On systems fitted with CCRs, the MSUP will require calibration to ensure the temperatures monitored are at the correct value.



On systems without CCRs fitted, the MSUP will NOT require calibration.

The value for **RT, 77K** and **4.2K** can be found on the **Magnet Acceptance document** (TDC Calibration Values)

All values are entered from the **command menu**.

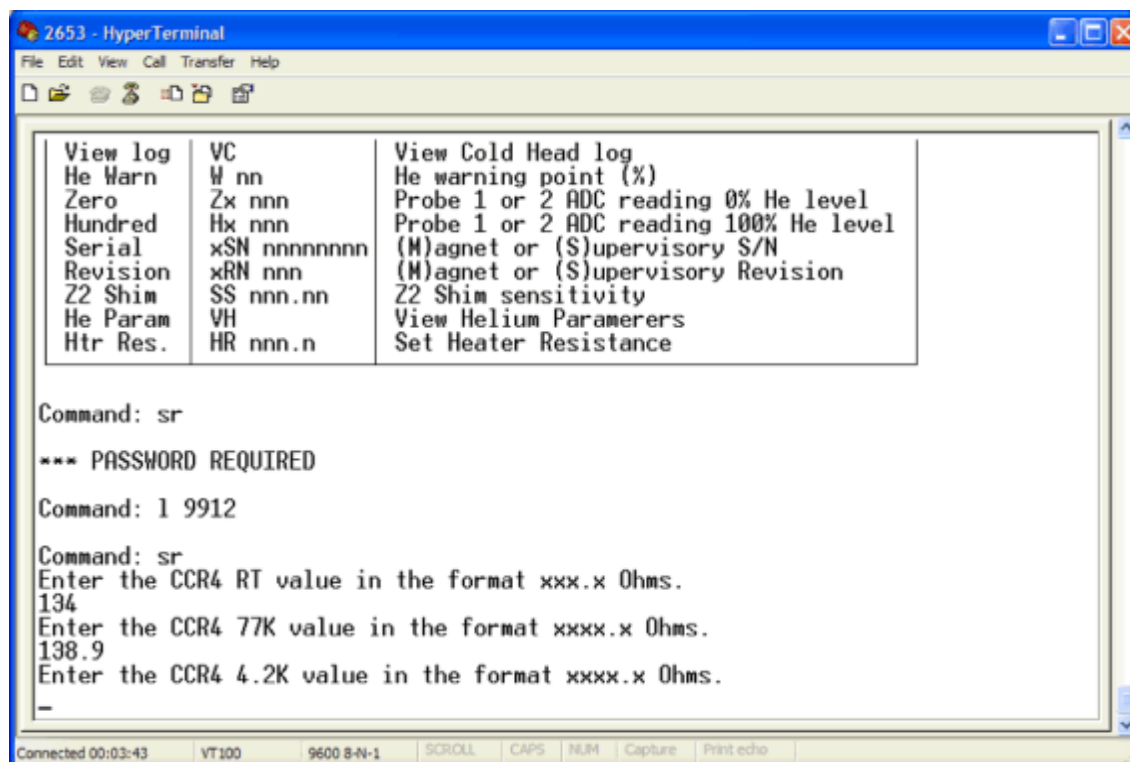
Fig. 256: OR70 Magnet Acceptance Document

SIEMENS		Magnet Acceptance Documentation Installation Data	
Siemens Magnet Technology Quality Regulations			
Cryostat number	21191434	SMT Reference	21192640 10.08
1.5T Actively Shielded Superconducting Twin Turret Magnet System		OR70	
Operating Parameters			
Description	Specification value	Measured value	
Operating current @ 1.4963	≤ 600Amps	584.4	
Field /current Ratio	N/A	10.9	
Magnetic Offset	≤ 7mm	0	
Carbon resistor values			
Position	Room Temperature	Nitrogen Temperature	Helium Temperature
CCR4 TDC D-E	942	1122	2651
CCR1 Top A-E	909	1143	2729
CCR2 Mid B-E	920	1194	2628
CCR3 Bottom C-	963	1230	2719
TDC Calibration Values:	907.6	1173.5	2679.8
Helium level probe reisitance when helium level is zero (%)			
When using a 250 mA supply			
Probe 1	101.02	Probe 2	101.02
	Ohms		Ohms
Probe 1	E43	Probe 2	E43
	HEX code		HEX code
Final temperature sensors values			
20K Bore	80K Bore	20K Link	80K Link
1436	1086	1458	1096
15.52456	52.61249	14.80261	45.56323
Millivolts ° K			
2557 Supervisory values Helium Level Probe Resistance(When using 250mA Supply			
@ 0 Resistance	Ohms	Probe 1	Probe 2
0 % Calibration		101.02	101.02
100 % Resistance	Ohms	916	916
100 % Calibration		0.11	0.11
Diode Table No		1	1
		-2	
Checked by	K Suraina		Signature
Date	09/10/2008		

- At the **Command** line:
 - Enter **SR <R>**

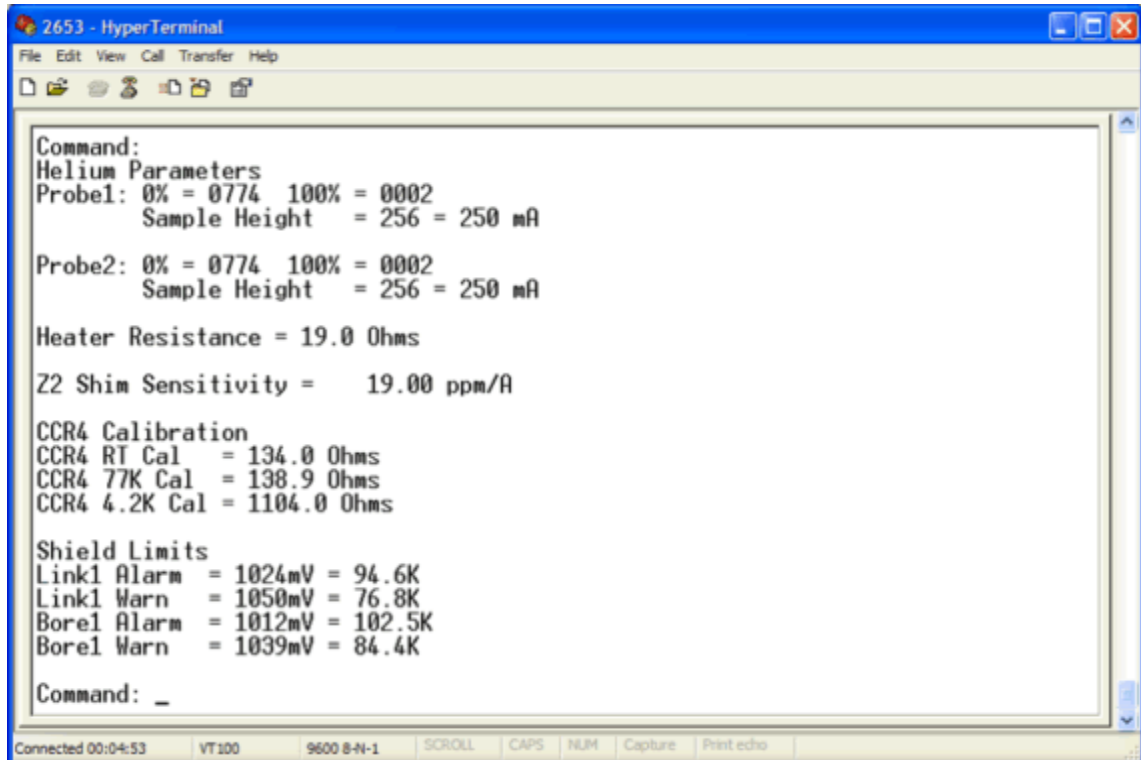
- **PASSWORD REQUIRED** is displayed
- Enter **L 9912** <R>
- Enter **SR** <R>
- Enter the CCR **RT**, **77K**, and **4.2K** values

Fig. 257: MSUP2657B CCR Calibration screen (example)



- Enter **VH** <R> to view, and verify data is correct.

Fig. 258: MSUP2657B View VH download screen (example)



The screenshot shows a HyperTerminal window titled "2653 - HyperTerminal". The window contains the following text:

```
Command:
Helium Parameters
Probe1: 0% = 0774 100% = 0002
        Sample Height = 256 = 250 mA
Probe2: 0% = 0774 100% = 0002
        Sample Height = 256 = 250 mA
Heater Resistance = 19.0 Ohms
Z2 Shim Sensitivity = 19.00 ppm/A
CCR4 Calibration
CCR4 RT Cal = 134.0 Ohms
CCR4 77K Cal = 138.9 Ohms
CCR4 4.2K Cal = 1104.0 Ohms
Shield Limits
Link1 Alarm = 1024mV = 94.6K
Link1 Warn = 1050mV = 76.8K
Bore1 Alarm = 1012mV = 102.5K
Bore1 Warn = 1039mV = 84.4K
Command: _
```

The status bar at the bottom of the window shows: Connected 00:04:53, VT100, 9600 8-N-1, SCROLL, CAPS, NUM, Capture, Print echo.

- Enter **R** <R> to return to the **Process Display**.

Tab. 15 MSUP 2657B Input Commands

Command	Input Data	Value 'n'	Example
0% Calibration	Z1 nnnn<R>	He Probe 1	Probe1: 88.8 ohms -enter as Z1 805<R>
	Z2 nnnn<R>	He Probe 2	Probe2: 88.8 ohms - enter as Z2 805<R>
100% Calibration	H1 nnnn<R>	He Probe 1	Probe1: 1.8 ohms - enter as H1 16<R>
	H2 nnnn<R>	He Probe 2	Probe2: 1.8 ohms - enter as H2 16<R>
Pressure Heater Resistance	*L 9912<R> HR nnn.n<R>	Pressure Heater Res	19.4 ohms - enter as HR 19.4<R>
Magnet Serial No.	*L 9912<R> MSN nnnnnnnn<R>	Magnet Serial No	Enter as MSN12345678<R>
Magnet Revision No.	*L 9912<R> MRN nnn<R>	Mag Rev No	Magnet Rev 4 - enter as MRN4<R>
Supervisory Serial No.	*L 9912<R> SSN nnnnnnnn<R>	Supervisory Serial No.	Supervisory Serial No 26570001 - enter as SSN 26570001<R>
Supervisory Revision No	L 9912<R> SRN nnn<R>	Supervisory Revision No	Supervisory Revision No 14 - enter as SRN14<R>
Magnet Operating Current	I nnn.nn<R>	Mag Operating Current	Operating current 508.23 - enter as I 508.23<R> This parameter is also automatically updated when the magnet is next ramped to field.
Date	D dd-mm-yy<R>	n/a	1st Jan 2006 - enter as D 01-01-06<R>
Time	T hh:mm:ss<R>	n/a	1 hour 30mins 30secs - enter as T 13:30:30<R>
Shield Bore Warning	L4 2947<R> SBW nnn<R>	Shield Bore Warning	Shield Bore warning of 77K - enter as SBW 77<R>
Shield Link Warning	L4 2947<R> SLW nnn<R>	Shield Link Warning	Shield Link warning of 80K - enter as SLW 80<R>

Command	Input Data	Value 'n'	Example
CCR Calibration	L 9912 <R> SR <R> XXX.X <R> YYYY.Y <R> ZZZZ.Z <R>	XXX.X = RT YYYY.Y = 77K ZZZZ.Z = 4.2K	L 9912 <R> SR <R> Enter CCR values as shown in (→ Fig. 257 Page 188)

Notes:

- <R> = 'Enter' or 'Return'
- Where a password is required, type **L 9912<R>** before the command is input. This will 'unlock' the option (otherwise it is not possible to change the value of the parameter)
 - » Only settable from supervisory loadware version **VA06D** onwards
- **CCR Calibration** - The MSUP will prompt the user to input values after the command **SR** has been used.
 - Refer to the **Magnet Acceptance Document** for TDC CCR4 Calibration Values.

4.3 Replacing the MSUP battery

4.3.1 Tools and time required

4.3.1.1 Time

40 min

4.3.1.2 Tools

- Non-magnetic tool kit: Screwdriver, 10 mm nut, 5 mm Allen key



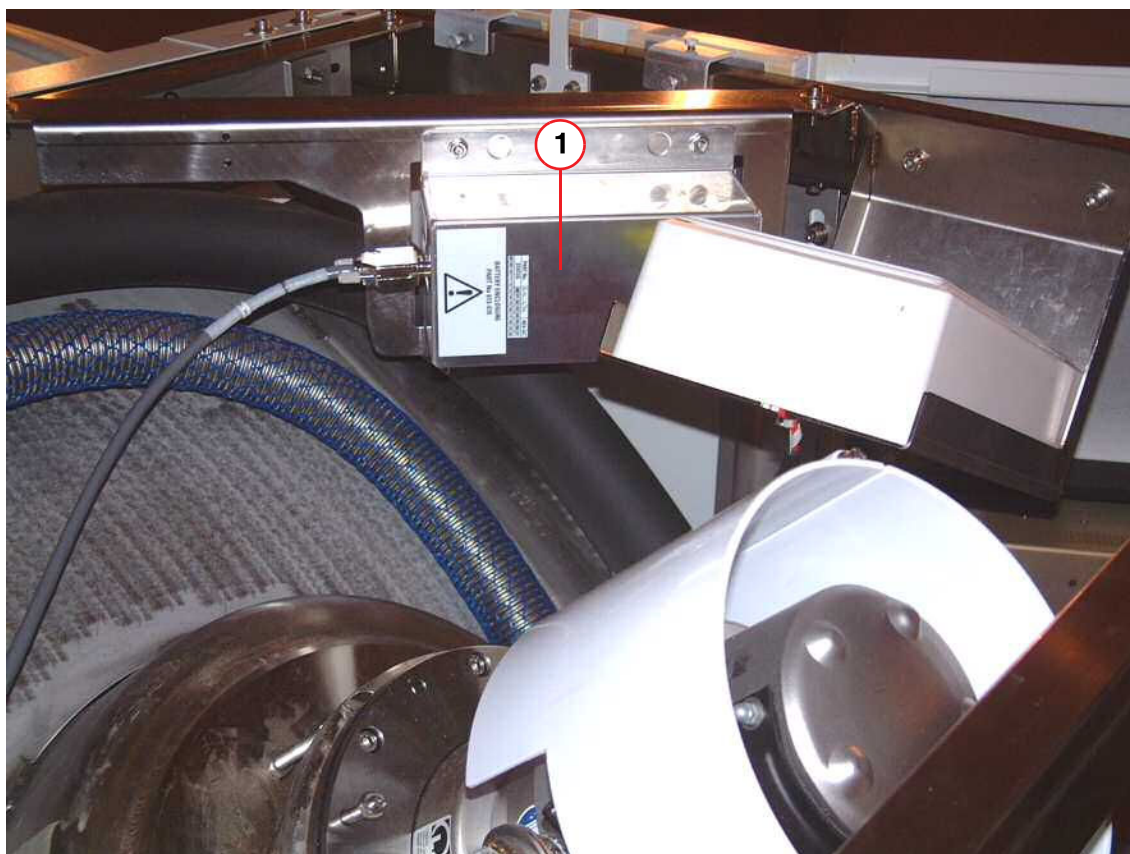
It is possible to disconnect the battery plug without turning off power to the MSUP.

The ERDU function will not be available if a MSUP power failure occurs when the battery is disconnected.

Exclude all non-service personnel from the magnet area.

- Remove the MSUP battery plug

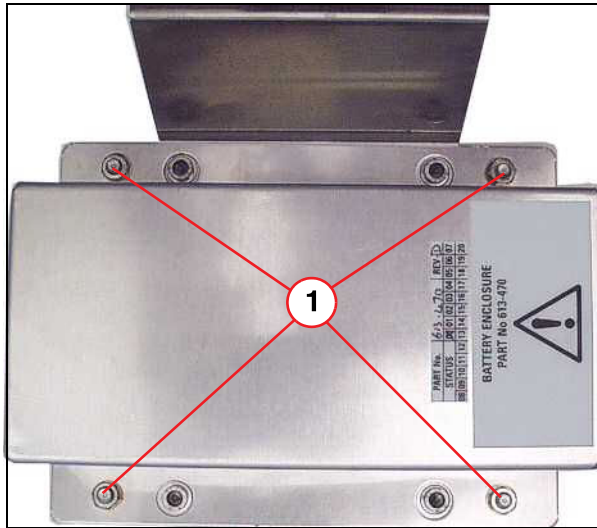
Fig. 259: MSUP Battery



(1) MSUP battery housing

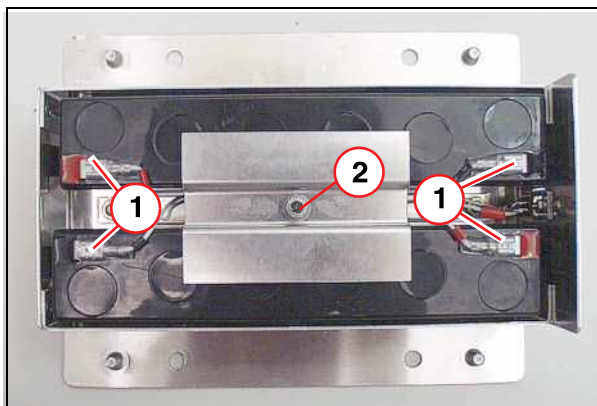
Disassembly:

- Unscrew the nuts (→ 1/ Fig. 260 Page 193)) and remove the cover.

Fig. 260: ERDU battery

(1) 10 mm nut

- Unscrew the Allen screw (→ 2/ Fig. 261 Page 193)) and remove the holder plate.
- Disconnect the battery plugs (→ 1/ Fig. 261 Page 193))

Fig. 261: ERDU battery

(1) Battery plugs

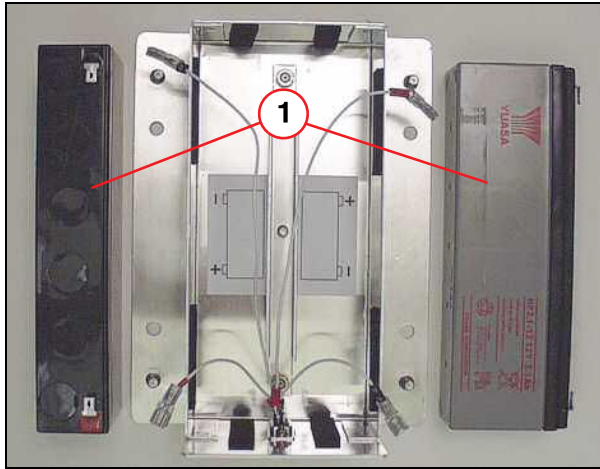
(2) Socked screw

- Remove the batteries. (→ 1/ Fig. 262 Page 194)

- The batteries contain lead.

Turn the batteries over to a disposal company in accordance with local regulations.

Fig. 262: ERDU batteries



Assembly:

1. Mount the new batteries in the tray. Observe the polarity.
2. Connect the battery plugs (→ 1/Fig. 261 Page 193). Observe the polarity once more.
3. Mount the cover and tighten the nuts (→ 1/Fig. 261 Page 193).
4. Connect the MSUP battery plug (→ Fig. 259 Page 192)

5.1 Replacing the Alarm box and cables with the Magnet at Field

5.1.1 Required time and tools

5.1.1.1 Time

60 min

5.1.1.2 Tools

- Non-magnetic tool kit.
- ERDU test load kit (**3861197**).

5.1.1.3 Spare parts

- Alarm box (part no **10096765**).

5.1.2 Preliminary



CAUTION

The replacement procedure will disable the magnet emergency shut down system for short intervals.

Failure to adhere to safety precautions may result in serious bodily injury or property damage.

- » For safety, set the MSUP to the ERDU test setting.
- » Exclude all non service personnel and all magnetic materials/tools from the magnet area.
- » Close the door of the RF room.

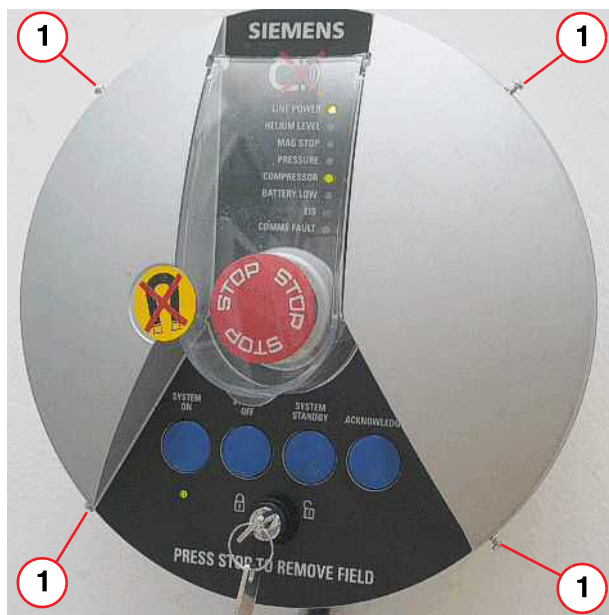
5.1.3 Replacement Procedure



It is important to replace the Alarm box along with the base plate, in order to keep the proper serial number information.

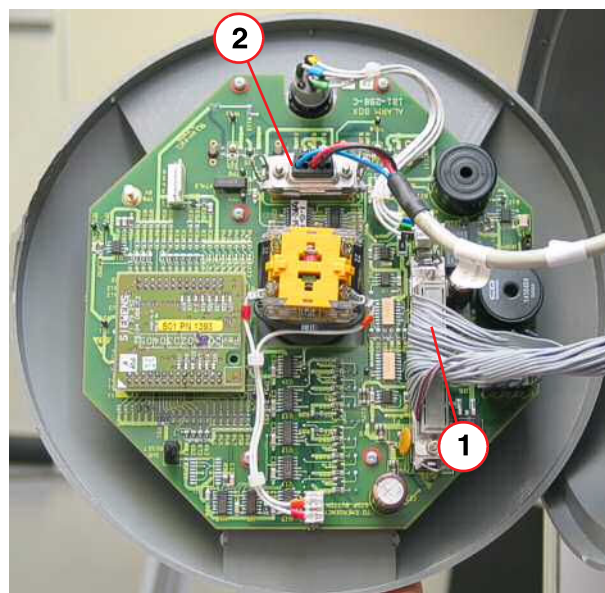
- **Isolate power** to the alarm box:
 - Switch **OFF** circuit breaker **F3** at the **ACC**.
- **Connect the ERDU Test Load** to the MSUP. **Refer to the ERDU safety related test.**

Fig. 263: Alarm box



(1) Securing screws

Fig. 264: Alarm box connectors

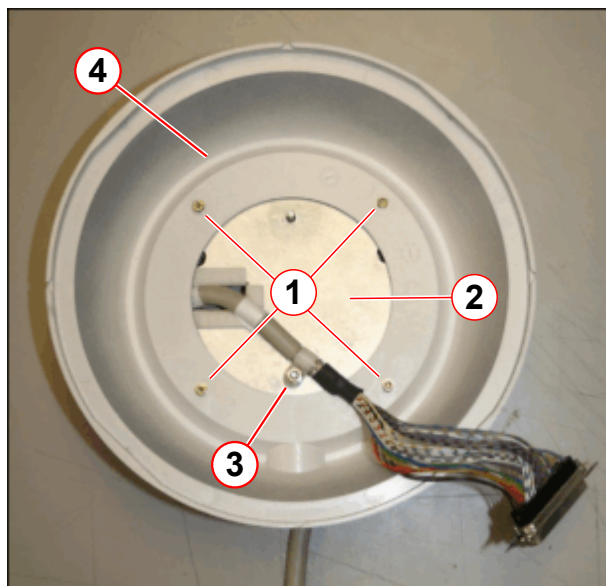


(1) Connector X7
(2) Connector X5

5.1.3.1 Removal

- **Remove** the 4 screws around the outer rim of the alarm box.
- **Separate** the 2 halves of the alarm box.
- **Disconnect** the control and RS232 communications cables (→ Fig. 264 Page 196).

Fig. 265: Back of Alarm Box (inside)

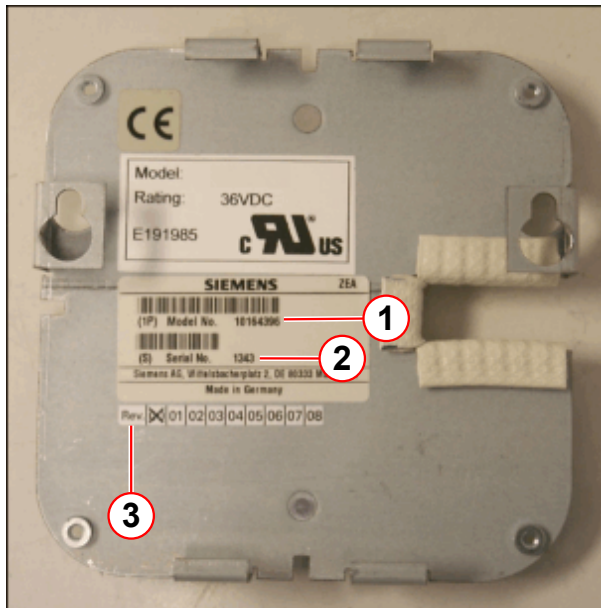


(1) Screws (4 off)
(2) Fixing plate
(3) 'P' clip and retaining nut
(4) Plastic base

- **Remove** the (4 off) mounting screws (1).
- Lift the **plastic base** clear (4).
- **Remove** the retaining **nut** and '**P**' clip (3).
- Loosen the (2 off) **fixing plate mounting screws**.
- **Remove metal fixing plate**.

Installation

Fig. 266: Alarm box identification

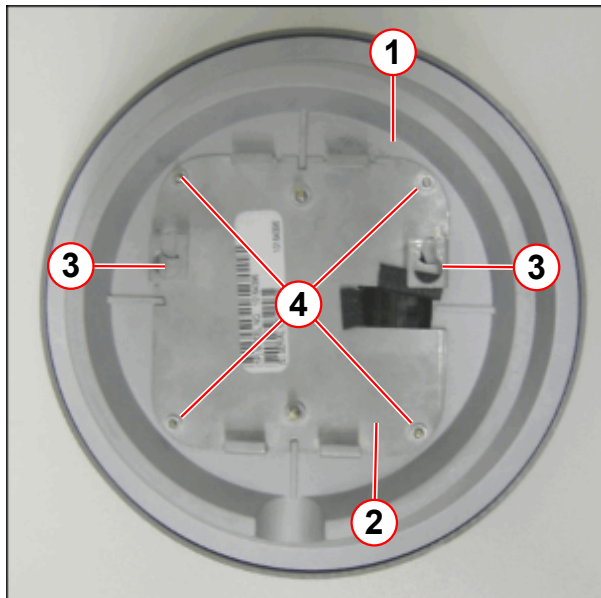


- (1) Revision number
- (2) Serial number
- (3) Part number

Note the alarm box **identification numbers** of the new unit.

- **Remove** the (4 off) screws around the outer rim.
- **Separate** the 2 halves of the alarm box.

Fig. 267: Alarm box base - fixing plate



- (1) Alarm box base
- (2) Fixing plate
- (3) Mounting slots (95mm apart)
- (4) Fixing plate screws (4 off)

- **Remove** the **metal fixing plate** from the plastic base.
 - **Unscrew** the (4 off) hex head socket screws.

- Feed the **control cable** through the slot in the fixing plate. (→ 2/Fig. 267 Page 197)
- **Locate** the fixing plate **mounting slots** over the **wall screws**, and tighten screws. (→ 3/Fig. 267 Page 197).
- Remove the **'P' clip** from the metal fixing plate (→ 3/Fig. 265 Page 196).
- **Secure the cable** through the **'P' clip** and refit to the **metal fixing plate**.
 - Ensure it makes **good grounding contact**.
- Ensure correct **cable channel orientation** for the mounting of the alarm box base.
- **Fit** the **plastic base** of the alarm box to the **fixing plate** with (4 off) screws.
- **Reconnect** the control and RS232 communications cables to **X7** and **X5**.
- **Refit** the alarm box **cover** using the (4 off) fixing **screws**.
- Turn **ON** circuit breaker **F3** at the **ACC**.



The compressor may take up to 20 minutes to restart.

5.1.4 Cable Removal

When **replacing cables** to the alarm box:

- **Isolate power** to the alarm box:
 - Switch **OFF** circuit breaker **F3** at the **ACC**.
- **Connect** the **ERDU Test Load** to the system. **Refer to the ERDU safety-related test**.
- Follow the **alarm box replacement procedure** when removing the covers.
- **Remove** the **'P' clip** on the fixing plate which secures the cable.
- **Remove** any other cable ties from the cables.
- **Carefully** remove the **cable** through the fixing plate. (→ 2/Fig. 265 Page 196)
- **Replace** the cable and **refit** the alarm box cover.

5.1.5 Final Checks

After replacing the alarm box, or one of the cables, an **operational check** must be performed on the new equipment.

- Turn **ON** circuit breaker **F3** at the **ACC**.
- Check the **LED functions** on the alarm box.
- Test the **ERDU circuit function** using the ERDU test load. **Refer to the ERDU safety-related test**.
- **Remove** ERDU test load from the 41 way turret connector and **refit the MSUP cable**.

6.1 Replacing the Shim Power Supply (SPSU)

6.1.1 Tools and time required

6.1.1.1 Time

60 min

6.1.1.2 Tools

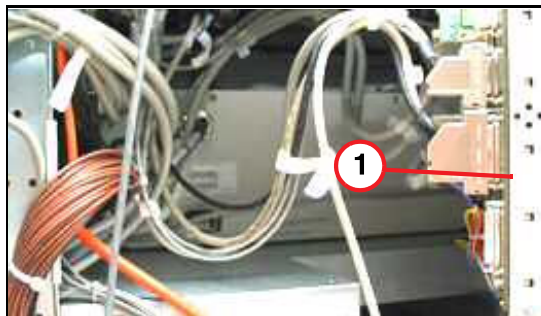
- Standard tool kit

6.1.2 Preliminary

- MR system is switched off

6.1.3 Replacement procedure

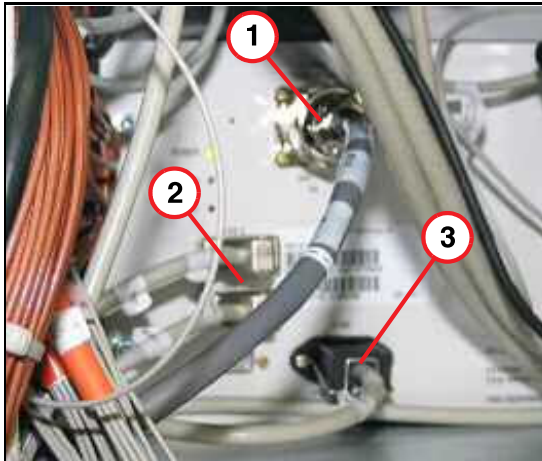
Fig. 268: Shim power supply



(1) AMC frame

- Switch off F21 in the ACC cabinet.
- Swing out the AMC frame.

Fig. 269: Shim power supply



- (1) Output shim cable
(2) Can1 / Can2
(3) X700

- Disconnect X700.
- Disconnect CAN 1 and CAN 2 cables.
- Disconnect the output shim cable.
- Remove and retain the four rack screws securing the shim power supply unit in the cabinet.
- Lift the SPSU clear of the cabinet.
- Place the new SPSU in the cabinet and secure it.
- Connect the cables at the front of the SPSU.
- Switch on F21.
- Switch on the MR system.

6.1.4 Final checks

- Check the function of the SPSU.
- Perform "Phantom Shim" under Tune up
- Perform "Shim Check" under Quality Assurance

6.2 Replacing the Shim Filter Unit

6.2.1 Tools and time required

6.2.1.1 Time

60 min additional ramping time

6.2.1.2 Tools

- Standard tool kit

6.2.2 Preliminary

- MR system is switched off



CAUTION

Magnetic components in the shim filter unit

Failure to adhere to safety precautions may result in serious bodily injury or property damage.

- » The magnet has to be ramped down before removing the shim filter unit.

- Magnet is ramped down



CAUTION

Weight hazard.

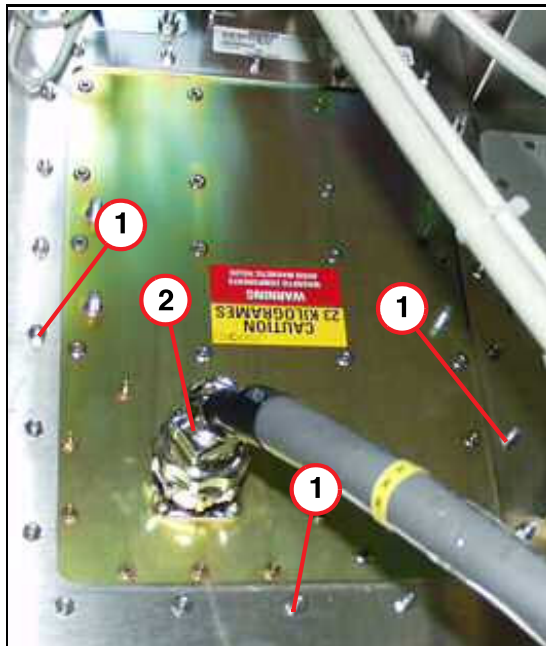
The filter unit weighs 23 kg (50.6 lbs.)

Failure to adhere to safety precautions may result in serious bodily injury or property damage.

- » Ensure sufficient people are available.

6.2.3 Replacement procedure

Fig. 270: Shim filter - equipment room



- (1) Studs
(2) Cable

- Switch off F21 in the ACC cabinet.
- Disconnect cable from the filter unit on the RF room side.
- Disconnect cable from the filter unit on the equipment room side.
- Support the filter unit at the RF room side; this requires a second person
- Remove the the nuts and washers from each of the 22 studs at the equipment room.
- Carefully lift the filter unit.



Before attempting to fit the filter unit in place, inspect the RF gasket. If any damage is determined, the gasket has to be replaced.

- On the RF room side, move the shim filter up to the opening in the RF filter panel and locate the 22 studs in the holes in the RF filter panel.
- Push the filter unit firmly against the gasket.
- At the equipment room, place the washer and nut on each of the 22 studs and tighten nuts to secure the filter unit in place.
- Connect the two cables to the filter unit.
- Switch on F21.
- Switch on the MR system.

6.2.4 Final checks

- Perform "Phantom Shim" under Tune up.
- Perform "Shim Check" under Quality Assurance.

- » The common print number (M6-015.841.01) from Replacement of Parts is divided into separate Component print numbers.

Magnet System	
All chapters	Graphics updated

Version 02

- Chapter 03
 - MSUP 2657B added.

Version 03

- Chapter 01
 - Hint: "Protective earth connection test" was added.
- Chapter 02
 - "Replacing the 10K Cold Head" added
- Chapter 04
 - Update of replacement procedure.

Version 04

- Chapter 05
 - Erdu test load, material number changed in 3861197.(→ Tools / Page 195)

Version 05

- Chapter 02
 - Updated 10K cold head displacer exchange (→ Replacing the Displacer - APD systems / Page 55). (Charm MR_00454054)

Version 06

- Replacing the Displacer - APD system reworked (→ Replacing the Displacer - APD systems / Page 55)

Version 07

- Chapter 01
 - FRU overview added.(→ FRU Overview / Page 6)
- Chapter 03
 - Replacing the F70 Compressor with Tooling kit, added. (→ Replacing the F70 Compressor with Tooling kit / Page 122)

Version 08

- Chapter 03
 - Replacing the Leypold displacer, tooling Cryo Service and Cryo Service Set NMR replaced with 10119990 Displacer Exchange Kit.

There are no Hazard IDs in this document.

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